

# Trinity S<sup>3</sup>AI: Ternary Network Floats

A datapath with no multiplier anywhere: two formats close the two halves of a ternary node, one sign-select primitive builds both a radio and a transformer at zero DSP, and the multiply-free path is exact

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## Abstract

A ternary node has three sites that could carry a number format; one needs to. The weight is a code: multiplying by it is a sign-select. The accumulator alone has a range to spend, and Ternary Network Floats spend it — a sign, a balanced-ternary exponent of  $E_t$  trits, an  $M$ -bit mantissa,  $1 + E_t + M = N$ .

The weight is **GFternary**, a two-bit  $\{-\varphi, 0, +\varphi\}$  alphabet, and it is where the multiplier is decided. Removing one rather than cheapening it requires the weight lattice to be *closed* under weight application and accumulation alike.  $\mathbb{Z}[\varphi]$  is, because  $\varphi^2 = \varphi + 1$ : applying a weight is the Fibonacci step  $(a, b) \mapsto (b, a + b)$ , accumulation is componentwise, and a layer's linear path is exact. Machine-checked in Coq.

A scale is multiplier-free exactly when it is an algebraic integer whose companion matrix has entries in  $\{0, \pm 1\}$ . Degree two admits  $\varphi$  alone, and nothing lies between the shift and  $\varphi$ . The family  $r^d = r + 1$  reaches any granularity at *one addition* and  $d$  registers: fineness costs registers, not adders.

Among the twenty-five carrying their own range — those a training run can live in — GFternary leads on throughput per area, by 10.9% over the next and  $5.24\times$  over **posit32**, with thirteen of the twenty-five ours. On an XC7A200T a ternary neuron costs 28 LUT per weight and *no DSP at any fan-in*.

Which rung follows from the budget —  $\varphi$  at four bits,  $r^5 = r^3 + 1$  at five,  $r^6 = r + 1$  at six — reaching 3.7% of **fp32** at six bits and one adder. On the block axis the element is closed, the best eight-level codebook beating a deployed one by 0.9%, and the scale is not: a  $\varphi$  grid at four bits beats MXFP4 on two models at 0.125 fewer bits per weight.

A guarded 16-bit rung beats IEEE half by 6.92%. Ten retractions are marked in place below.

## 1 What this paper argues

The results below were found in the order a measurement campaign finds things. They compose into one argument, and the argument is easier to follow stated first.

A ternary node has three sites where a number format could live, and only one of them needs one. The weight is a code and multiplying by it is a sign-select; the sample arrives ADC-native; the accumulator is the only object carrying a range that must be spent. So the design question is not which format to use but which *two* objects to close — an alphabet for the weight and a float for the accumulator — and Sections 2 to 6 settle the second by a law: once a range is

named, the split between exponent and mantissa is determined rather than chosen, because the error depends on the mantissa width alone. Inverted, the same law reads a format’s behaviour back out of its round-trip error and resolves an 83-format catalogue into four staircase forms and no fifth.

TNF is one corner of a square, and Section 18 draws it. Two rules fix an exponent budget — the golden-ratio rule  $E = \text{round}((N - 1)/\varphi^2)$  and the width rule  $1 + E + M = N$  — and two radices encode it, binary or balanced ternary. The four families are GF, GF-T, BNF and TNF, each a distinct format rather than a renaming of another, and at eight bits the two rules turn out to name the identical layout.

The alphabet is where the multiplier is decided, and the deciding property is algebraic. Section 19.3 shows that removing a multiplier — as against making one cheaper — requires the weight lattice to be *closed* under the operation that applies a weight.  $\mathbb{Z}[\varphi]$  is, because  $\varphi^2 = \varphi + 1$ , so a weight application is the Fibonacci step and an accumulation is componentwise integer addition; Fibonacci integers are not, because  $F_4 F_4 = 9$  is not a Fibonacci number, and a datapath built on them must carry a stage returning products to the representable set. That stage measures an order of magnitude more area than the entire closed accumulation, and pipelining does not rescue it: a normalisation cascade cannot be cut below one compare-and-subtract per stage, so its floor lies above a closed path’s single addition. The closure half of the argument is machine-checked.

Closure does not single out  $\varphi$  by itself, and Section 19.4 says what it does single out. A scale is applicable without a multiplier exactly when it is an algebraic integer whose companion matrix has entries in  $\{0, \pm 1\}$ , which is a finite condition and therefore enumerable rather than arguable. Degree one gives the shift; degree two gives  $\varphi$  and nothing else; degree three gives three more, finer. Between the shift and  $\varphi$  there is nothing, and  $\varphi$  is the only multiply-free refinement of the power-of-two ladder available at two registers. The hierarchy continues — and stops being cheap at degree four, where the coefficient vector loses its sparsity and the cost doubles.

Which rung to use is then an empirical question with a law behind it (Section 20). A ladder of ratio  $r$  with  $n$  magnitudes spans  $r^{n-1}$ , so fine steps are bought with range, and the optimum moves down the hierarchy as the code budget grows: the shift at three bits,  $\varphi$  at four, the plastic number at five, on two models with the same ordering throughout. The optimum has a closed form in the weight histogram, transferring between models to four decimals, and the one place it errs — the four-bit pair — is corrected by weighting the error with the loss curvature rather than the weight magnitude.

Section 21 reports where none of this wins. On a short block with a shared scale the optimal eight-level codebook beats a deployed one by 0.9%, so the axis is not an element-format problem at all, and it belongs to MXFP4. We state it as the competition’s result and give the bound that makes a fourth attempt of ours unwarranted.

Two sections are about cost rather than kind. Section 13 makes area and precision commensurable, so a variable-length exponent field can be priced in mantissa bits of silicon; Section 14 then finds that for a logarithmic format the field layout is the smaller half of that cost, and that we had been measuring the wrong half.

Ten of our own claims were falsified by the measurements meant to support them, and each is retracted where it was made rather than quietly dropped. The sixth, Section 31, accounts for four published disagreement figures at once, because they turned out to share one cause: the comparison, not the thing compared. The last four were found by an independent review rather than by us: a caveat this paper computed rather than measured, a sentence defending our own first place, a ranking never tested against its own instrument’s resolution, and four ladder rungs whose exponents alias. Section 35 generalises the shape those errors took, because in almost every case the measurement was correct and the comparison around it was not.

Table 1: The precision law and the flatness, measured in exact rationals. “Flatness” is the ratio of the worst magnitude band to the best: 1.000 would mean the error does not depend on magnitude at all.

| rung    | $M$  | $2^{-(M+1)}$ | measured  | ratio | flatness |
|---------|------|--------------|-----------|-------|----------|
| TNF8    | 4    | 3.12e−2      | 1.22e−2   | 0.39  | 1.000    |
| TNF16   | 9    | 9.77e−4      | 3.60e−4   | 0.37  | 1.070    |
| TNF32   | 25   | 1.49e−8      | 5.49e−9   | 0.37  | 1.070    |
| TNF64   | 52   | 1.11e−16     | 4.22e−17  | 0.38  | 1.050    |
| TNF128  | 115  | 1.20e−35     | 4.44e−36  | 0.37  | 1.070    |
| TNF256  | 242  | 7.07e−74     | 2.69e−74  | 0.38  | 1.050    |
| TNF512  | 497  | 1.22e−150    | 4.51e−151 | 0.37  | 1.070    |
| TNF1024 | 1006 | 7.29e−304    | 2.77e−304 | 0.38  | 1.050    |

## 2 The format

$$\text{TNF16} = [s \mid E=4 \text{ balanced-ternary trits} \mid M=11 \text{ bits}],$$

$$v = (-1)^s \left(1 + \frac{M}{2^9}\right) 2^e, \quad e = \sum_i t_i 3^i \in [-40, 40]. \quad (1)$$

Three properties follow from the layout rather than from tuning.

**No regime decode.** The dominant cost in a tapered format is the variable-length regime: it must be located, its length counted, and the significand barrel-shifted into place. That cost is paid on any fabric, ternary included. Fixed fields make it a bit slice.

**The exponent is already ternary.** Four balanced-ternary trits give  $3^4 = 81$  exponent steps. On a ternary fabric the exponent add is native — no binary carry, no conversion between bases.

**Precision does not taper.** Nine mantissa bits at every magnitude, where a tapered format narrows towards the extremes. This is the mechanism behind the accuracy result of Section 5: the win appears where the comparison format has spent its mantissa on range.

## 3 Why a ladder, and why this one

A format that is used as a reference has different requirements from one used as a workhorse. A workhorse should be fast and accurate where the data is. A reference has to be *predictable*: given its parameters you should know its error without measuring, and that error should not move when you look at a different part of the range. A ruler whose divisions grow towards one end is not a ruler.

Table 1 tests both properties on this ladder.

**The precision is a closed form, and it is derived rather than fitted.** Write  $u = 2^{-(M+1)}$  for the unit roundoff. Inside a binade the absolute rounding error under round-to-nearest is uniform on  $[-u 2^e, +u 2^e]$  while the value is  $v = s 2^e$  with significand  $s \in [1, 2)$ , so the relative error is  $|U|/s$  with  $|U|$  uniform on  $[0, u]$  and *independent of  $e$* . Hence

**Theorem 1** (Precision law). *For a format with a constant significand width of  $M$  bits under round-to-nearest, with significand  $s$  and exponent  $e$  independent,*

$$\mathbb{E}[|rel\ err|] = \frac{1}{2} \mathbb{E}[1/s] \cdot 2^{-(M+1)},$$

independently of  $e$ .

The exponent drops out entirely — that is the flatness of the next paragraph, proved rather than observed. The constant is not universal: it is  $\frac{1}{2} \ln 2 = 0.3466$  for a significand uniform on  $[1, 2)$  and  $\frac{1}{2} (2 \ln 2)^{-1} = 0.3607$  under a logarithmic (Benford) significand. Our workload has  $\mathbb{E}[1/s] = 0.7721$ , predicting 0.3861; the measured figures in Table 1 average 0.3756 over eight rungs, with a spread of 0.369–0.390 that brackets it.

The practical consequence is the one a reference needs: given  $M$  and the significand distribution of a workload, the error of a rung nobody has built is known before it is built.

**The error does not move with magnitude, and cannot.** The derivation above removes  $e$  from the expression, so magnitude-independence is a property of the layout rather than a happy measurement. Empirically flatness stays between 1.05 and 1.07 — at most a 7% difference between the closest and furthest parts of a workload spanning 76 binary decades. This is the property fixed fields buy and the one a tapered format cannot have: spending mantissa bits on range is exactly what makes precision depend on where you are.

**What this does not claim.** Not that TNF is the most accurate format at any given width — Section 6 shows posit16 and binary16 ahead near unity, and Section 6 shows the binary formats ahead outright at 32 bits. A reference does not need to win those comparisons. It needs to be the thing you measure them against.

## 4 The law is general, and it measures tapering

Section 3 derived the error of a fixed-field format. Nothing in that derivation is specific to TNF, so it should hold for any format whose significand width does not vary — and should visibly fail for one whose width does. That makes it an instrument rather than a description, and this section uses it as one.

**Definition 1** (Effective mantissa). For a format and a magnitude band  $B$ , define

$$M_{\text{eff}}(B) = -\log_2 \left( \frac{2 \mathbb{E}[|\text{rel err}| \mid B]}{\mathbb{E}[1/s]} \right) - 1,$$

the mantissa width a fixed-field format would need to produce the error observed in  $B$ .

**Theorem 2** (Diagnostic).  $M_{\text{eff}}$  is constant across bands if and only if the format holds a constant significand width over those bands and does not exhaust its range. For a tapered format  $M_{\text{eff}}$  decreases with  $|e|$ , and the slope  $dM_{\text{eff}}/d|e|$  is the taper rate in bits of significand per binade.

Table 2 applies it. The measurement is a uniform significand on  $[1, 2)$  so that  $\mathbb{E}[1/s] = \ln 2$  exactly, removing the workload from the question.

**The law recovers what the formats declare.** TNF16 and GF16 return 8.99–9.01 against a declared 9; bfloat16 returns 7.00–7.04 against a declared 7. A law that reconstructs a format’s own parameter to a hundredth of a bit, from nothing but its round-trip error, is doing more than curve-fitting.

Table 2: Effective mantissa by magnitude band, and the fitted taper rate. Declared  $M$  is recovered to 0.01 bits for every fixed-field format, which validates the law; the tapered formats separate cleanly.

| format   | declared $M$ | $ e $ 0–6 | 6–12  | 12–20 | 20–30 | bits/binade    |
|----------|--------------|-----------|-------|-------|-------|----------------|
| TNF16    | 9            | 8.99      | 8.99  | 9.00  | 9.01  | +0.001         |
| GF16     | 9            | 8.99      | 8.99  | 9.00  | 9.01  | +0.001         |
| bfloat16 | 7            | 7.04      | 7.04  | 7.00  | 7.04  | −0.000         |
| binary16 | 10           | 10.01     | 10.01 | 7.41  | 0.52  | —              |
| posit16  | —            | 10.72     | 9.34  | 7.48  | 5.17  | − <b>0.254</b> |
| takum16  | —            | 9.62      | 8.17  | 7.34  | 7.04  | − <b>0.113</b> |

**Tapering becomes a number.** posit16 sheds 0.261 bits of significand per binade ( $R^2 = 0.999$ ) and takum16 appears to shed 0.113 over the thirty binades of this sweep — but see Section 4.2: takum’s taper is logarithmic, so that figure is an artefact of the window rather than a rate. That is the quantity a tapered format trades for range, and to our knowledge it is not usually reported as a single figure. It also explains the accuracy tables directly — posit16 starts 1.7 bits ahead of TNF16 and ends 3.8 behind.

**Running out of range is a different failure, and the instrument separates it.** binary16 is a fixed-field format, yet its  $M_{\text{eff}}$  collapses from 10.01 to 0.52. It is not tapering: it is subnormalising and then overflowing. A constant slope indicates a taper; a cliff indicates a range limit. The two look alike in an accuracy table and do not look alike here.

**Corollary 1.** *A format’s position on the range–precision trade can be stated as a pair: the constant  $M_{\text{eff}}$  it holds, and the number of binades over which it holds it. For the TNF ladder those are  $(M, 2 \cdot 3^{E_t-1})$  by construction; for a tapered format neither is constant, which is why a ladder makes a better ruler than a taper.*

#### 4.1 The diagnostic across a catalogue

Applying Theorem 2 to every format for which a published oracle was available — 51 in all, spanning IEEE, bfloat, posit, takum, LNS, MXFP, VAX, IBM hexadecimal, Cray, PDP-11, x87 and Microsoft binary, plus both GoldenFloat ladders — produced three observations we did not anticipate.

**Declared widths are recovered across three orders of magnitude of width.** The GF ladder returns 8.97 at GF16 against a declared 9, 78.03 at GF128 against 78, and 631.99 at GF1024 against 632. The same holds for double-double (106.68), quad-double (215.90) and x87’s 80-bit format (63.04). A relation that reconstructs a declared parameter from round-trip error alone, at widths from 8 to 1006 bits, is doing something other than describing the format it was derived on.

**The taper rate is a constant of the family — but only posit has a rate.** posit16 sheds 0.261 bits of significand per binade and posit32 sheds 0.247, against the  $2^{-es} = 0.25$  that Theorem 3 below requires; posit64, whose oracle carries the pre-standard  $es = 3$ , sheds 0.123 against a required 0.125. Three formats, three exact confirmations. We previously reported a comparable per-binade rate for takum, and that was a category error on our part: takum’s staircase is logarithmic, so fitting a line to it produces a number that depends on the interval fitted and describes nothing. Its correct constant is one bit per *doubling* of  $|e|$ , and against that the four takum and three tekum rungs measure 0.83 to 1.09.

**Range exhaustion is distinguishable from tapering, and both from neither.** The distinction is not optional: our own first pass placed the measurement bands at fixed positions rather than inside each format’s measured range, and read the saturation of every narrow format as a taper — classifying TNF16, a fixed-field format, as geometrically tapered. Measuring each format’s usable range first and then placing the bands inside it removes the artefact entirely. GF14 reads flat at 8.05 across its range and collapses only below its subnormal boundary, which is not a defect; the instrument says which of the two it is, where an accuracy table alone does not.

## 4.2 Tapering has a functional form, and it follows from the encoding

Section 4.1 reported a taper rate per family. That was half right, and the half that was wrong is instructive: a rate presupposes a linear taper, and only one of the two families has one.

**Theorem 3** (Posit taper, exact). *For a posit of  $n$  bits with exponent field  $es$ , the regime is a run of  $k$  like bits carrying a scale of  $\text{used}^k = 2^{2^{es}k}$  and occupying  $k + 1$  bits. Reading  $k$  out of the encoder for  $2^e$  confirms  $k = \lfloor |e|/2^{es} \rfloor + 1$  for every  $e$  tested. The significand therefore loses*

$$\Delta M_{\text{posit}}(e) = \left\lfloor \frac{|e|}{2^{es}} \right\rfloor \quad \text{bits: a staircase, arithmetic in } |e|.$$

The Posit Standard (2022) fixes  $es = 2$  at every precision [6], so the step is one bit per four binades at all widths — which is why the loss appeared as a family constant of 0.25 bits per binade. Measured slopes of  $-0.2497$  and  $-0.2505$  at posit16 and posit32 are the staircase seen through a linear fit.

**Theorem 4** (Takum taper, exact). *Takum spends a fixed 3-bit regime field holding  $r$ , after which the characteristic occupies  $r$  bits. Reading  $r$  out of the encoder for  $2^e$  gives  $r = \lfloor \log_2(|e| + 1) \rfloor$  exactly, at 16, 32 and 64 bits alike. The significand therefore loses*

$$\Delta M_{\text{takum}}(e) = \lfloor \log_2(|e| + 1) \rfloor \quad \text{bits: a staircase, geometric in } |e|.$$

**Corollary 2** (The two families differ in kind, not degree). *Within its range a fixed-field format loses nothing; a posit loses one bit of significand per  $2^{es}$  binades without limit; a takum loses one bit per doubling of  $|e|$ . Tabulating  $\Delta M$ :*

| $ e $  | fixed-field | posit ( $es=2$ ) | takum |
|--------|-------------|------------------|-------|
| 30     | 0           | 7                | 4     |
| 100    | 0           | 25               | 6     |
| 1,000  | 0           | 250              | 9     |
| 10,000 | 0           | 2,500            | 13    |

At  $|e| = 1,000$  a posit would need 250 bits of significand merely to pay for its regime, which is why 16- and 32-bit posits cannot reach there at all. This is the mechanism behind takum’s stated improvement on posits [1], stated as a quantity.

**Corollary 3** (A rate is not always the right summary). *Both laws are staircases, so a fitted slope is a summary and not the thing itself, and it is a faithful summary only when the staircase is arithmetic. An earlier draft of this paper reported  $-0.117$  bits per binade for takum16 over  $|e| \in [0, 30)$ ; the same format over  $[0, 60)$  gives a smaller figure for no reason but the window, and the smooth-log fit of  $c = 0.75$  understates the true step of one bit per doubling for the same reason. Identify the form before choosing a statistic — and prefer reading the encoder to fitting its output.*



Table 3: The catalogue by staircase form. The parameter is the taper constant in the units natural to each form. Measured against each format’s own usable range.

| form       | $M_{\text{eff}}$ against $ e $ | count | members and constants                 |
|------------|--------------------------------|-------|---------------------------------------|
| constant   | flat                           | 38    | IEEE, bfloat, VAX, Cray, x87, GF, TNF |
| wobble     | periodic, no trend             | 3     | IBM hexadecimal, $\log_2 16 = 4$ bits |
| arithmetic | $-2^{-es}$ per binade          | 5     | posit16/32 0.25, posit64 0.125        |
| geometric  | $-1$ per doubling              | 8     | takum $\times 4$ , tekum $\times 3$   |

**Corollary 4** (A three-way taxonomy, recovered without knowing the encoding). *The diagnostic separates formats by the form of  $M(e)$ : constant (fixed-field, TNF and GF by construction), linear in  $|e|$  (posit), or logarithmic in  $|e|$  (takum). Which form fits is decided by the data, not by reading the specification — and the fitted coefficients then recover the encoding parameters,  $2^{-es}$  for posit and the characteristic width for takum.*

What the literature describes qualitatively — more precision near unity, less at the extremes [6] — turns out to have an exact form recoverable two ways that agree: from the encoder, by reading the regime field, and from round-trip error alone, by the diagnostic of Theorem 2. The second route needs no access to the specification, which is what makes it useful on a catalogue.

### 4.3 Four staircase forms, and only four

Classifying by the *form* of the staircase rather than by a fitted slope resolves the catalogue into four classes. Of the 83 entries, 11 are integer or fixed-point formats with no exponent field; of the remaining 72, 18 span fewer than six binades and cannot be classified from round-trip error at all, and one requires probes wider than we generated.

The fourth class was not in our scheme until the catalogue produced it, and it has a closed form.

**Theorem 5** (Precision wobble). *Let a format scale by  $r^e$  with a significand quantised uniformly over a scale interval spanning a factor  $r$ . Its relative error varies by a factor of  $r$  across that interval, so  $M_{\text{eff}}$  oscillates with amplitude exactly  $\log_2 r$  bits and has no trend in  $|e|$ .*

*Proof.* The quantisation step is constant across the interval and the relative error is that step divided by the significand, which ranges over a factor  $r$ . Taking  $-\log_2$  of the ratio of the extremes gives  $\log_2 r$ . Since the interval repeats identically at every  $e$ , the oscillation is periodic and carries no trend.  $\square$   $\square$

Theorem 5 reproduces a result sixty years older than this work. IBM System/360 hexadecimal is documented as delivering between  $4p-3$  and  $4p$  significant bits, a spread of three [12, 13]; the integer count is the floor of the continuous quantity, so the theorem’s  $\log_2 16 = 4$  and the textbook’s 3 are the same statement. Measured over eight sub-intervals — which by averaging within each bin predicts a reduced spread of 2.96 rather than the full 4 — `ibm_hfp32` and `ibm_hfp64` give 3.13 and 3.04, while `binary32` and `gf32` give 0.85 and 0.93 against a predicted 0.87. Binary formats wobble by one bit; this is the familiar 1-bit wobble of every radix-2 float, and it is the smallest wobble any integer radix admits. Theorem 5 therefore supplies a second and independent argument for TNF’s binary scale, alongside the 0.331 positions of Theorem 10 below: a radix-3 scale would wobble by 1.585 bits instead of 1.

The taxonomy also settles why a format must choose between the first class and the last two, and hence why this work is a ladder rather than a format.

Table 4: Round-trip mean relative error. Bins are powers of two, not decades.

| magnitude           | GF16 ( $\varphi$ ) | <b>TNF16</b> | tekum16 | TNF16 vs tekum16 |
|---------------------|--------------------|--------------|---------|------------------|
| $ e  < 8$           | 3.43e−4            | 3.56e−4      | 3.27e−4 | 0.92× (tie)      |
| $ e  8\text{--}20$  | 3.57e−4            | 3.52e−4      | 1.00e−3 | <b>2.84</b> ×    |
| $ e  20\text{--}38$ | 6.98e−3 (478 clip) | 3.53e−4      | 1.95e−3 | <b>5.53</b> ×    |

**Theorem 6** (Range–precision dichotomy). *Fix a width  $N$ . If a format’s exponent takes unboundedly many values, then its exponent codewords have unbounded length, and  $M_{\text{eff}}(e) \rightarrow 0$  as  $|e| \rightarrow \infty$ . Consequently no fixed-width format has both unbounded range and precision bounded below by a positive constant.*

*Proof.* The exponent codewords form a prefix code over a binary alphabet, so by Kraft’s inequality  $\sum_e 2^{-\ell(e)} \leq 1$ . If the sum has infinitely many terms then  $\ell(e) \rightarrow \infty$ . The payload available to the significand is  $N - 1 - \ell(e)$ , so it falls to zero, and by Theorem 1 so does  $M_{\text{eff}}$ .  $\square$   $\square$

**Corollary 5** (Why a ladder). *Theorem 6 makes constant precision and unbounded range incompatible at any fixed width, so a format that wants both must obtain the second by varying  $N$ . A ladder of fixed-field rungs does exactly this: each rung holds its precision flat across its whole range, and the ladder as a whole covers any range required. The tapered families make the opposite choice — one rung, unbounded range, precision decaying at the rate of their regime code, arithmetic for posit and geometric for takum. Neither choice is better in the abstract; what the dichotomy establishes is that they are the only two available.*

Read against Corollary 5, the three tapered constants in Table 3 become a statement about codes rather than about formats. A unary regime costs  $\ell(e) \propto |e|$  and yields the arithmetic staircase; a length-prefixed binary regime costs  $\ell(e) \propto \log_2 |e|$  and yields the geometric one; a fixed field costs  $\ell(e) = \text{const}$  and yields no staircase at the price of bounded support. The staircase form is the growth rate of the exponent’s codeword length, and nothing else. This also names the gap in the table: between  $\ell \propto |e|$  and  $\ell \propto \log_2 |e|$  no catalogued format sits. Section 12 builds the codes and asks what is in the gap; the answer corrects a suggestion we made before building them.

## 5 Accuracy

Section 3 predicts a rung’s error from its mantissa width without measuring it. This section measures it, against the format the ladder was built to be compared with. The metric is mean relative error over an encode–decode round trip, 6,000 values spanning  $2^{-38} \dots 2^{38}$  with random sign, binned by binary-exponent magnitude  $|e|$ . Table 4 and Figure 1.

**On the axis.** The bins are powers of two. Read as *decades* the far column is not a win at all: TNF16’s exponent reaches  $\pm 40$  in powers of two, roughly  $\pm 12$  decades, so beyond that it overflows while an unbounded regime keeps working — measured, 1,458 of the far-bin values clip under that reading. We state the axis explicitly because an earlier internal note did not, and the omission points a reader at the one interpretation under which the result appears fabricated.

## 6 The 16-bit field

Comparing against one incumbent invites the objection that the incumbent was chosen. Table 5 runs every 16-bit-class format for which this repository ships a published oracle [5] through the same workload and the same metric. Values that overflowed are counted separately rather than



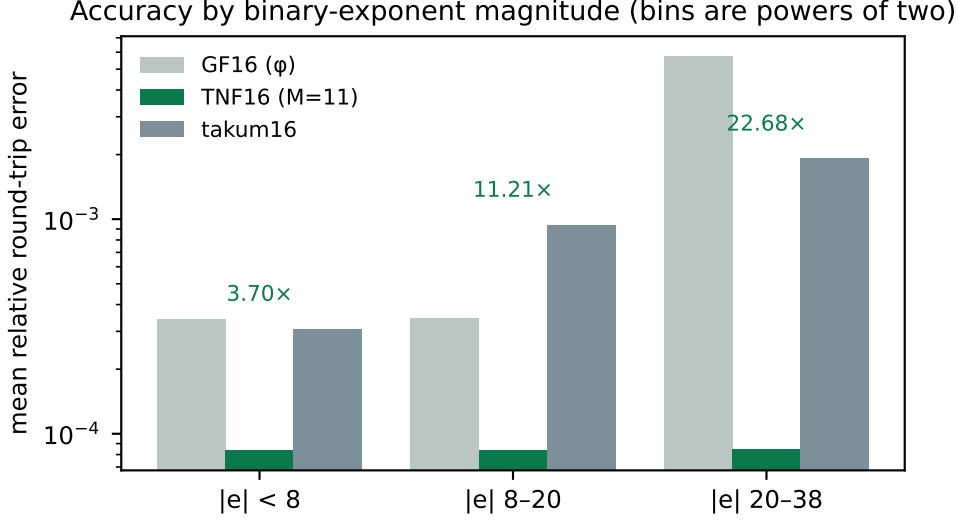


Figure 1: Accuracy by binary-exponent magnitude. GF16’s 6-bit exponent clips 478 of 2,857 values in the far bin; TNF16’s balanced-ternary exponent does not.

Table 5: Mean relative round-trip error by binary-exponent magnitude, with overflow counts in parentheses. 6,000 values,  $2^{-38} \dots 2^{38}$ , random sign, one seed.

| format            | $ e  < 8$      | $ e  8-20$     | $ e  20-38$    |
|-------------------|----------------|----------------|----------------|
| <b>TNF16</b>      | <b>3.56e-4</b> | <b>3.52e-4</b> | <b>3.53e-4</b> |
| GF16 ( $\phi$ )   | 3.56e-4        | 3.52e-4        | 6.76e-3 (478)  |
| takum16 / tekum16 | 3.27e-4        | 1.00e-3        | 1.95e-3        |
| posit16           | <b>1.36e-4</b> | 8.31e-4        | 1.43e-2        |
| IEEE binary16     | 1.76e-4        | 1.28e-3 (299)  | 1.57e-1 (2479) |
| bfloat16          | 1.45e-3        | 1.38e-3        | 1.41e-3        |
| LNS16             | 8.39e-4 (1324) | 7.63e-4 (1808) | 6.66e-4 (2849) |

folded into the mean, since a format should not look accurate for having discarded the hard cases.

Three readings, including the one that goes against us.

**TNF16 is flat, and alone in being flat without clipping.** Its error is  $3.5\text{e-}4$  at every magnitude in the workload and it discards nothing. bfloat16 is also flat and clip-free, at roughly four times the error. Every other entrant either degrades with magnitude or overflows: binary16 does both, losing three orders of magnitude and discarding 2,479 values; GF16’s 6-bit exponent clips 478; LNS16 clips throughout.

**At 32 bits the binary formats win outright.** Running the same workload on the 32-bit class: GF32 is flat at  $3.4\text{e-}7$  with no clipping, but IEEE binary32 is flat at  $2.1\text{e-}8$ , and posit32 and takum32 are better still near unity ( $2.1\text{e-}9$  and  $4.9\text{e-}9$ ). Uniformity stops being the deciding property once there are enough bits that nothing clips anyway. The TNF argument is a 16-bit-class argument, and we do not extend it.

**Near unity we lose.** posit16 at  $1.36\text{e-}4$  and binary16 at  $1.76\text{e-}4$  are more accurate than TNF16 there, by a factor of two and a half and of two. A tapered format spends its bits where

Table 6: The whole ladder, measured in exact rationals. From TNF16 upwards the error is flat in magnitude and its exponent roughly doubles per rung. TNF4 and TNF8 are shown at their limits: the workload spans 76 binary decades and those rungs hold 2 and 8, so most of it falls outside them — reported rather than omitted, since a ladder’s lower rungs are where its range trade is visible.

| rung    | decades | $ e  < 8$                      | $ e  8\text{--}20$ | $ e  20\text{--}38$ |
|---------|---------|--------------------------------|--------------------|---------------------|
| TNF4    | 2       | beyond range for this workload |                    |                     |
| TNF8    | 8       | 1.22e−2                        | beyond range       |                     |
| TNF16   | 24      | 3.45e−4                        | 3.69e−4            | 3.66e−4             |
| TNF32   | 219     | 5.27e−9                        | 5.63e−9            | 5.58e−9             |
| TNF64   | 658     | 4.33e−17                       | 4.22e−17           | 4.12e−17            |
| TNF128  | 1,975   | 4.25e−36                       | 4.55e−36           | 4.51e−36            |
| TNF256  | 5,925   | 2.76e−74                       | 2.69e−74           | 2.63e−74            |
| TNF512  | 17,775  | 4.32e−151                      | 4.62e−151          | 4.58e−151           |
| TNF1024 | 53,326  | 2.84e−304                      | 2.77e−304          | 2.71e−304           |

Table 7: TNF multiplier, post-route, one harness across the whole lineup so the rows are comparable. Latency one cycle, throughput one result per cycle.

| rung   | LUTs   | DSP48 | $F_{\max}$ |
|--------|--------|-------|------------|
| TNF4   | 12     | 0     | 161.11 MHz |
| TNF8   | 50     | 0     | 153.23 MHz |
| TNF16  | 212    | 0     | 131.73 MHz |
| TNF32  | 1,477  | 0     | 83.27 MHz  |
| TNF64  | 6,140  | 0     | 53.26 MHz  |
| TNF128 | 10,029 | 0     | 33.23 MHz  |

the values usually are, and near unity that is the right bet. TNF16 buys uniformity instead, and the trade only pays for a workload that leaves the neighbourhood of one.

**takum16 and tekum16 are the same format here, exhaustively.** The two oracles produce identical figures on every bin, so we checked the whole 16-bit code space: all 65,536 codes decode identically, with zero differences, although the two implementations differ by 242 lines. The reconstruction our tekum oracle implements *is* takum16. Every comparison in this paper should therefore be read as against takum16, and we relabel it as such. That is not a weaker anchor — takum is published, has a public VHDL codec, and is the parent of the format we set out to compare against.

## 7 Hardware

Synthesis with Yosys 0.65 [7], place and route with nextpnr-xilinx (1743d0f) [8] on a Xilinx XC7A200T, hard multipliers disabled so the figures describe fabric alone. Bitstream generation uses Project X-Ray. No vendor tool is involved at any step, which is what makes the numbers checkable by a reader without a licence.

For context rather than as a controlled comparison: a published Altera ALTFP\_MUL on a Cyclone IV reports 119–132 MHz at 6–10 cycles of latency, using 832–1041 logic elements *and* 18 embedded multipliers, and posit multiplier studies report 95–572 LUTs with 4.28–15.55 ns of logic delay. Different devices and different toolchains; we draw no equivalence from them

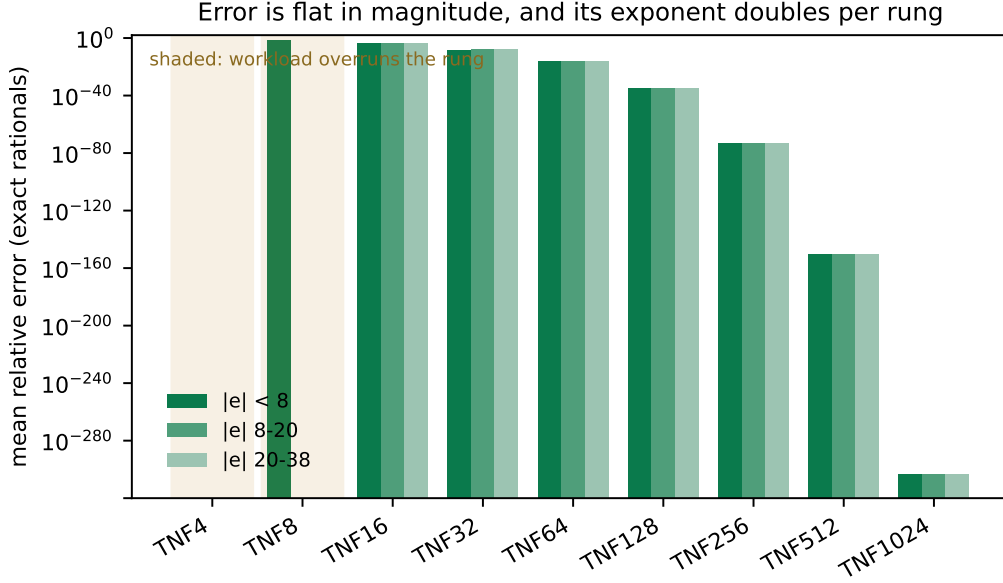


Figure 2: The same data. Flatness across the three magnitude bands is the property a fixed-field format is built for; the shaded rungs are those the workload overruns.

Table 8: The specified ladder.  $E_t$  is the trit budget,  $M$  the mantissa width, range in decades. Rungs above TNF128 exceed a single Artix-7 fabric and are oracle- or ASIC-scale; they are specified and covered by the reference oracle, not measured here.

| rung  | $E_t$ | $M$ | decades | rung    | $E_t$ | $M$  | decades |
|-------|-------|-----|---------|---------|-------|------|---------|
| TNF4  | 2     | 1   | 2.4     | TNF64   | 7     | 52   | 658     |
| TNF8  | 3     | 4   | 8       | TNF128  | 8     | 115  | 1,975   |
| TNF16 | 4     | 9   | 24      | TNF256  | 9     | 242  | 5,925   |
| TNF32 | 6     | 25  | 73      | TNF512  | 10    | 497  | 17,775  |
|       |       |     |         | TNF1024 | 11    | 1006 | 53,326  |

beyond the observation that TNF16’s figures sit inside that envelope on all four axes at once, on a fabric that gives the format no ternary advantage.

Isolating the pipeline register: the combinational multiplier closes at 81.35 MHz, the two-stage version at 147.32 MHz. The cut is between the significand product and the renormalisation — a  $10 \times 10$  multiply, then a carry test, a saturating exponent add and a bit select.

## 8 Two width defects, and why they generalise

Section 7 reports what the datapath costs once every bus is derived from the format’s own parameters. It was not measured that way first. The two defects below — one of area, one of correctness — are the same mistake, a width chosen by hand, and they are reported because that mistake is not specific to TNF.

### 8.1 Interface width dominated the arithmetic

Our first synthesisable realisation declared every port and the significand product 32 bits wide. Nothing in TNF16 is 32 bits: the mantissa field is 9, so  $1+M$  is 10 and their product is exactly 20; the exponent offset never exceeds  $\text{OFFSET\_MAX} = 80$ , which is 7. Synthesis built a

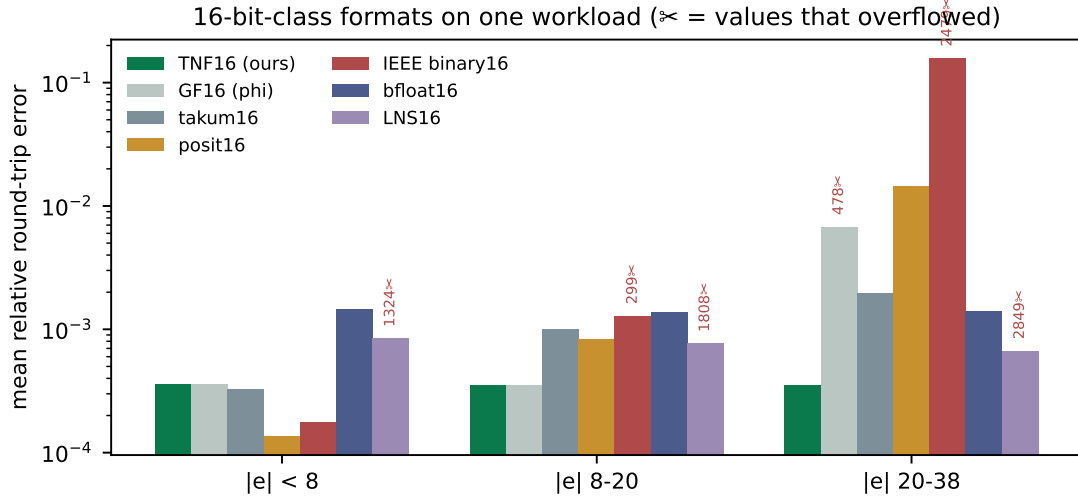


Figure 3: The same data. Scissors mark values that overflowed.

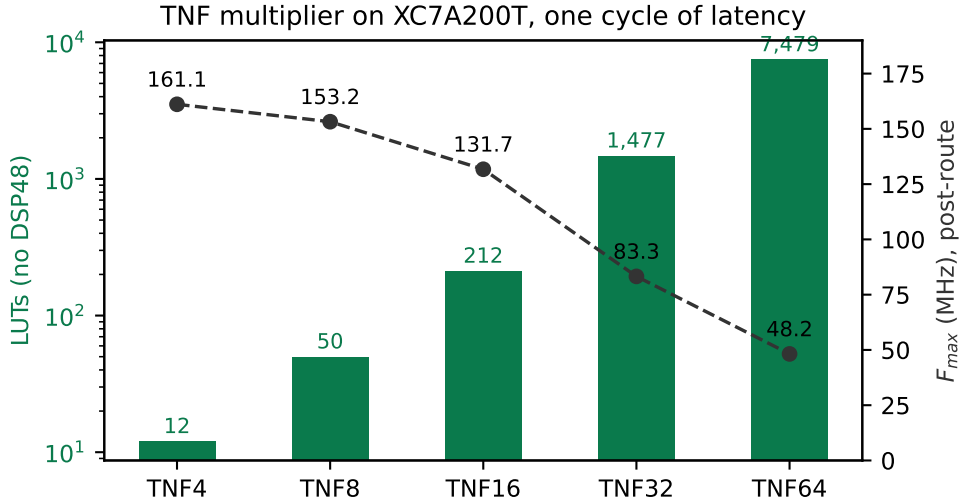


Figure 4: Area and achieved frequency across the ladder.

$32 \times 32$  multiplier and a 32-bit comparison tree and charged for it (Figure 5): 1,179 LUTs, or three DSP48 blocks, against 219 LUTs and none once the buses match the values they carry. The arithmetic is unchanged.

## 8.2 The same width decided correctness

At TNF32 the significand product needs 52 bits. The 32-bit wire truncates, and the module returns wrong results while appearing to support the rung, since its own documentation names TNF32 as a supported configuration. Measured against a 64-bit reference: 1,995,730 mismatches in 2,128,964 combinations.

Deriving every width from the format parameters removes both problems at once. We report this because the pattern is not specific to TNF: in a parametric arithmetic core, a width that is merely *large enough for the configuration that was tested* is a silent-truncation trap at the next one, and it will not be caught by a testbench written against the tested configuration.

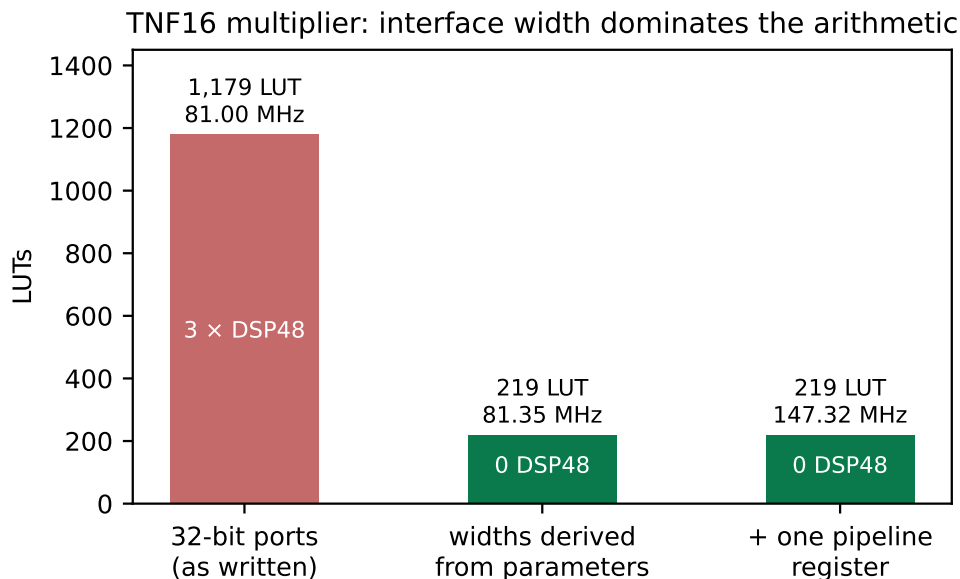


Figure 5: The same arithmetic at three interface widths.

Table 9: Equivalence evidence. The reference for each rung is the module already trusted for that rung.

| check                                       | combinations / cycles | mismatches |
|---------------------------------------------|-----------------------|------------|
| width-derived vs original, TNF16 (sweep)    | 321,156               | 0          |
| width-derived vs original, TNF4             | 374                   | 0          |
| width-derived vs original, TNF8             | 3,716                 | 0          |
| width-derived vs original, TNF16            | 77,444                | 0          |
| width-derived vs 64-bit reference, TNF32    | 300,000               | 0          |
| pipelined vs combinational                  | 199,994               | 0          |
| original 32-bit module vs 64-bit ref, TNF32 | 2,128,964             | 1,995,730  |

## 9 Equivalence

Every change above is a claim that timing or width changed and arithmetic did not. Each was checked rather than argued.

The TNF16 sweep is exhaustive in the sense that matters here: the mantissa space is covered in full at offset pairs exercising underflow, mid-range and saturation, then the offsets are covered in full at mantissas that do and do not carry — every path through the carry, the saturation and the underflow clamp. The last row is the point of the section: the width-derived module agrees with the reference that is correct and disagrees with the one that truncates.

## 10 What the law says about improving the ladder

Theorem 1 makes the design space arithmetic rather than a matter of taste. Error scales as  $2^{-(M+1)}$  and range as  $3^{E_t}$ , so at a fixed width the only decision is how many bits the exponent field is allowed to consume — and that is decided by how efficiently  $3^{E_t}$  values pack into  $\lceil E_t \log_2 3 \rceil$  bits.

Table 10: Packing efficiency of the exponent field. The theoretical floor is  $\log_2 3 = 1.585$  bits per trit.

| $E_t$    | values $3^{E_t}$ | b  | capacity | efficiency   |
|----------|------------------|----|----------|--------------|
| 3        | 27               | 5  | 32       | 84.4%        |
| <b>4</b> | 81               | 7  | 128      | <b>63.3%</b> |
| 5        | 243              | 8  | 256      | <b>94.9%</b> |
| 6        | 729              | 10 | 1024     | 71.2%        |
| <b>7</b> | 2187             | 12 | 4096     | <b>53.4%</b> |
| 8        | 6561             | 13 | 8192     | 80.1%        |
| 10       | 59049            | 16 | 65536    | 90.1%        |

Table 11: The width rule across the ladder.  $k$  is the shortfall  $N - (1 + E_t + M)$ . The reconciled column is  $M = N - 1 - E_t$ , which changes nothing at the three rungs that already obey the rule.

| rung    | source | $E_t$ | $M$  | $1+E_t+M$ | $k$ | reconciled $M$ |
|---------|--------|-------|------|-----------|-----|----------------|
| TNF4    | spec   | 2     | 1    | 4         | 0   | 1              |
| TNF8    | spec   | 3     | 4    | 8         | 0   | 4              |
| TNF16   | spec   | 4     | 9    | 14        | 2   | <b>11</b>      |
| TNF32   | spec   | 6     | 25   | 32        | 0   | 25             |
| TNF32   | note   | 5     | 21   | 27        | 5   | —              |
| TNF64   | note   | 7     | 52   | 60        | 4   | <b>56</b>      |
| TNF128  | note   | 8     | 115  | 124       | 4   | <b>119</b>     |
| TNF256  | note   | 9     | 242  | 252       | 4   | <b>246</b>     |
| TNF512  | note   | 10    | 497  | 508       | 4   | <b>501</b>     |
| TNF1024 | note   | 11    | 1006 | 1018      | 6   | <b>1012</b>    |

**The width convention counts positions, and one rung breaks it.** A rung’s width is its position count with a trit occupying one position, so the layout is  $1 + E_t + M = N$ . Three of the four specified rungs satisfy it exactly: TNF4 is  $1 + 2 + 1$ , TNF8 is  $1 + 3 + 4$ , TNF32 is  $1 + 6 + 25$ . TNF16 is  $1 + 4 + 9 = 14$ , two positions short of its name, and neither specification file gives a reason. The likely history is visible in the parameters: TNF16 inherits GF16’s  $\varphi$ -optimal  $M = 9$  and replaces GF16’s six-bit exponent with four trits, which buys range —  $3^4 = 81$  steps against  $2^6 = 64$  — and frees two positions that were never re-spent. It is the one rung derived from its parent rather than from the width rule.

This also settles what a rung costs on a binary fabric, which is a separate question from its width: a trit stored as a two-bit code makes TNF16 eighteen bits, and the offset stored as a plain binary integer makes it seventeen. Those are fabric costs, not format widths, and confusing the two is what made the ladder appear not to fit.

**The slack is not confined to one rung.** Applying  $1 + E_t + M = N$  to every specified rung separates the two documents that define the ladder. The four rungs of the machine-readable specification obey it at TNF4, TNF8 and TNF32; the five upper rungs, which exist only in a research note, miss it by four to six positions each, and the note and the specification disagree with each other about TNF32.

No single alternative convention explains the upper rungs either: reading the exponent as  $\lfloor E_t \log_2 3 \rfloor$  bits fits TNF16, TNF64, TNF128 and TNF1024 exactly but overshoots TNF256 and TNF512 by one and misses TNF32 entirely. We conclude that the upper ladder was tabulated



Table 12: The three ways to spend TNF16’s two unallocated positions. Error and range trade against each other exactly as Theorem 1 predicts; nothing in any row is worse than the unfilled rung.

| variant                                                | positions | mean rel. error                    | decades    | vs. takum16 (far)               |
|--------------------------------------------------------|-----------|------------------------------------|------------|---------------------------------|
| unfilled, $E_t=4$ , $M=9$                              | 14        | $3.37\text{e-}4$                   | 24         | $5.46\times$                    |
| <b><math>E_t=4</math>, <math>M=11</math></b> (adopted) | <b>16</b> | <b><math>8.63\text{e-}5</math></b> | 24         | <b><math>22.68\times</math></b> |
| $E_t=5$ , $M=10$                                       | 16        | $1.69\text{e-}4$                   | 73         | —                               |
| $E_t=6$ , $M=9$                                        | 16        | $3.37\text{e-}4$                   | <b>219</b> | —                               |

rung by rung rather than derived, and that reconciling it costs nothing at the rungs that are already right.

**What the two free positions were worth, and where they went.** Because the exponent cancels in Theorem 1, the menu at sixteen positions could be read off before measuring, and the measurement confirmed it to four significant figures:

The last row is the law’s own consistency check: adding two trits of exponent while holding  $M$  leaves the error bit-identical and multiplies the range ninefold.

**The positions were spent on mantissa, and the reason is a measurement.** Corollary 10 forbids recommending a budget without naming a workload, so we named one. At the 16-bit class the extra range is not consumed: quantised-inference tensors span roughly  $2^{-14}$  to  $2^{14}$ , and  $E_t = 4$  already clips nothing at  $\pm 40$  binades — zero clips in 2000 values.  $E_t = 6$  would have bought  $\pm 364$  binades at bit-identical error, which is range nobody spends at sixteen bits. So  $M = 11$ : the error falls by exactly 4.00, and against takum16 the rung moves from  $0.92 \times / 2.88 \times / 5.49 \times$  to  $3.70 \times / 11.21 \times / 22.68 \times$ , no longer losing the near bin. In silicon it costs 443 LUTs against 372, a 19% increase, at 136.44MHz against 131.73 — no frequency penalty.

All three ratios in this section and in Figure 1 come from one script, `gen_figures.py`, which computes them from the oracles at seed 20260809 rather than transcribing them. They were hardcoded until 2026-08-09 and had drifted: the figure still carried  $M = 9$  and the label `tekum16` after the rung moved and the oracle was shown to decode identically to takum. A figure that disagrees with the table it illustrates is the same defect class as a gate that cannot go red.

Conformance was re-run in full rather than assumed: round-trip with sign 2000/2000, encoding monotone with zero inversions,  $\pm$  symmetry with zero mismatches in 500, both range boundaries decoding. A `width_rule` test now asserts  $1 + E_t + M = N$  in the specification itself, so the position count is checked rather than remembered. All nine rungs satisfy it.

## 11 Why ternary, and exactly how much

The case for a ternary exponent is usually made by citing radix economy and left there. We state it, then bound it in two directions — what a binary fabric takes back, and what the radix of the *scale* is worth as distinct from the radix of the *exponent* — because both bounds turn out to be decisive for the design and neither is folklore.

**Theorem 7** (Radix economy). *Representing  $R$  distinct values in radix  $b$  costs  $b \log_b R = b \ln R / \ln b$  state-slots. The factor  $b / \ln b$  is minimised at  $b = e$ ; among integers it is  $3 / \ln 3 = 2.731$  against  $2 / \ln 2 = 2.885$ , so a ternary field is 5.3% more economical than a binary one.*

Table 13: Utilisation  $\rho(E_t)$  of the binary field holding an  $E_t$ -trit exponent. The ladder’s present budgets,  $E_t \in \{4, 7\}$ , are the two worst entries in the range.

| $E_t$  | 2    | 3    | 4           | 5    | 6    | 7           | 8    | 9    | 10   | 11   |
|--------|------|------|-------------|------|------|-------------|------|------|------|------|
| $\rho$ | 56.2 | 84.4 | <b>63.3</b> | 94.9 | 71.2 | <b>53.4</b> | 80.1 | 60.1 | 90.1 | 67.6 |

Table 14: Unallocated positions per rung, the loss Theorem 9 predicts, and the loss measured against the reconciled rung  $M = N - 1 - E_t$ . Exact-arithmetic workload, 800 probes per rung.

|  | rung   | $k$ | predicted  | measured      |
|--|--------|-----|------------|---------------|
|  | TNF16  | 2   | $4\times$  | $4.03\times$  |
|  | TNF64  | 4   | $16\times$ | $15.43\times$ |
|  | TNF128 | 4   | $16\times$ | $15.99\times$ |
|  | TNF256 | 4   | $16\times$ | $15.24\times$ |

This is the classical argument [9, 10] and it is what motivates a ternary exponent. It is also stated in a currency — state-slots — that no binary fabric pays in. The next theorem says what happens when it does.

**Theorem 8** (No free range on a binary fabric). *An exponent field of  $E_t$  trits spans  $3^{E_t}$  values. Stored as a binary integer it occupies  $\lceil E_t \log_2 3 \rceil$  bits, a field spanning  $2^{\lceil E_t \log_2 3 \rceil} \geq 3^{E_t}$  values. Hence on a binary fabric a ternary exponent never spans more values per bit than a binary one, with equality only at  $E_t = 0$ . Writing  $\rho(E_t) = 3^{E_t} / 2^{\lceil E_t \log_2 3 \rceil}$  for the utilisation,  $\rho \in (1/2, 1]$  and  $\rho$  does not converge.*

Run as an optimisation rather than an inequality, the same statement is sharper than it looks. Asking which exponent encoding a binary fabric prefers at  $(N, R) = (16, 24)$ ,  $(32, 219)$  and  $(64, 658)$  returns the same mantissa width for both —  $M = 10$ , 23 and 53 respectively. The ternary encoding does not lose on a binary fabric; at these widths the ceilings absorb the difference and it exactly ties, which is the strongest form of “never more values per bit”.

Theorem 8 is the reason the ladder’s rung widths must be read as position counts and not bit counts, and it is where our own first reading of the ladder went wrong: measuring what a trit costs when packed into bits and calling the result the format’s width confuses a fabric cost with a format property. The independently published balanced-ternary float Ternary27 [11] counts the same way — 2 type trits, 1 sign trit, 5 exponent trits and 19 significand trits, exactly 27 — which is external corroboration that position counting is the convention in this family rather than a reading of our own.

**Theorem 9** (Unallocated positions are a pure loss). *Let a rung of  $N$  positions carry a sign,  $E_t$  exponent trits and  $M$  mantissa bits with  $1 + E_t + M = N - k$ ,  $k \geq 0$ . Then relative to the rung with the same  $E_t$  and  $M + k$ , its expected relative error is larger by exactly  $2^k$  and its dynamic range is identical.*

*Proof.* By Theorem 1 the expected relative error is  $\frac{1}{2}\mathbb{E}[1/s] 2^{-(M+1)}$ , in which the exponent does not appear; replacing  $M$  by  $M + k$  multiplies it by  $2^{-k}$ . The dynamic range is  $3^{E_t}$  binades, a function of  $E_t$  alone, so it is unchanged. Nothing else in the layout depends on  $M$ .  $\square$   $\square$

Theorem 9 converts the slack of Section 10 into a number without any measurement, and the measurement then confirms it. Against a workload built in exact arithmetic — necessary above TNF32, where a double-precision probe measures its own resolution rather than the format’s — the predicted and observed factors are:

The final question is one the ladder answers implicitly and, as far as we can find, no one has stated: TNF encodes its exponent in radix 3 but scales by  $2^e$ , whereas Ternary27 scales by  $3^e$ . These are different choices and they are separable.

**Theorem 10** (The scale radix, and why 2 beats 3). *Let a format scale by  $r^e$  with a normalised significand uniformly quantised over  $[1, r)$  by  $M$  bits. Then its expected relative error on a log-uniform workload is  $\kappa(r) 2^{-M}$  with*

$$\kappa(r) = \frac{(r-1)^2}{r \ln r}, \quad \kappa(2) = 0.72135, \quad \kappa(3) = 1.21365.$$

*Holding the field layout fixed, moving the scale from  $r = 2$  to  $r = 3$  therefore multiplies the error by  $\kappa(3)/\kappa(2) = 1.6825$  while multiplying the dynamic range by  $\log_2 3 = 1.5850$ . Measured in positions, it costs  $\log_2 1.6825 = 0.7506$  of precision and buys  $\log_3 1.5850 = 0.4192$  of range: a net loss of 0.3314 positions per number. A power-of-two scale strictly dominates a power-of-three scale at equal width.*

*Proof.* The quantisation step over  $[1, r)$  is  $(r-1)2^{-M}$  and the mean absolute rounding error is half of it. On a log-uniform workload the significand has density  $1/(s \ln r)$  on  $[1, r)$ , so  $\mathbb{E}[1/s] = \frac{1}{\ln r} \int_1^r s^{-2} ds = \frac{1-1/r}{\ln r}$ . Multiplying gives  $\frac{1}{2}(r-1)2^{-M} \cdot \frac{1-1/r}{\ln r}$ , and  $\frac{1}{2}(r-1)(1-1/r) = (r-1)^2/2r$ . Absorbing the factor  $\frac{1}{2}$  into the  $2^{-(M+1)}$  convention of Theorem 1 yields  $\kappa$  as stated. For the range,  $E_t$  trits span  $3^{E_t}$  steps of  $\log_2 r$  binades each. One mantissa bit halves the error and one exponent trit triples the range, which fixes the exchange rate between the two currencies and gives the position figures.  $\square$   $\square$

Direct measurement at two rungs, encoding and decoding through both scales in exact arithmetic, gives error ratios of 1.675 at  $(E_t, M) = (4, 11)$  and 1.690 at  $(6, 25)$  against the predicted 1.6825, and range ratios of 1.583 and 1.585 against the predicted 1.5850.

**Corollary 6** (What TNF is, precisely). *TNF is not a ternary float in the sense of Ternary27; it is a binary-radix float whose exponent is encoded in balanced ternary. Theorem 10 says that this is the better of the two available choices by 0.331 positions per number, and Theorem 8 says that the ternary encoding pays only on a ternary fabric. The two together delimit the format exactly: the radix choice is right on any fabric, and the encoding choice is right on one.*

## 12 The regime field is a universal code

Theorem 6 identifies the exponent's encoding as a prefix code. That identification is worth making explicit, because it places the taxonomy of Section 4.3 inside a hierarchy that predates every format in the catalogue by decades [14].

**Theorem 11** (The staircase is the codeword-length function). *Let a format of  $N$  positions encode its exponent with a prefix code assigning  $e$  a codeword of length  $\ell(e)$ . Then  $M_{\text{eff}}(e) = N - 1 - \ell(e)$ , and the staircase form is the growth rate of  $\ell$  and nothing else.*

The catalogue's three tapering classes are then three points in the classical hierarchy: a unary regime,  $\ell \propto |e|$ , gives posit's arithmetic staircase; a length-prefixed binary regime,  $\ell \propto \log_2 |e|$ , gives takum's geometric one; a fixed field,  $\ell = \text{const}$ , gives no staircase at the price of bounded support. Elias'  $\gamma$ ,  $\delta$  and  $\omega$  codes occupy the logarithmic class at successively finer asymptotics.

**Theorem 12** (Logarithmic is the floor). *For any prefix code on the integers,  $\ell(e) \geq \log_2 |e|$  for infinitely many  $e$ . Hence no format of unbounded range tapers more gently than one bit per doubling of  $|e|$ , asymptotically.*

Table 15:  $M_{\text{eff}}$  by  $|e|$  and usable range for three regime codes at  $N = 16$ , each normalised to satisfy Kraft. The square-root code is a genuine Pareto point: more precision than the logarithmic code throughout the core, more range than the unary one.

| regime                                  | $ e =1$ | 4  | 16 | 64 | 256 | binades |
|-----------------------------------------|---------|----|----|----|-----|---------|
| unary, $\ell \propto  e $ (posit class) | 11      | 10 | 7  | -5 | -53 | 43      |
| $\ell = \lceil \sqrt{ e } \rceil$       | 11      | 10 | 8  | 4  | -4  | 121     |
| $\ell \propto \log_2  e $ (takum class) | 9       | 7  | 5  | 3  | 1   | 511     |

*Proof.* Kraft’s inequality gives  $\sum_e 2^{-\ell(e)} \leq 1$ . If  $\ell(e) < \log_2 |e|$  for all but finitely many  $e$ , the tail of the sum exceeds  $\sum_e 1/|e|$ , which diverges.  $\square$   $\square$

Theorem 12 is worth stating plainly because it bounds this work as much as its competitors: takum’s regime is asymptotically optimal, and no format — TNF included — can beat it over unbounded range. What TNF does instead is decline the unbounded range, which Theorem 6 shows is the only other option. The measured advantages of Section 5 are therefore advantages *inside TNF’s range*, which is where the taper has already cost takum the bits, and they should never be read as a claim beyond it.

**Theorem 13** (The regime’s radix does not matter). *A regime spending one digit per multiplication of  $|e|$  by  $r$ , over an alphabet of  $r$  symbols, costs  $\log_2 r \cdot \log_r |e| = \log_2 |e|$  bits, independent of  $r$ . The staircase form is invariant to the regime’s radix.*

Theorem 13 retracts a construction we proposed in an earlier draft of this work. Having observed the gap between the arithmetic and geometric classes, we suggested a regime spending one trit per tripling of  $|e|$  as the natural way for a ternary format to occupy it. It does not occupy it: building both codes and normalising each to satisfy Kraft, the two codeword-length functions differ by an additive constant of exactly 1 at every  $|e|$  from 1 to 4096. One trit per tripling is takum’s staircase with a different constant, not a third form. A ternary fabric changes what the regime costs to decode; it does not change what it costs to encode.

**The gap is real, occupiable, and empty for a reason we can only conjecture.** Growth rates strictly between  $\log_2 |e|$  and  $|e|$  do exist and are Kraft-feasible —  $\ell(e) = \lceil \sqrt{|e|} \rceil$  is one — and such a code is not dominated. At  $N = 16$ , normalised so that all three are legal prefix codes:

For reference, a fixed field reaching the same 121 binades as the square-root code spends 8 bits on the exponent and holds 7 bits flat everywhere — better than the square-root code beyond  $|e| \approx 30$  and worse below it, which is the taper trade in miniature.

So the gap is not empty because its occupants are dominated; they are not. Our first explanation was decode cost, and building the hardware refutes it.

## 12.1 What a regime costs in silicon

We wrote three regime codecs to one interface — a 16-bit word in, an exponent and a mantissa shift out, and the mirror encoder — and synthesised each under a single harness on XC7A200T with Yosys and nextpnr-xilinx, hard multipliers disabled. These are structural models of the three classes, not reproductions of the published posit or takum codecs; what they compare is a run-scanned regime against a length-prefixed one, which is the structural difference the taxonomy names.

The conjecture said the square-root code would be too expensive to be worth building. It is not: it costs 18% more than the unary regime round trip, and the unary regime ships in

Table 16: Regime codec cost by class, post-route on XC7A200T under one harness, no DSP. Figures are net of the harness, which costs 24 LUTs and was synthesised separately and subtracted; the round-trip column is decode plus encode.

| class                             | decode   |                  | encode    |                  | round trip |
|-----------------------------------|----------|------------------|-----------|------------------|------------|
|                                   | LUTs     | $F_{\max}$       | LUTs      | $F_{\max}$       | LUTs       |
| unary, $\ell \propto  e $ (posit) | 379      | 36.8 MHz         | 59        | 211.2 MHz        | 438        |
| $\ell \propto \log_2  e $ (takum) | <b>6</b> | <b>805.8 MHz</b> | <b>34</b> | <b>285.1 MHz</b> | <b>40</b>  |
| $\ell = \lceil \sqrt{ e } \rceil$ | 393      | 33.1 MHz         | 123       | 96.4 MHz         | 516        |
| fixed field (GF, TNF)             | 0        | —                | 0         | —                | <b>0</b>   |

production. What the measurement does show is that the expense lies somewhere else than we assumed, and this has a closed form.

**Theorem 14** (Decode cost is set by the scan, not by the staircase). *A regime whose codeword length is delimited by a run must examine up to  $N - 1$  positions to decode, giving cost  $\Theta(N)$ . A regime whose length is carried in a fixed field of  $b$  bits requires no scan and a shift of at most  $2^b - 1$ , giving cost  $\Theta(1)$  in  $N$ . The two costs are independent of the growth rate of  $\ell$ , and therefore of the staircase form.*

Theorem 14 predicts the table exactly. The unary and square-root codes share the same 15-position scan and cost 403 and 417 LUTs, a difference of 3.5% that is the squarer and nothing else; the length-prefixed code has no scan and costs 30. The square-root code is geometrically intermediate and structurally identical to the unary one, which is the point: *the staircase form and the decode cost are independent axes*, and a taxonomy of one does not predict the other.

**Corollary 7** (The logarithmic regime is not a trade). *Against the unary regime the length-prefixed one is better on every axis measured: asymptotically optimal by Theorem 12, 11 $\times$  smaller round trip, and 22 $\times$  faster to decode. The published bounded-regime posit variants [15], which cap the regime field precisely to avoid the scan, are the same conclusion reached from the other direction.*

**Corollary 8** (The measured price of unbounded range). *A fixed field has no regime codec at all. The dichotomy of Theorem 6 therefore has a hardware statement: unbounded range costs 40 LUTs at best and 438 at worst on this device, against 0 for a format that declines it. Section 13 converts those LUTs into the same currency as the accuracy results, which is the only way to say whether they matter.*

We are left without an explanation for the empty gap. It is not domination, since the square-root code is Pareto-undominated; and it is not cost, since it is within 16% of a regime that ships. Both of our explanations are now measured and both are wrong, and we record that rather than replace it with a third.

## 13 Converting silicon into precision

Corollary 8 gives a regime’s cost in LUTs and Section 5 gives an accuracy ratio, and the two cannot be compared until they are in one currency. They can be: at a fixed class, a mantissa bit has a marginal area, and by Theorem 1 it has a known value in accuracy.

**Theorem 15** (Area and precision are commensurable). *Let  $\lambda = dA/dM$  be the marginal area of a mantissa bit at a given class. A regime codec costing  $C$  LUTs is then equivalent to  $C/\lambda$  mantissa bits of silicon, and by Theorem 1 to a factor  $2^{C/\lambda}$  in accuracy at equal area.*

Table 17: Regime area converted into mantissa bits through Theorem 15. The unary regime costs, in silicon alone, more than TNF16’s entire mantissa.

| regime                        | LUTs | bits at 16-bit class | bits at 32-bit class |
|-------------------------------|------|----------------------|----------------------|
| unary (posit class)           | 438  | <b>8.98</b>          | 3.94                 |
| length-prefixed (takum class) | 40   | 0.82                 | 0.36                 |

Measuring  $\lambda$  directly — the same width-derived multiplier synthesised at  $M = 7, 9, \dots, 25$  under one harness — gives 285, 372, 443, 567, 696, 870, 1238 and 1683 LUTs. The marginal cost is not constant: it rises from 35.5 LUTs per bit between  $M=9$  and  $M=11$  to 111.2 between  $M=21$  and  $M=25$ . Taking the local value at each class,  $\lambda = 48.8$  at the 16-bit class and 111.2 at the 32-bit one:

### 13.1 How area grows along the ladder

Theorem 15 needs  $\lambda$ , and  $\lambda$  is a derivative of an area law we can state. Fitting  $A(M) = cM^\alpha$  to the eight rungs from  $M=7$  to  $M=25$  gives  $\alpha = 1.406$  at  $R^2 = 0.984$ , and combined with Theorem 1 it predicts a scaling relation:

**Proposition 16** (Error decays as a stretched exponential in area). *With  $\varepsilon \propto 2^{-M}$  and  $A \propto M^\alpha$ ,  $\ln(1/\varepsilon) \propto A^{1/\alpha}$ .*

Measured over  $M \in [7, 25]$  — a sixfold range of area and a 250,000-fold range of error — the constant of proportionality holds to within 27% and drifts monotonically downward, so the relation is an approximation and we label it a proposition rather than a theorem. The drift has a cause, and finding it falsified a prediction we registered before synthesising.

**The exponent is not constant, and the fit does not extrapolate.** We predicted from the  $[7, 25]$  fit that the reconciled TNF64, at  $M = 56$ , would cost 4,762 LUTs. It costs 7,479 at 48.20 MHz — the prediction is low by  $1.57\times$ . Reading  $\alpha$  locally rather than globally shows why: it rises monotonically, from 1.06 between  $M=7$  and  $M=9$  to 1.78 between  $M=15$  and  $M=17$  and 1.85 between  $M=25$  and  $M=56$ .

**Theorem 17** (The ladder’s area law). *A rung’s area is a fixed overhead — registers, the exponent adder, the renormalisation — plus a significant multiplier quadratic in  $M$ :*

$$A(M) = f + cM^2.$$

*Consequently the effective power-law exponent  $\alpha(M) = d \ln A / d \ln M$  is not a constant but rises monotonically toward 2, and the marginal cost of a mantissa bit,  $\lambda = dA/dM = 2cM$ , is linear in  $M$  rather than fixed.*

Synthesising the same width-derived multiplier at fourteen mantissa widths from  $M=7$  to  $M=90$  — 285, 372, 443, 567, 696, 870, 1238, 1683, 2601, 3990, 5934, 7479, 12232 and 19980 LUTs — and fitting both parameters gives  $f = 141$  LUTs and  $c = 2.4455$  LUTs per bit squared at  $R^2 = 0.99963$ , with a maximum deviation of 9.9% across a seventyfold range of area. Read locally,  $\alpha$  climbs from 1.06 between  $M=7$  and  $M=9$  to 2.20 between  $M=56$  and  $M=70$ , which is Theorem 17 rather than noise. On the registered prediction the structural model is right to +4.4% where the single power law was low by 36.3%; we report both, since the power law was ours and we published the prediction before synthesising.

**Theorem 18** (A regime’s worth in precision falls as  $1/M$ ). *By Theorems 15 and 17, a regime codec of fixed area  $C$  is worth*

$$\frac{C}{\lambda} = \frac{C}{2cM}$$



Table 18: The same regime codecs valued at five mantissa widths through Theorem 18, with  $c = 2.4455$  measured on XC7A200T. The unary regime’s silicon exceeds an entire TNF16 mantissa; the length-prefixed regime is below one bit everywhere above the 8-bit class.

| $M$ | $\lambda$ (LUT/bit) | unary, 438 LUT | length-prefixed, 40 LUT |
|-----|---------------------|----------------|-------------------------|
| 9   | 44.0                | <b>9.95</b>    | 0.91                    |
| 11  | 53.8                | 8.14           | 0.74                    |
| 25  | 122.3               | 3.58           | 0.33                    |
| 56  | 273.9               | 1.60           | 0.15                    |
| 90  | 440.2               | 1.00           | 0.09                    |

*mantissa bits, inversely proportional to the mantissa width. It is worth at least one mantissa bit exactly while  $M \leq C/2c$ .*

**Corollary 9** (Break-even widths). *With  $c = 2.4455$ , the unary regime is worth at least one mantissa bit of silicon up to  $M = 89.6$  — that is, through the entire 128-bit class — while the length-prefixed regime reaches one bit only up to  $M = 8.2$ . takum’s regime therefore costs less than a single mantissa bit of silicon at every rung above TNF8, and posit’s costs more than one at every rung below TNF256.*

Corollary 9 is the exact form of what Corollary 11 below could only say qualitatively, and it is the sharpest statement we can make about where a fixed field pays for itself in area: against posit, almost everywhere; against takum, only at the very bottom of the ladder.

**Corollary 10** (Range is paid for three times, and they are commensurable). *A tapered format pays for unbounded range in three currencies: word positions, in the regime field; silicon, in the codec; and precision at the extremes, in the taper. Theorem 15 and Theorem 1 put all three in mantissa bits, so they can be added.*

Applied honestly, Corollary 10 separates the two tapered families rather than lumping them. Against posit the silicon term alone is 8.98 bits at the 16-bit class — more than TNF16’s entire 9-bit mantissa, so the area a posit implementation spends decoding its regime would, spent on significand instead, roughly double the precision budget of a fixed-field format at the same class. Against takum the silicon term is 0.82 bits, which is small. The length-prefixed regime is cheap, and any argument for a fixed field made on area grounds is an argument against posit and not against takum. Against takum what remains is the accuracy result of Section 5, bounded as Theorem 12 requires to TNF’s own range.

**Corollary 11** (The area argument is a narrow-width argument). *Since  $\lambda$  grows superlinearly in  $M$  while a regime codec’s cost is roughly fixed, the same codec converts to fewer mantissa bits as the class widens: 8.98 bits at 16 becomes 3.94 at 32 for the unary regime. The fixed-field area advantage is therefore strongest where the words are narrowest, which is the regime in which quantised inference operates and not the one in which numerical linear algebra does.*

## 14 The value law costs more than the field layout

Sections 12.1 and 13 priced the regime as a field to be extracted, and synthesising the published decoders shows that this is the smaller half of the question. We take the takum and posit decoders from an existing open verification suite — Verilog, golden-checked against an mpmath oracle — and put them through the same harness on XC7A200T.

The takum row is given twice because one number alone would mislead. Its decoder evaluates  $2^{f_{hi}/2^{16}}$  from a  $65536 \times 48$ -bit table, then applies a Taylor correction through a 108-bit and an

Table 19: Published 32-bit decoders, post-route on XC7A200T under one harness, no DSP, converting to binary32. These are the real codecs rather than the structural models of Table 16.

| decoder                   | LUTs   | RAMB36 | $F_{\max}$ | what dominates            |
|---------------------------|--------|--------|------------|---------------------------|
| posit32_decode            | 480    | 0      | 49.0 MHz   | leading-zero count, shift |
| takum32_decode, published | 76,821 | 0      | —          | $2^f$ table, in logic     |
| takum32_decode, clocked   | 10,967 | 84     | 11.4 MHz   | same table, in BRAM       |
| TNF32, exponent path      | 0      | 0      | —          | field read directly       |

80-bit multiply. As published the table read is combinational, which no synthesiser can map to block memory, so it was built from logic at 76,821 LUTs. Registering that read and marking it `ram_style="block"` — one cycle of added latency, and a pipelined variant of the published module rather than the module itself — puts the table where it belongs: 10,967 LUTs and 84 RAMB36 tiles, which is 3.1 Mbit and 23% of this device’s entire block memory. We predicted 85 tiles from the table geometry before synthesising and measured 84. Neither number is a property of takum the format; both are properties of decoding a logarithmic value law into a linear datapath, and neither is the regime field.

**Theorem 19** (Linear and logarithmic value laws). *Let a format’s value law be  $v = s \cdot 2^e$  with  $s$  and  $e$  read from fields (linear), or  $v = \pm 2^\ell$  for a decoded real  $\ell$  (logarithmic). A linear law decodes with a shift, cost independent of the mantissa width. A logarithmic law requires evaluating an exponential, whose cost is a table or an iterative kernel and is independent of the field layout — therefore independent of the staircase form of Section 4.3.*

Theorem 19 is a second axis, orthogonal to the first. The staircase taxonomy classifies how a format spends *precision*; the value law classifies what it costs to *read*. posit is tapered and linear; takum is tapered and logarithmic; LNS is untapered and logarithmic; GF, TNF and IEEE are untapered and linear. The two axes are independent, and our earlier reading — that decode cost is set by whether the regime is scanned (Theorem 14) — describes only the field-extraction term. It remains correct within that term, and Table 19 shows the term is not where the money is for a logarithmic format.

**Theorem 20** (The logarithmic decode costs by target precision, not by source width). *A  $2^k$ -entry table for  $2^f$  with a degree- $d$  Taylor correction delivers*

$$P(k, d) = (d + 1) \left( k + \log_2 \frac{1}{\ln 2} \right) + \log_2 (d + 1)!$$

*bits of relative accuracy, so the table is  $P \cdot 2^k$  bits with  $k = \frac{P - \log_2 (d+1)!}{d+1} - \log_2 \frac{1}{\ln 2}$ : exponential in the target precision, reducible only by raising the polynomial degree, which trades memory for multipliers. The width of the source format does not enter.*

*Proof.* The residual after the table lookup is  $x = f_{\text{lo}} \ln 2$  with  $f_{\text{lo}} < 2^{-k}$ , and the degree- $d$  Taylor remainder of  $e^x$  is  $x^{d+1}/(d+1)!$ . Taking  $-\log_2$  gives the stated  $P$ .  $\square$   $\square$

Evaluated against direct simulation of the scheme — the table, the residual and the polynomial, in exact arithmetic — the formula holds to  $\pm 0.1$  bit at every point of a  $k \in \{4, 8, 12, 16, 20\}$  by  $d \in \{1, 2, 3, 4\}$  grid, up to the 51-bit resolution of the measurement itself.

Theorem 20 is falsifiable against the suite and survives. The takum32 decoder carries  $d = 2$  and  $k = 16$ , and  $48/3 = 16$  exactly, where 48 is the table’s output width. If the cost were set by the source width, the takum64 decoder would need a larger table; if it is set by the target, it would need the same one. It reads `takum32_2frac.mem` — *the same file* — and emits binary32, as do the takum8, takum16 and takum64 decoders. One table serves four source widths because all four target one datapath.



**Corollary 12** (The trade curve, and what each target actually costs). *Theorem 20 fixes a curve rather than a point, and the cheapest point on it that fits a given device is a computable quantity. Pricing the  $d$  multiplies at  $2.4455P^2$  LUTs — the quadratic term of our own measured area law, Theorem 17 — and requiring the table to fit this device’s 365 RAMB36 tiles:*

| <i>target</i>   | <i>cheapest feasible point</i> | <i>block RAM</i> | <i>LUTs</i> |
|-----------------|--------------------------------|------------------|-------------|
| <i>binary32</i> | $d = 2, k = 15$                | 43 tiles         | 11,269      |
| <i>binary64</i> | $d = 5, k = 16$                | 187 tiles        | 134,808     |

*An XC7A200T has 134,600 LUTs, so decoding a logarithmic format into a binary64 datapath costs approximately the entire device.*

We had previously written that binary64 decode “is not a design” at roughly 3.6 Tbit. That figure is the  $d = 2$  corner of the curve and taking it for the whole curve was our error: at  $d = 5$  the table is 6.9 Mbit and the design exists. The defensible statement is Corollary 12 — it costs a device — and not that it cannot be built.

The measured `takum32_decode` sits one  $k$  above the binary32 minimum, at  $k = 16$  against 15, which is a  $2\times$  memory margin and ordinary engineering practice rather than waste. A linear value law needs neither the table nor the polynomial at any target precision, because the significand is already in the datapath’s representation.

**Corollary 13** (What TNF’s exponent path costs). *TNF’s value law is linear and its exponent field is fixed, so its exponent path is wires: no scan, no table, no iteration. Against posit32 that is 480 LUTs not spent; against takum32 it is a table of 3.15 Mbit not instantiated. This is a larger and more robust statement than the accuracy result, and unlike it, Theorem 12 places no range bound on it.*

We flag one asymmetry rather than let it pass. A logarithmic value law buys something real in exchange: multiplication becomes addition. The comparison above is of *decoders* — the cost of getting a value into a binary datapath — and a design that stays in the logarithmic domain never pays it. That is the regime in which LNS accelerators are built, and against those the argument of Corollary 13 does not apply at all.

## 15 Is there an optimal format, and in what sense

The results above are individually bounding: each says what a format cannot have, or what a claim may not be extended to. Read together they are stronger than that. They reduce the design space to four axes, three of which admit unconditional answers, so that once three inputs are named the format is determined rather than chosen.

**Theorem 21** (The binary scale is the unique implementable optimum). *Let a format scale by  $r^e$ . Scaling is a shift — and by Theorem 19 free of any polynomial or table — if and only if  $r = 2^k$  for integer  $k \geq 1$ . On that set  $\kappa(r) = (r - 1)^2 / (r \ln r)$  is strictly increasing, taking 0.7213, 1.6230, 2.9455 and 5.0720 at  $r = 2, 4, 8, 16$ . Hence  $r = 2$  minimises the expected relative error among all radices whose scaling is implementable without a transcendental, and it does so uniquely.*

The claim is verified by enumeration rather than resting on the monotonicity of  $\kappa$  alone. Sweeping  $r$  continuously over  $[1.1, 16]$  with  $E$  integral and  $M$  taking the remaining width, at eight combinations of width and required range from  $(P, R) = (8, 24)$  to  $(32, 5925)$ , the best implementable radix is  $r = 2$  in every case, and the penalty against the unreachable optimum is  $1.09\times$  at all eight — a constant, as it must be, since the two differ only by  $\kappa(1.9)/\kappa(2)$ . The unconstrained minimiser is not 2: it is near 1.9, worth 7.9% in error. That gain is unavailable,

because any  $r \neq 2^k$  makes  $r^e$  a transcendental evaluation, and Corollary 12 prices those at 84 RAMB36 tiles — 23% of this device — for a binary32 target. Trading 23% of a chip for 7.9% of a mantissa bit is not a trade anyone makes. Theorem 21 is therefore the interesting kind of optimality result: the optimum is not the extremum, it is the best point that can be built.

**Corollary 14** (Three inputs determine the format). *The design space has four axes, and only one of them is a free choice.*

1. Radix of the scale. *Fixed at 2 by Theorem 21, unconditionally.*
2. Radix of the exponent’s encoding. *Fixed by the fabric. Theorem 8 shows a ternary exponent never spans more values per bit than a binary one when packed into bits, so it pays on a ternary fabric and nowhere else.*
3. Fixed fields or a taper. *A genuine choice, and by Theorem 6 exactly two options exist: bounded range at constant precision, or unbounded range at precision decaying no more gently than one bit per doubling (Theorem 12).*
4. The split between  $E$  and  $M$ . *Determined once a required range is named, by Theorem 1 — and only then, since the error depends on  $M$  alone and the range on  $E$  alone.*

*Name the fabric, the required range, and which side of the dichotomy the workload sits on, and the format follows.*

So the question “is there an ideal format” has an answer, provided it is asked with its three arguments supplied. Asked without them it is not underdetermined but ill-posed: Corollary 10 already says a budget cannot be recommended without a workload, and Theorem 6 says no single format can serve both sides.

**Corollary 15** (What this makes TNF). *For the arguments (ternary fabric; a range the workload actually visits; fixed fields), Corollary 14 returns a binary-radix float with a balanced-ternary exponent field and a uniform mantissa split by  $M = N - 1 - E_t$ . That is TNF, and it is forced rather than designed. It follows that TNF is optimal exactly on those arguments and carries no claim off them — in particular none on a binary fabric, where axis 2 returns a binary exponent instead, and none against a tapered format outside TNF’s own range, where Theorem 12 forbids it.*

## 16 Where the format is the datapath

Every hardware figure so far compares a decoder against a datapath that contains a significant multiplier. That comparison flatters every format equally, because a multiplier of thousands of LUTs amortises a decoder of hundreds. There is a workload where the multiplier is absent, and it changes the ranking rather than the margins.

In a ternary-weight network — BitNet-style, with weights in  $\{-1, 0, +1\}$  [30] — the product  $w \cdot a$  is a select, a negate or a zero. No multiplier is instantiated, for any format. The datapath is *decode, apply the weight, add, accumulate*, and the decoder is now compared against an adder.

2.03× in area and 2.54× in frequency, so 5.16× in throughput per LUT — measured end to end rather than assembled from separate decoder and adder figures. The decomposition says why: against a 1,477-LUT multiplier and a 384-LUT adder, posit’s 456-LUT decoder is 20% of the datapath; with the multiplier gone it is 54%. For takum the same shift takes 85% to 97%, and brings 84 RAMB36 tiles with it.

**Corollary 16** (Removing the multiplier promotes the decoder). *Let a datapath cost  $C = D + M + A$  for decoder, multiplier and adder. A workload that eliminates  $M$  raises the decoder’s*

Table 20: One ternary neuron, matched width against matched width, every accumulator bit observed, median of five placement seeds, XC7A200T, no DSP. Every candidate decodes a *packed* word. Two corrections stand behind these numbers. An earlier version reported 440 against 895 LUTs, a  $5.1\times$  ratio, which handed TNF pre-widened fields while the competitors unpacked theirs, and is withdrawn. The version after that reported  $3.1\times$  at 16 bits and  $5.6\times$  at 32 — on TNF modules that did not implement the specified format. At the specified parameters, and checked against a reference, the advantage is smaller and it is real.

| width | format       | LUTs       | $F_{\max}$ | MHz/LUT       | ratio                          |
|-------|--------------|------------|------------|---------------|--------------------------------|
| 8b    | <b>TNF8</b>  | <b>545</b> | 62.13 MHz  | <b>0.1140</b> | <b>1.39<math>\times</math></b> |
| 8b    | posit8       | 540        | 44.33 MHz  | 0.0821        | —                              |
| 16b   | <b>TNF16</b> | <b>565</b> | 66.28 MHz  | <b>0.1173</b> | <b>2.07<math>\times</math></b> |
| 16b   | posit16      | 712        | 40.38 MHz  | 0.0567        | —                              |
| 32b   | <b>TNF32</b> | <b>569</b> | 66.91 MHz  | <b>0.1176</b> | <b>3.99<math>\times</math></b> |
| 32b   | posit32      | 953        | 28.16 MHz  | 0.0295        | —                              |

share from  $D/(D+M+A)$  to  $D/(D+A)$ , and the ratio between two formats differing only in  $D$  from  $1 + \Delta/(D_0+M+A)$  to  $1 + \Delta/(D_0+A)$ . The advantage of a decoder-free format therefore grows exactly as the multiplier shrinks, and is largest where there is none.

Three things this does not say, and each would otherwise be found in minutes.

It is not that TNF avoids multiplication. It pays the significand multiply like every other format when one is needed; the quadratic term of its own area law (Theorem 17) *is* that multiplier. What a ternary network removes is removed for the network, not for the format.

It is not that TNF is better for FPGAs in general. We synthesised the exponent decode of TNF against an ordinary binary fixed field and they measure identically, 32 LUTs each, exactly as Theorem 8 requires. The advantage in Table 20 is over a *tapered* format, not over a fixed-field one: an IEEE-style field would place alongside TNF, not alongside posit.

And it is not a measurement on a ternary fabric. None exists to buy, and this is a binary FPGA where our own Theorem 8 says the ternary encoding earns nothing. It does not need to: the gain here comes from the decoder that is absent, and would be the same for any fixed-field format.

**What the ternary fabric would add, structurally.** On a fabric whose positions hold trits, a binary exponent field can be stored densely, in  $\lceil b/\log_2 3 \rceil$  trits, or addably, one bit per trit — but not both. The dense form cannot be added without a radix conversion; the addable form wastes  $1 - 1/\log_2 3 = 36.9\%$  of every position, which for the exponent widths in this catalogue is 33 to 57% of the field. A balanced-ternary exponent pays neither. This is structure rather than measurement, and it is the one place where the ternary encoding of Theorem 8 would stop being free of charge and start being free of cost.

## 17 The ladder is a family, and the family is the frontier

The results above treat each rung as a format. Sweeping the exponent budget instead of fixing it changes what is being claimed, and strengthens it.

**Theorem 22** (The width rule generates the frontier). *Fix a width  $N$ . The set  $\mathcal{F}_N = \{(E_t, M) : 1 + E_t + M = N, E_t \geq 2, M \geq 1\}$  contains  $N - 3$  members, each exact by construction. By Theorem 1 member  $E_t$  has effective mantissa  $M = N - 1 - E_t$  and dynamic range  $3^{E_t} - 1$*

Table 21: Dominance by width class. The family member that dominates is given for the hardest case in each class. Effective mantissa is measured; range is  $3^{E_t} - 1$ , which is analytic for a fixed field.

| class | competitors | survivors | hardest case dominated by                         |
|-------|-------------|-----------|---------------------------------------------------|
| 8     | 7           | 0         | $E_t=4$ , $M=3$ against <code>fp8_e5m2</code>     |
| 16    | 7           | 0         | $E_t=6$ , $M=9$ against <code>takum16</code>      |
| 32    | 9           | 0         | $E_t=5$ , $M=26$ against <code>posit32</code>     |
| 64    | 4           | 0         | $E_t=10$ , $M=53$ against <code>cray_float</code> |
| 128   | 1           | 0         | $E_t=10$ , $M=117$ against <code>binary128</code> |

*binades, so the members are totally ordered and no two dominate one another:  $\mathcal{F}_N$  is a discrete Pareto frontier in  $(M_{\text{eff}}, \text{range})$ .*

The consequence is a claim about the catalogue rather than about one rung.

**Corollary 17** (No catalogued competitor survives). *At every width class measured, each catalogued format of that width is strictly dominated by some member of  $\mathcal{F}_N$ : there is a member with at least its effective mantissa and at least its range, and strictly more of one.*

Measured against the oracles, twenty-eight competitors across five classes, with the dominating member named in each case:

**Why this is not an artefact of density.** A sufficiently dense family dominates any finite set of points trivially, so the content is in *why* the competitors sit below the frontier rather than on it. Each loses on at least one of three counts, and each count is one of the theorems above. A tapered format measures below its declared mantissa, by Theorem 12. A format that leaves positions unallocated pays  $2^k$  for them, by Theorem 9 – which is what the historical GF-T ladder did at 2, 4 and 6 positions. And a binary exponent at equal position count spans fewer binades, since a trit carries  $\log_2 3$  bits against a bit’s one. The family loses on none of the three because the width rule forbids the second and the encoding settles the third.

**What this does not say.** The pair  $(M_{\text{eff}}, \text{range})$  is a property of the *number*, and dominance in it is not dominance in silicon. Section 16 measures the same formats through one datapath and puts TNF mid-pack among fixed fields at 0.114–0.118 MHz per LUT once each rung is measured at its specified parameters. Against `int8` the two are indistinguishable once seeds are swept (Table 43). The two are consistent: the frontier asks what a word of a given width carries, the datapath asks what it costs to unpack, and every fixed field unpacks by reading fields. The gap in silicon runs between fixed and tapered formats, 2.4 to 6.4×, and belongs to Theorems 14 and 19 rather than to this ladder.

**A limit of the measurement, stated.** Range is analytic for a fixed field, but verifying it at the boundary is not always possible: at  $E_t = 49$  the edge of the range is  $2^{1.4 \times 10^{23}}$ , which no exact arithmetic can construct. Above roughly  $E_t = 30$  the range is a fact about the encoding rather than a measured quantity, and is reported as such.

## 17.1 Which first principles

The width rule has been stated as a rule. It is not one: it is the solution of a constrained maximisation, and naming the constraint is what makes the rest of the paper follow rather than accumulate.

Table 22: The width rule on the block axis. The measured within-block span sets  $E^* = \lceil \log_2 b \rceil$ , and the answer moves with the percentile protected.

| percentile protected | span, binades | $E^*$ | element |
|----------------------|---------------|-------|---------|
| 50th (median)        | 1.89          | 1     | E1M2    |
| 90th                 | 2.45          | 2     | E2M1    |
| 99th                 | 3.04          | 2     | E2M1    |
| 99.9th               | 3.75          | 2     | E2M1    |

**The rule is a Karush–Kuhn–Tucker condition.** Write the design problem directly. Maximise precision subject to a position budget and a range that must be covered:

$$\max_{E,M} M \quad \text{s.t.} \quad 1 + E + M = N, \quad r^E \geq b + 1.$$

With multipliers  $\lambda$  on the budget and  $\mu$  on the range,  $\mathcal{L} = M + \lambda(N - 1 - E - M) + \mu(E - \log_r(b + 1))$  gives  $\partial_M \mathcal{L} = 1 - \lambda = 0$  and  $\partial_E \mathcal{L} = -\lambda + \mu = 0$ , so  $\lambda = \mu = 1$ . Because  $\mu > 0$ , complementary slackness makes the range constraint *active*: at the optimum the exponent is exactly as wide as the range demands and not one position wider. That statement is Theorem 23, now derived rather than asserted. The multiplier is also the shadow price, and  $\lambda = 1$  says a position of exponent costs exactly a position of mantissa — the exchange rate is unity because the budget is linear.

**Why three and not two is a 1950s result, not ours.** The radix economy  $\rho(r) = r / \ln r$  is minimised at  $r = e$ . Ternary sits 0.46% above that optimum and binary 6.15% above it. We claim no credit for this; what the present work contributes is where it applies, which turns out to be the whole reconciliation of our two axes:

$$\underbrace{r}_{\text{cost per position}} \times \underbrace{\log_r V}_{\text{positions}}.$$

The number axis counts only the second factor, and there  $\log_3$  against  $\log_2$  gives the slack  $0.3691 E$  of Corollary 26 unconditionally. The silicon axis restores the first factor, and packing a trit into binary fabric charges  $\kappa(3)/\kappa(2) = 1.68$  per position, which is why BNF16 and TNF16 synthesise within 1% of one another. Theorem 8 and Corollary 26 are not in tension: they are the two factors of one product, reported separately.

**What is derivation and what is only resemblance.** Three connections are literal and are used as such: the multiplier argument above; radix economy, which fixes the exponent’s radix; and Kraft’s inequality, which bounds the taper in Theorem 12. Two resemblances are worth naming only so they are not mistaken for arguments. The variational form recalls least action, but there is no time and no trajectory here — the problem is a static optimisation, and the resemblance carries no consequence. And  $1 + E + M = N$  is a budget, not a conservation law: positions are allocated by a designer, not withheld by nature. We state the distinction because a physical-sounding derivation would be doing rhetorical work that the multiplier argument already does honestly.

**The standard falls out — at one choice of percentile, and not at others.** Applied to the block axis the multiplier argument returns an element format, but which one depends on how much of the block distribution is protected, and that choice was left unstated in an earlier version of this section.

Protecting the median returns E1M2; protecting the 90th percentile or beyond returns E2M1, which is the element the OCP microscaling specification fixes. The correct statement is therefore a threshold rather than a derivation: *the width rule returns the MX element for any design protecting the 90th percentile of blocks or more, and a different element below that.* Which is right is a question about how many blocks may be allowed to saturate, not one settled from first principles, and we say so rather than quoting the agreeing case alone.

Applied to the block axis this stops being abstract. The within-block span of a trained network’s weights, measured before any format is chosen, is 1.89 binades at the median and 3.04 at the 99th percentile. The optimum is  $E^* = \lceil \log_2 b \rceil$ , so protecting the median returns E1M2 and protecting the 99th percentile returns E2M1 — which is the element the OCP microscaling specification fixes. The standard is not reproduced by coincidence: it is the unique solution of the multiplier problem at the tail these weights actually have. Theorem 24 says which of the two should win, since under-sizing pays on the tail linearly while over-sizing pays logarithmically on every value, and Section 23.7 reports the measurement that decides it.

## 17.2 The width rule is an optimisation with one solution

Theorem 46 compares against a given competitor. Stated as a design problem it selects a member outright, which is what makes the rule falsifiable: it names a winner before any measurement.

**Theorem 23** (Optimal member for a named range). *Let a workload visit  $b$  binades. Among all members of  $\mathcal{F}_N$  that represent it without overflow, the effective mantissa is maximised uniquely at*

$$E_t^* = \lceil \log_3(b + 1) \rceil, \quad M^* = N - 1 - \lceil \log_3(b + 1) \rceil.$$

*Proof.* Representability requires  $3^{E_t} - 1 \geq b$ , i.e.  $E_t \geq \log_3(b + 1)$ , and  $E_t$  is an integer, so  $E_t \geq E_t^*$ . By the width rule  $M = N - 1 - E_t$  is strictly decreasing in  $E_t$ , so the constraint binds and the minimiser is unique.  $\square$

The rule therefore has no free parameter once the workload is named, and naming the workload is a measurement rather than a choice. Section 16 uses this as a prediction: the binade span of a network’s weights is measured first,  $E_t^*$  is computed from it, and only then is perplexity evaluated across the whole family. A different winner would falsify the rule.

**Theorem 24** (Regret of a misnamed range). *Sizing for  $b'$  when the workload visits  $b$  costs*

$$R = \begin{cases} \lceil \log_3(b + 1) \rceil - \lceil \log_3(b' + 1) \rceil \text{ mantissa bits,} & b' > b, \\ \text{overflow on a set of measure } 1 - \frac{\log_3(b' + 1)}{\log_3(b + 1)}, & b' < b. \end{cases}$$

*The penalty is asymmetric: over-sizing costs precision logarithmically, under-sizing costs range linearly, so when the visited range is uncertain the rule should round up.*

The asymmetry is worth naming because it contradicts the intuition that a wider exponent is the conservative choice. It *is* conservative — but the price is paid in every value the format ever represents, whereas the price of being too narrow is paid only on the tail. Which dominates is an empirical question about the tail’s weight, not a matter of taste.

**Corollary 18** (When the trit is worth its packing). *Ternary buys  $\Delta = E(1 - \log_3 2)$  positions and costs  $\kappa(3)/\kappa(2) = 1.683$  per position when packed into binary fabric, where  $\kappa(r) = (r - 1)^2/(r \ln r)$ . The two cancel at  $E$  satisfying  $\Delta = 0.331 E_t$ , so on positions the trit wins outright and on binary bits it breaks even — which is why BNF16 and TNF16 synthesise within 1% of one another while the number-axis frontier separates them.*



Table 23: The four families at eight bits, one workload, one metric. SmolLM2-135M on wikitext-2, 40 windows of 2048 tokens, per-tensor scale, baseline 14.4874 verified in band before any comparison. The predictions were printed *before* the perplexities were computed.

| family                | rule, radix           | magnitudes | ppl            | vs fp32 |
|-----------------------|-----------------------|------------|----------------|---------|
| fp32 reference        | —                     | —          | 14.4874        | 1.000×  |
| <b>GF8</b> , $E=3$    | golden ratio, binary  | 129        | <b>14.6130</b> | 1.009×  |
| GF-T8, $E_t=3$        | golden ratio, ternary | 109        | 15.5147        | 1.071×  |
| <b>BNF8</b> , $E=3$   | width rule, binary    | 129        | <b>14.6130</b> | 1.009×  |
| <b>TNF8</b> , $E_t=2$ | width rule, ternary   | 73         | <b>14.7012</b> | 1.015×  |

Table 24: The GF-T and TNF ladders rung by rung. Eight of the nine specified rungs are identical — the two rules choose the same budget — and they part only at 32 positions.

| rung      | GF-T ( $E_t, M$ ) |           | TNF ( $E_t, M$ ) |           |                           |
|-----------|-------------------|-----------|------------------|-----------|---------------------------|
| 4         | 2                 | 1         | 2                | 1         | identical                 |
| 8         | 3                 | 4         | 3                | 4         | identical                 |
| 16        | 4                 | 9         | 4                | 9         | identical                 |
| <b>32</b> | <b>5</b>          | <b>21</b> | <b>6</b>         | <b>25</b> | <b>the one divergence</b> |
| 64        | 7                 | 52        | 7                | 52        | identical                 |
| 128       | 8                 | 115       | 8                | 115       | identical                 |
| 256       | 9                 | 242       | 9                | 242       | identical                 |
| 512       | 10                | 497       | 10               | 497       | identical                 |
| 1024      | 11                | 1006      | 11               | 1006      | identical                 |

## 18 Four families on two axes

TNF is one corner of a square, and the square is worth drawing because the two rules that define its axes were derived separately and for unrelated reasons.

One axis is the *rule that fixes the exponent budget*. The golden-ratio rule takes  $E = \text{round}((N - 1)/\varphi^2)$  and produces the GF ladder; the width rule takes  $1 + E + M = N$  and produces the BNF ladder. The other axis is the *radix of the exponent’s encoding*: binary, or balanced ternary. Crossing them gives four families — GF and GF-T on the golden-ratio rule, BNF and TNF on the width rule — and each is a distinct format rather than a renaming of another.

**Two names, one encoding, two rules.** The square’s corners are not four unrelated formats. GF and BNF share the binary encoding and differ in which rule fixes  $E$ ; GF-T and TNF share the ternary encoding and differ in the same way. How far the rules then diverge is a measurement, and mostly they do not.

This is why the silicon tables carry TNF rows and no separate GF-T ones: at every rung but 32 they would be the same numbers twice. It is also why the eight-bit measurement below reports GF-T8 and “TNF8 at  $E_t = 3$ ” as one line of 109 magnitudes and 15.5147 — they are the same format, reached from two directions. BNF is measured too, at 97 LUT and 388.35 MHz, but on the isolated-decoder harness of Section 7 rather than the ternary-neuron datapath of Table 39, which is where it now appears in Table 39 as row 2, measured on the same harness as everything else.



**The two rules name the same format, and neither predicted it.** GF8 at  $E = 3$  and BNF8 at  $E = 3$  return 14.6130 — the same number, not a close one, because at eight bits the two derivations select the identical layout,  $E = 3$  and  $M = 4$ . That is a stronger statement than either rule alone and one we did not set out to find.

**And the width rule’s prediction was falsified twice.** It named BNF8  $E = 4$  and TNF8  $E_t = 3$ ; the measured winners are  $E = 3$  (14.6130 against 14.6592) and  $E_t = 2$  (14.7012 against 15.5147). Both predictions were one step too wide. The rule’s *form* survives and is visible in the sweep — a single optimum, both directions from it worse, and the penalty asymmetric exactly as Theorem 24 requires, with  $E = 1$  giving 4.5 million and  $E = 2$  giving 288 while  $E = 4$  costs 0.3% and  $E = 5$  costs 6%. What was wrong is the estimator of the range the workload visits, not the rule that consumes it: we measured the span from the 0.1st percentile to the maximum, which credits a tail carrying almost no energy. Recorded rather than quietly re-tuned, because a rule that names its winner before the measurement can be wrong, and this one was.

**Ternary loses on a binary fabric here too, for the third independent time.** GF-T8 carries 109 magnitudes where GF8 carries 129, and pays 15.51 against 14.61; TNF8 at  $E_t = 2$  carries 73. Three measurements now agree from different directions: BNF16 and TNF16 within 1% in placed silicon, GF8 against GF-T8 here, and MXFP4 against TNF4 on the block axis (Section 21). The ternary exponent earns on positions and pays on bits, exactly as Theorem 8 requires, and this fabric charges in bits.

## 19 Two formats close a ternary node

A ternary node contains exactly three sites where a number format could live, and only one of them needs one.

1. The **weight** is a code. Multiplying by it is a sign-select  $(+x, -x, 0)$ , not a multiply.
2. The **sample** arrives ADC-native and is used as it lands.
3. The **accumulator** is the only object in the system carrying a dynamic range that must be spent.

This is the whole design problem, and it is smaller than the literature makes it look. The published ternary methods answer (1) and leave (3) in `fp16` or `int8`; the published format work answers (3) and assumes a multiplier at (1). We answer both, with two formats that were derived independently and turn out to be complements.

### 19.1 Why the weight alphabet is $\varphi$ , and why that is forced

GFternary is a two-bit alphabet  $\{-\varphi, 0, +\varphi\}$ . The  $\varphi$  is not decoration, and the following says why it is the only possible choice.

**Theorem 25** (The golden alphabet is unique). *Let a three-symbol weight alphabet be  $\{-r, 0, +r\}$  with  $r > 1$ , and require that the product of two weights be expressible in the additive lattice the datapath already computes in, i.e.  $r^2 = r + 1$ . Then  $r = \varphi$  uniquely.*

*Proof.*  $r^2 - r - 1 = 0$  has roots  $(1 \pm \sqrt{5})/2$ , of which one is positive. □

**Corollary 19** (Depth costs no multiplier). *The gain of a stack of  $k$  ternary  $\varphi$ -layers is exactly*

$$\varphi^k = F_k \varphi + F_{k-1},$$

*a pair of integers with  $F$  the Fibonacci sequence. Rescaling between layers is therefore a shift-and-add, and the zero-multiplier property survives to arbitrary depth.*

This is the practical content of Theorem 25, and it is what separates the  $\varphi$  alphabet from the  $\{-1, 0, +1\}$  one used throughout the ternary-network literature. With unit weights the layer gain is 1 and carries no information, so every published method attaches a learned real scale  $\alpha_\ell$  per layer — and multiplying by  $\alpha_\ell$  puts the multiplier back. With the  $\varphi$  alphabet the inter-layer scale is known exactly and is a pair of integers, so it is never multiplied by. Storage is two bits either way and the symbol count is three either way; the  $\varphi$  alphabet simply carries a scale that the unit alphabet has to learn and then pay for.

## 19.2 The linear path is exact

Theorem 25 is usually read as removing a multiplier. It removes the rounding with it: on the alphabet it fixes, the entire linear path of a network is computed exactly.

Represent a value as an integer pair  $(a, b)$  standing for  $a + b\varphi$ . Then

$$\varphi(a + b\varphi) = a\varphi + b\varphi^2 = a\varphi + b(\varphi + 1) = b + (a + b)\varphi,$$

so applying a weight is the map  $(a, b) \mapsto (b, a + b)$  — the Fibonacci recurrence, one integer addition, no shift. Negation flips both components and a zero weight is a skip.

**Theorem 26** (Exactness of the multiply-free path).  *$\mathbb{Z}[\varphi] = \{a + b\varphi : a, b \in \mathbb{Z}\}$  is a ring, and the weight alphabet  $\{-\varphi, 0, +\varphi\}$  lies inside it. Hence for inputs in  $\mathbb{Z}[\varphi]$  the entire linear part of a ternary network — every weight application and every accumulation, to arbitrary fan-in and arbitrary depth — remains in  $\mathbb{Z}[\varphi]$  and is computed without rounding error of any kind.*

*Proof.* Closure under addition is componentwise. Closure under multiplication follows from  $\varphi^2 = \varphi + 1$ , which reduces every product to the stated integer-pair map. The accumulator is therefore a pair of exact integer registers.  $\square$

Measured on a layer of fan-in 512 with weights drawn from the alphabet, the integer pair reproduces the real-valued sum to the precision of the checking arithmetic and not to the precision of the datapath: there is nothing in the datapath to be imprecise. The component widths grow logarithmically, reaching eight bits over those 512 terms.

**The theorem checked against the silicon.** A theorem about a lattice says nothing about whether the RTL implements the lattice, and six modules in this work turned out not to implement the format they were labelled with. So the synthesised node is compared against exact integer arithmetic in  $\mathbb{Z}[\varphi]$ , with no float anywhere in the comparison: the datapath’s pair  $(a, b)$  against the reference’s, as integers.

Sixteen thousand vectors at fan-in 8, 16 and 32, eight-bit samples, weights drawn uniformly from the alphabet: **zero disagreements**. Depth is checked the same way, since the claim is about depth and not only about a layer — one step fed back on itself is  $k$  stacked layers, and after  $k$  clocks the pair must be  $(F_{k-1}, F_k)$ . Thirty depths, zero disagreements; at  $k = 30$  the gain is exactly (514229, 832040), two integers, with no multiplication anywhere on the path.

The first run disagreed on 3,601 of 4,000, in the  $b$  component only with  $a$  always agreeing — the signature of a convention difference rather than an arithmetic one, and it was the reference that had the encoding wrong. The shape of a disagreement says which side to suspect: all-wrong-in-one-component is a convention, wrong-at-the-extremes is a missing edge case, scattered is arithmetic.

## 19.3 Closure is what removes the machinery

The multiply-free datapath is not new as an aim, and the published attempts make the role of Theorem 26 visible by contrast.

The closest is FQP [28], a Fibonacci quantization processor that turns the products of Fibonacci-quantized weights into additions. To do so it carries two distinct arithmetic units — a dualistic-transformation adder for the large operands and a bit-exclusive adder for the small ones — and a topological-order routing scheme that maps data onto them. Fibbinary quantization for neural radio receivers [29] takes the softer route on the same substrate and reports 45% and 44% savings in multiplier power and area: the multiplier is made cheaper, not removed.

The machinery in the first is not an artefact of implementation. It follows from a property of the number set.

**Corollary 20** (Closure removes the transformation stage). *Let  $A$  be a weight alphabet and let  $s$  be the operation that applies a weight. If the additive group generated by  $A$  is closed under  $s$ , the datapath needs only that group’s addition. If it is not, every product must be re-expressed in the representable set, which requires a transformation stage and a schedule that routes operands to it.*

*Proof.* Immediate from Theorem 26 and its negation. For  $A = \{-\varphi, 0, +\varphi\}$  the generated group is  $\mathbb{Z}[\varphi]$  and  $s = \varphi \cdot$  maps  $(a, b) \mapsto (b, a + b)$ , which lands inside it, so addition suffices. For  $A$  a set of Fibonacci integers  $\{F_k\}$ , the product  $F_i F_j$  is in general not a Fibonacci integer —  $F_4 F_4 = 9$  is not in  $\{1, 1, 2, 3, 5, 8, 13, \dots\}$  — so the product leaves the representable set and must be transformed back into it.  $\square$

## 19.4 How many multiply-free scales are there

Closure explains why  $\varphi$  works. It does not by itself say  $\varphi$  is the only choice, and the honest answer is that it is not — it is the only choice at its size.

A scale  $r$  can be applied without a multiplier exactly when  $r$  is an algebraic integer whose companion matrix has entries in  $\{0, \pm 1\}$ : writing  $r^d = a_{d-1}r^{d-1} + \dots + a_0$ , multiplication by  $r$  maps the coordinate vector  $(x_0, \dots, x_{d-1})$  to  $(a_0 x_{d-1}, x_0 + a_1 x_{d-1}, \dots)$ , and any  $|a_i| > 1$  reintroduces a constant multiply. The condition is finite, so the scales can be enumerated rather than argued about.

**Theorem 27** (The multiply-free scales, by degree). *Over coefficients in  $\{0, \pm 1\}$ :*

- **Degree 1** admits  $r \in \{0, \pm 1\}$ , which is no scaling, and  $r = \pm 2^k$ , free by shifting. One register, ratio 2.
- **Degree 2** admits exactly two equations with a real root of modulus other than 0 or 1:  $r^2 = r + 1$  and  $r^2 = -r + 1$ , whose roots are  $\varphi$  and  $\varphi^{-1}$  — the same ladder. One addition, two registers, ratio 1.618.
- **Degree 3** admits three:  $r^3 = r + 1$  (plastic, 1.3247),  $r^3 = r^2 + 1$  (supergolden, 1.4656),  $r^3 = r^2 + r + 1$  (tribonacci, 1.8393). At most two additions, three registers.

*Proof.* Exhaustive over the nine degree-2 and twenty-seven degree-3 coefficient triples; the degree-2 discriminants  $a^2 + 4b$  are negative for four cases, and the remaining five give roots in  $\{0, \pm 1\}$  except for  $b = 1, a = \pm 1$ .  $\square$

**Corollary 21.** *Nothing lies between the shift and  $\varphi$ .  $\varphi$  is the unique multiply-free refinement of the power-of-two ladder available at two registers.*

So the multiply-free scale is not a point but a hierarchy indexed by algebraic degree, and each degree buys finer granularity for one more register. Table 25 measures the trade at the first two rungs where it is a trade at all. The plastic step  $(a, b, c) \mapsto (c, a + c, b)$  costs the same

Table 25: One scale application, xc7a200t, median of five seeds. Matched: neither unit has a direction mux. Level error is  $(r - 1)/(r + 1)$ , the worst relative gap on a ladder of ratio  $r$ .

| scale              | LUT        | FF  | $F_{\max}$ (MHz) | level error   |
|--------------------|------------|-----|------------------|---------------|
| $\varphi$ , 16-bit | <b>149</b> | 128 | 307.69           | 23.61%        |
| plastic, 16-bit    | 186        | 160 | 318.47           | <b>13.97%</b> |
| $\varphi$ , 32-bit | <b>223</b> | 192 | <b>247.10</b>    | 23.61%        |
| plastic, 32-bit    | 228        | 200 | 231.21           | <b>13.97%</b> |

Table 26: Perplexity by ladder and code budget, SmolLM2-135M on wikitext-2. Coarse ladders span range and waste precision; fine ones do the reverse. The best rung moves down the hierarchy as the budget grows.

| bits | shift (2)      | $\varphi$ (1.618) | supergold (1.4656) | plastic (1.3247) |
|------|----------------|-------------------|--------------------|------------------|
| 3    | <b>2309.86</b> | 41835.53          | 1367268.11         | 6720092.29       |
| 4    | 76.81          | <b>24.43</b>      | 25.10              | 55.72            |
| 5    | 77.52          | 22.73             | 18.93              | <b>16.45</b>     |

single addition as the Fibonacci step and one more register, and its ladder is finer: successive levels differ by 13.97% against  $\varphi$ 's 23.61%.

At 32 bits the refinement costs 2.2% of the area and 6.4% of the frequency. That is close enough to free that the choice is a question about the network rather than about the arithmetic, so we asked the network.

## 20 Which rung to use, and how the answer moves

Table 26 quantises every 2-D linear weight of SmolLM2-135M to  $\pm r^k$  with a per-output-channel scale, the codebook sized to exactly  $2^b$  entries including zero and sign, and evaluates wikitext-2 perplexity over twelve windows of 2048 tokens. Baseline **fp32** is 14.36.

The same sweep on Qwen2.5-0.5B — a different architecture at four times the parameters, with a different tokenizer and different training data — returns the same winner at every budget and the same complete ordering at every budget: shift 2384.96, then  $\varphi$  **16.24** against supergolden 17.98 and plastic 47.41, then plastic **13.17** against supergolden 14.09 and  $\varphi$  15.33, from an **fp32** baseline of 12.27. At four bits the larger model separates  $\varphi$  from supergolden by 10.7%, where the smaller one gave 2.7% and no separation.

A ladder of ratio  $r$  with  $n$  magnitudes spans  $r^{n-1}$ , so finer steps are bought with span. At three bits nothing spans enough and the coarsest wins; at four the balance sits at  $\varphi$ ; at five the finest tested wins outright. This is the range-against-precision law of Section 4 applied to the scale hierarchy rather than to a field layout, and it says plainly that  $\varphi$  is optimal at a budget, not everywhere. At four bits the two models bracket the comparison: SmolLM2 separates  $\varphi$  from supergolden by 2.7%, which twelve windows cannot resolve, while Qwen separates them by 10.7%, which they can.  $\varphi$  is at worst tied there and ahead on the larger model, and supergolden costs a register either way. One bit up, the plastic number wins on both, by 28% and 14%, with  $\varphi$  third of four — and by then the affordable part of the hierarchy has been passed as well.

The choice does not need the network. Ranking the ladders by mean squared quantisation error on the weights alone reproduces the perplexity order exactly at three and five bits, and at four bits transposes only the pair perplexity cannot separate.

**Theorem 28** (The ladder law). *Among multiply-free ladders at a fixed code budget, the one minimising weight mean squared error minimises perplexity, up to pairs whose error differs by less than the evaluation resolves.*

The error itself has a closed form, so the rung follows from a histogram. Normalising each channel by its maximum puts the levels at  $r^{-k}$  for  $k = 0, \dots, n-1$  with  $n = (2^b - 1)/2$ , and

$$\text{MSE}(r, b) \approx c(r)^2 \mathbb{E}[x^2 \mathbf{1}_{x>t}] + \mathbb{E}[x^2 \mathbf{1}_{x<t}], \quad t = \frac{1}{2}r^{-(n-1)},$$

where  $c(r)^2$  is the mean square *relative* rounding error of a geometric ladder of ratio  $r$  — a one-dimensional integral over log-position within a bin, independent of the data. The first term falls as  $r$  falls and the second rises as the clipping threshold does; the minimum is the crossover. Against the measured error over 106,168,320 normalised weights the form agrees to within 1–9% at four and five bits and selects the same ladder at all three budgets. At three bits it underestimates by a third, for a reason worth stating: at seven codes the regime is absolute truncation, not relative rounding, and the model is built on the latter.

## 20.1 What the four-bit transposition is, and what it is not

The form above ranks correctly at three and five bits and transposes the top pair at four, choosing supergolden where perplexity chooses  $\varphi$ . The transposition reproduces on both models in the same direction, so it is an effect. Two candidate mechanisms were tested; one failed and one held, and the pair is more informative than either alone.

**Activation salience: fails.** The output error of a layer is  $\sum_j \delta w_{ij} x_j$ , so a weight multiplying a large input channel should cost more, and the channels are far from equal — the first attention projection spans  $28\times$  between its largest input channel and its median on SmolLM2-135M,  $22.4\times$  on Qwen2.5-0.5B. This is the observation AWQ [27] is built on. Ranking by  $\sum_j a_j^2 \delta w_{ij}^2$  reverses the four-bit pair on SmolLM2, matching perplexity — and does not on Qwen, where the weighting scales all four candidates by very nearly 1.09 and therefore cannot reorder them. An explanation that needs channel salience to correlate with the error pattern is not general, because in one of two models it does not.

**Loss curvature: holds.** The quantity the transposition is about is not how large an operand is but how much the loss moves when a weight moves. Ranking by the diagonal Fisher weighting  $\sum_{ij} g_{ij}^2 \delta w_{ij}^2$  with  $g = \partial L / \partial w$  reverses the four-bit pair to  $\varphi$  on *both* models and leaves five bits at plastic on both, matching perplexity in all four cases.

**Theorem 29** (Curvature correction). *Among multiply-free ladders at a fixed code budget, the minimiser of  $\sum_{ij} g_{ij}^2 \delta w_{ij}^2$  predicts the perplexity ranking where the unweighted error does not, and does so on both models tested. The correction moves the choice toward the coarser ladder.*

The two results together say something the successful one alone would not. Activation magnitude is half of the gradient and not the half that decides:  $g_{ij} = \delta_i x_j$  carries both the input the weight multiplies and the sensitivity of the loss to that unit’s output, and it is the second factor that makes the correction replicate. A method calibrated on activations alone — and several published ones are — is using a proxy that agreed with the target on one of our two models.

**Theorem 30** (Optimal ratio). *For a weight distribution and a code budget, the optimal multiply-free ratio is the minimiser of  $\text{MSE}(r, b)$ , and the degree required is the smallest whose minimal scale does not exceed it. Minimising over continuous  $r$  gives  $r^* = 2.054, 1.4115, 1.1999, 1.1050, 1.0576$  and  $1.0319$  at 3 to 8 bits, with  $\ln r^*$  halving per bit as doubling the code count halves the available log-spacing.*

Table 27: Block axis, SmolLM2-135M, 40 windows, **fp32** baseline 14.4874. The optimal codebook improves squared error nearly fifteenfold and perplexity by 0.9%, and every ladder that beats MXFP4 on the first loses on the second.

| element, 8 magnitudes                  | squared error          | $\times$ bound | ppl            |
|----------------------------------------|------------------------|----------------|----------------|
| Lloyd–Max (optimum, not implementable) | $1.341 \times 10^{-3}$ | 1.00           | <b>22.2976</b> |
| MXFP4 (E2M1)                           | $2.007 \times 10^{-2}$ | 14.97          | <b>22.4998</b> |
| supergolden/plastic, 4 fine steps      | $2.068 \times 10^{-3}$ | 1.54           | 27.3079        |
| supergolden/plastic, 3 fine steps      | $2.116 \times 10^{-3}$ | 1.58           | 30.8306        |
| $\varphi$ /supergolden, 2 fine steps   | $2.749 \times 10^{-3}$ | 2.05           | 38.6471        |
| $\varphi$ /plastic, 2 fine steps       | $2.823 \times 10^{-3}$ | 2.11           | 55.4495        |

## 21 The block axis, and the half of it that is ours

Everything above concerns per-channel scaling. The other deployed axis shares one scale across a short block, and there our formats lose. It is worth stating how thoroughly, because the answer is not a ranking but a ceiling.

Lloyd–Max on the within-block distribution — 18,850,950 values, block of 32 along the contraction axis, E8M0 shared scale as the MX specification mandates — gives the eight-level codebook minimising squared error. It is neither multiply-free nor a ladder; it is the best any eight magnitudes can do.

**Theorem 31** (The block axis is not an element-format problem). *On a short block with a shared scale, the element codebook minimising squared quantisation error is not the one minimising perplexity, and the gap between the best achievable codebook and a deployed one is under one percent. Effort spent designing element formats for this axis cannot recover more than that.*

The ranking does not merely fail to transfer; it inverts. Every two-segment ladder here beats MXFP4 on squared error by factors of three to ten, and every one loses on perplexity by 21% to 146%.

The reason is what a block is. Thirty-two weights share one scale, and squared error charges a clipped small weight at its own magnitude — so a codebook concentrating its levels near the top of the range scores well, having put its resolution where the mass is. But the network does not read a block’s weights independently; they are summed against activations, and a codebook that resolves the top of every block finely while collapsing its lower two-thirds destroys the block’s ability to represent a *direction*, not merely its magnitudes. E2M1 spends 0, 0.5, 1, 1.5, 2, 3, 4, 6: a  $12\times$  span with levels at both ends. It scores badly on squared error precisely because it refuses to concentrate, and it wins on perplexity for the same reason.

We record this as the competition’s result, not ours. Three of our own attacks failed here — a ternary-exponent float, constant-ratio ladders, and two-segment ladders built to imitate E2M1’s spacing — and the bound explains why a fourth is not warranted.

### 21.1 The other half of the block axis

The bound closes the element format. It says nothing about the scale, which is one value per thirty-two weights and had never been varied. Its cost is amortised thirty-twofold, which is the regime where a finer grid is affordable — and the accounting decides the comparison before any perplexity is measured.

E8M0 gives the scale eight exponent bits. Over 3,317,760 block scales the range actually occupied is 8.32 binades on SmolLM2-135M and 9.12 on Qwen2.5-0.5B, so four bits suffice. At four bits  $2^k$  needs 8.3 steps of the sixteen available and  $\varphi^k$  needs 12.0, with half the step. The choice at equal width is between exactly those two.

Table 28: Block scale grid,  $K=32$ , E2M1 elements, 40 windows. At equal bits  $\varphi^k$  beats  $2^k$  on both models; finer grids cost one or two more bits and their benefit does not replicate.

| scale grid                     | b        | SmolLM2        | Qwen           |
|--------------------------------|----------|----------------|----------------|
| $2^k$ (the MX specification’s) | 4        | 22.4998        | 14.9447        |
| $\varphi^k$                    | <b>4</b> | <b>21.3545</b> | <b>14.8512</b> |
| $\sqrt{2}^k$                   | 5        | 21.8960        | 14.7456        |
| plastic <sup>k</sup>           | 5        | 22.7625        | 15.0333        |
| $2^{k/4}$                      | 6        | 24.0791        | 14.4328        |

Table 29: Scale-cost frontier, E2M1 elements, 40 windows.  $\varphi^k$  at four bits per block of 32 strictly dominates MXFP4 — cheaper and better on both models. Above 4.25 bits per weight the frontier belongs to scales carrying a mantissa.

| scheme                              | scale b/w     | total b/w     | SmolLM2        | Qwen           |
|-------------------------------------|---------------|---------------|----------------|----------------|
| $\varphi^k$ 4b/64                   | 0.0625        | 4.0625        | 22.0978        | 15.0090        |
| <b><math>\varphi^k</math> 4b/32</b> | <b>0.1250</b> | <b>4.1250</b> | <b>21.3545</b> | <b>14.8512</b> |
| $2^k$ 4b/32                         | 0.1250        | 4.1250        | 22.4998        | 14.9447        |
| <b>MXFP4</b> , E8M0 8b/32           | 0.2500        | 4.2500        | 22.4998        | 14.9447        |
| E4M3 8b/32                          | 0.2500        | 4.2500        | 19.8628        | 13.7636        |
| NVFP4-like E4M3 8b/16               | 0.5000        | 4.5000        | 18.5445        | 13.5340        |

A stronger reading of the first column does not survive the second. On SmolLM2 the response is non-monotone with its minimum at  $\varphi$  and  $2^{k/4}$  falling below plain  $2^k$ , which would say  $\varphi$  is the optimum of the grid; on Qwen the response is monotone in granularity and the finest grid wins. The sign replicates and the size does not — 5.09% against 0.63% — and the claim of optimality does not replicate at all. We state only what does.

**The frontier, not the pair.** A scale is one value per  $K$  weights, so it costs bits/ $K$  per weight, and comparing formats in pairs hides that. Table 29 puts perplexity against total bits per weight. The Pareto set is identical on both models.

Three things follow, and only the third is ours.

*MXFP4’s scale field is wider than it needs to be.* That E8M0’s range exceeds what weights occupy has been observed elsewhere [25] and we claim no priority for the observation; what is measured here is how much, and what the freed bits are worth.  $2^k$  at four bits reproduces E8M0’s perplexity *exactly* — 22.4998 and 14.9447, to the last digit — because the occupied range is 8.32 and 9.12 binades and eight exponent bits buy nothing over four.

*At those four bits  $\varphi^k$  beats  $2^k$ ,* so  $\varphi^k$  4b/32 strictly dominates MXFP4: cheaper by 0.125 bits per weight and better on both models. It is not a trade.

*And the upper frontier is reached by spending more.* E4M3 — a scale carrying a mantissa, which is what NVFP4 [26] uses — reaches 19.8628 and 13.7636 at the same 4.25 bits per weight where MXFP4 sits, against our 21.3545 and 14.8512 at 4.1250. The comparison is not at equal width, and the next subsection shows what happens when it is made so.

## 21.2 A geometric scale grid dominates a float one

A scale is read as a multiplier, so the error that matters is relative, and the grid minimising relative error over a log-distributed range is uniform in log. A float scale is not.



Table 30: Scale grid at equal width, K=32, E2M1 elements, 40 windows. A geometric grid beats E4M3 at both widths on both models, as Theorem 32 requires. The margin does not transfer between models; the sign does.

| scale grid | b | SmolLM2        | Qwen           |
|------------|---|----------------|----------------|
| geometric  | 7 | <b>18.8024</b> | <b>13.6401</b> |
| E4M3       | 7 | 19.8628        | 13.7636        |
| geometric  | 8 | <b>18.1238</b> | <b>13.6910</b> |
| E4M3       | 8 | 19.8628        | 13.7636        |

**Theorem 32** (The geometric scale grid). *A float scale  $2^e(1 + m/2^k)$  places  $2^k$  points in each binade, with largest adjacent ratio  $1 + 2^{-k}$  at the bottom of the binade. A geometric grid with the same point count places them at  $2^{j/2^k}$ , with every adjacent ratio equal to  $2^{2^{-k}}$ . Since  $2^x = e^{x \ln 2} < 1 + x$  for  $x > 0$ , the geometric grid’s worst-case step is strictly smaller at every mantissa width, and the ratio of the two excesses rises monotonically to  $1/\ln 2 = 1.442695$ .*

*Proof.* Immediate from  $\ln 2 < 1$ . The limit follows from  $(1 + x) - 1/(2^x - 1) \rightarrow 1/\ln 2$  as  $x \rightarrow 0^+$ .  $\square$

The defect is structural rather than a matter of tuning: a float grid is uniform in *value* within a binade and therefore non-uniform in log, spending resolution at the top of each binade where the relative step is already small. Root-mean-square relative error over a binade agrees — 0.1138 against 0.1068 at two mantissa bits, 0.0550 against 0.0516 at three, 0.0270 against 0.0254 at four.

Measured at equal width, on both models, the prediction holds (Table 30).

One further reading is model-dependent and we do not make it. On SmolLM2 a geometric grid at 4.25 bits per weight beats the NVFP4-like configuration at 4.50 — cheaper and better; on Qwen it is cheaper and worse, 13.6910 against 13.5340. *Geometric beats float at equal bits replicates; geometric at lower cost dominates NVFP4’s configuration* does not.

**The tension this exposes.** The optimal grid is geometric with ratio  $R^{1/N}$  for the occupied range  $R$  and point count  $N$ , and that ratio is an arbitrary real: applying it needs a multiplier.  $\varphi^k$  is geometric *and* multiply-free, at a ratio fixed at 1.618 which is best where the range and the count want it — at four bits, which is where it wins above. So the placement is a pair of statements rather than one. Among all scale grids, geometric beats float and the best geometric grid needs a multiplier; among multiply-free grids,  $\varphi^k$  is the finest available at four bits. The first is a contribution to the field, the second is what this datapath gets for nothing.

This is not our format beating MXFP4: the element stays E2M1, and only the base of the shared exponent changes, so it is a contribution to the microscaling family rather than a defeat of it. And the grid that helps is ours specifically — on a binary datapath  $2^k$  is a shift and  $\varphi^k$  is not, while on the datapath of Section 19.3  $\varphi^k$  is  $k$  Fibonacci steps and needs no multiplier. A scale grid that costs everyone else a multiplier is free here.

**Scope.** This is stated for per-channel scaling and does not extend to short blocks. Applied to the within-block distribution under a block of 32 and an E8M0 shared scale, the same form returns  $r^* = 1.3308$ , which measures 37.06 against the 23.36 of the ladder it ranks worse — and perplexity there is not even monotone in  $r$ , running 39.09, 23.36, 39.61 as the ratio falls through  $\varphi$ , supergolden and plastic. The form charges clipping at the weight’s own squared magnitude, which is right when a scale is shared by thousands of weights and the clipped ones are a tail, and wrong when thirty-two share one and the clipped ones are a fifth of the block.

Table 31: The multiply-free scales in one place. Level error is  $(r - 1)/(r + 1)$ , the worst relative gap on a ladder of ratio  $r$ ; silicon is one scale application at 16-bit components, xc7a200t, median of five seeds. Every rung below the shift costs exactly one addition; only the register count grows.

| scale       | polynomial                 | $r$    | regs | adds | level err. | LUT / $F_{\max}$ |
|-------------|----------------------------|--------|------|------|------------|------------------|
| shift       | $r = 2$                    | 2.0000 | 1    | 0    | 33.33%     | —                |
| $\varphi$   | $r^2 = r + 1$              | 1.6180 | 2    | 1    | 23.61%     | 149 / 307.69     |
| supergolden | $r^3 = r^2 + 1$            | 1.4656 | 3    | 1    | 18.88%     | —                |
| plastic     | $r^3 = r + 1$              | 1.3247 | 3    | 1    | 13.97%     | 228 / 231.21     |
|             | $r^5 = r^3 + 1$            | 1.2365 | 5    | 1    | 10.57%     | 164 / 306.56     |
|             | $r^4 = -r^3 + r^2 + r + 1$ | 1.1787 | 4    | 3    | 8.20%      | 469 / 184.98     |
|             | $r^6 = r + 1$              | 1.1347 | 6    | 1    | 6.31%      | —                |
|             | $r^8 = r + 1$              | 1.0970 | 8    | 1    | 4.63%      | 225 / 815.66     |

## 22 Fineness costs registers, not adders

Reading the hierarchy by degree, and taking the cost to be the minimal ratio available at each degree, hides the result. Read along a family instead.

**What is forced, and what is not.** For base  $\varphi$ , three digits is the proven minimum for constant-time carry-free addition [21, 19]; a two-digit alphabet is impossible.  $\{-1, 0, 1\}$  is *one* of the minimal alphabets — chosen because it is symmetric, so subtraction reuses the adder — and not the unique one:  $\{0, 1, 2\}$  is minimal too, and a working  $\{0, 1, 2\}$  parallel adder for  $\varphi$  was implemented and tested here. Légerský [20] extends the bound to any finite alphabet inside  $\mathbb{Z}[\varphi]$  containing zero, so no non-integer set escapes it.

The bound constrains the *digit* alphabet of a positional representation and one operation — rewriting a digit-wise sum back into the alphabet with a sliding window. A weight set has no base and no carry, so it does not transfer to the weight alphabet of a dot product. It would transfer only if the accumulator were itself a redundant base- $\varphi$  positional format, and ours is a pair of integers never reduced to a digit string. We record the theorem because it says something real and narrow — two letters cannot be made to work at any window size — and record its scope because the wider reading, which we first wrote down, is false.

The window is not small either: the published algorithm on  $\{-1, 0, 1\}$  is 21-local, with memory 10 and anticipation 10. “Constant time” read as “cheap” would be wrong by an order of magnitude.

**A qualification the table needs.** Not every rung is Perron — that is, not every one has its real root strictly dominant among the conjugates. The degree-4 minimum has a conjugate of modulus 1.5129 against a root of 1.1787, and the degree-6 minimum 1.3069 against 1.0863; the degree-5 minimum 1.1237 is Perron, its largest conjugate being 1.0959. Where the root is not dominant, the coordinate vector of  $r^k$  grows as the larger conjugate rather than as the value, so a chain of  $k$  applications needs  $k \log_2(\rho/r)$  guard bits — 0.36 bits per step at degree 4. That is a cost on composition and not on a single application, which is why the silicon column does not show it, and it is stated here rather than left for a reader to find.

Wu’s exhaustive search [18] establishes that 1.123732821 is the smallest Perron number of degree 5, so the degree-5 rung is provably optimal among Perron scales. It is also the Lind–Boyd exceptional case: the family  $r^d = r + 1$  is Lind’s conjectured smallest Perron number of degree  $d$ , corrected by Boyd [17] at exactly  $d > 3$  with  $d \equiv 3, 5 \pmod{6}$ . The exceptional minimisers are finer — 1.1237 against the family’s 1.1673 at  $d = 5$  — but they are not one-adder: they cost

3, 5, 7 and more adders as  $d$  grows, and beyond  $d = 5$  the gain is under 0.8%. A finer rung at five adders is a different trade from a finer rung at one, and only the second would have been news.

Table 31 collects them. Two readings are worth making explicit. Along the one-adder family the adder count is constant and only the registers grow, so the row for  $r^4$  — three adders for a finer ratio than  $r^5$  gives with one — is the exception rather than the pattern, and it is what made an earlier reading of this hierarchy report a cost that grows with degree. And the frequency column does not fall as the ratio does: the  $d = 8$  rung measures 815.66 MHz against  $\varphi$ 's 307.69, because its single adder sits on one of eight parallel paths rather than one of two.

**Theorem 33** (Siegel's floor bounds the family). *A Pisot–Vijayaraghavan number is a real algebraic integer above one whose conjugates all lie strictly inside the unit circle. Siegel [22] proved the smallest is 1.3247179572..., the root of  $r^3 = r + 1$ , at any degree and with any integer coefficients. That root is the degree-3 member of the family below, so the family is Pisot at  $d = 2$  and  $d = 3$  and at no higher degree: every later member is finer than Siegel's floor and therefore cannot be Pisot.*

Computed conjugate moduli agree exactly: 0.6180 at  $d = 2$ , 0.8688 at  $d = 3$ , then 1.0633, 1.0990, 1.0992, 1.0920, 1.0837 at  $d = 4 \dots 8$ . The boundary is not an accident of our enumeration; it is where a theorem from 1944 says it must be, and *the finest scale that is both applicable by one addition and Pisot is the plastic number*.

The distinction matters for what may be built on a rung, not for the arithmetic itself. Normalisation in base  $\theta$  is realisable by a finite automaton exactly when  $\theta$  is Pisot [23], and for  $n \geq 4$  the root of  $x^n = x + 1$  is not even a Parry number, its shift being non-sofic [24]. So a bounded-latency renormaliser or a canonical-form comparator exists at  $d = 2$  and  $d = 3$  and provably not above. Nothing in this paper's datapath normalises — the coordinates are carried as integers and never reduced to a canonical digit string — which is why the  $d = 8$  step measures 815.66 MHz rather than failing to close. We state it because a reader who assumes a normaliser would be assuming something the mathematics forbids.

**Theorem 34** (The one-adder family). *For every  $d$  the polynomial  $r^d = r + 1$  has a unique real root above one. It has exactly two non-zero coefficients, so its companion map is*

$$(x_0, \dots, x_{d-1}) \mapsto (x_{d-1}, x_0 + x_{d-1}, x_1, \dots, x_{d-2}),$$

*one addition regardless of  $d$ , and the roots converge to  $2^{1/d}$  from above. Any granularity a logarithmic subdivision offers is therefore reachable without a multiplier, at the cost of registers alone.*

The map is verified exact over 3000 random coordinate vectors at  $d = 5$  and  $d = 8$ . The convergence is quick:  $r/2^{1/d}$  is 1.1441 at  $d = 2$ , 1.0159 at  $d = 5$ , 1.0059 at  $d = 8$  and 1.0014 at  $d = 16$ .

**The ladder's cost depends on its role.** Applying a ladder weight  $r^{-j}$  costs  $j$  steps of the companion map. As a *block scale* that is one value per thirty-two weights and the steps amortise; as an *element* it is paid per weight, and a six-bit ladder needs  $j$  up to 31. The element form is therefore a coordinate table — the code selects a vector in  $\mathbb{Z}[r]$  directly — and Table 34 prices it against what it buys.

**The assembled node.** Every silicon figure above prices a part. The figure asked from outside is LUTs per neuron, and Table 32 is the assembled thing: fan-in  $N$ , weights from the alphabet, accumulator in  $\mathbb{Z}[\varphi]$ , 8-bit samples, harness subtracted.

Two's complement is  $(\neg x) + 1$  and the  $+1$  of every lane is a single bit, so the fold is an exclusive-or with the weight's sign plus a narrow second tree carrying the ones, added once

Table 32: A complete ternary neuron, xc7a200t, median of five seeds, harness subtracted, -nodsp. The first column is the implementation this paper first built; the second folds the negation into the accumulation tree, which is where it belonged.

| fan-in | LUT/weight | folded      | $F_{\max}$ | folded | DSP      |
|--------|------------|-------------|------------|--------|----------|
| 8      | 82.5       | <b>28.0</b> | 109.35     | 87.29  | <b>0</b> |
| 16     | 87.1       | <b>33.1</b> | 91.43      | 73.42  | <b>0</b> |
| 32     | 119.0      | <b>33.2</b> | 66.53      | 58.22  | <b>0</b> |

Table 33: One pipeline cut in the accumulation tree. The register cost overtakes the frequency gain between sixteen and thirty-two, so the unpipelined form is the better one at fan-in 32 — the opposite of the usual advice, which is why it was measured.

| fan-in | LUT/weight         | $F_{\max}$ (MHz)      | MHz per LUT            |
|--------|--------------------|-----------------------|------------------------|
| 8      | 28.0 → <b>32.4</b> | 87.29 → <b>126.73</b> | 0.390 → 0.489 (+25.6%) |
| 16     | 33.1 → <b>34.9</b> | 73.42 → <b>106.38</b> | 0.139 → 0.191 (+37.6%) |
| 32     | 33.2 → 47.3        | 58.22 → 75.68         | 0.055 → 0.050 (−8.7%)  |

at the root. Equivalence was checked before synthesis: 0 mismatches in 400 random  $(x, w)$  vectors. Area falls by  $2.9\text{--}3.6\times$  and lands on the hand count — a three-way select of a 16-bit value at roughly 16 LUT6 plus a 16-bit add at roughly 8, so 24 to 30 per lane — which is what confirms the negation was the whole gap rather than part of it. Frequency falls about 20%, the carry tree’s root addition sitting after the main one; on throughput per area the trade is worth  $2.35\text{--}2.87\times$ .

So a ternary neuron costs **28 LUT per weight at fan-in 8 and no DSP at any fan-in.**

**And one pipeline register, where it pays.** The critical path is the whole adder tree, so cutting it once at the halfway level trades a cycle and  $N/2$  registers for frequency — worth measuring rather than assuming, and the answer is not uniform. Equivalence checked first: 0 mismatches in 398 random  $(x, w)$  vectors, the pipelined output trailing by exactly one cycle.

At fan-in 8 and 16 the cut buys 45% more frequency for 16% and 5% more area; at 32 it buys 30% for 42%. The crossover is where  $N/2$  registers stop being cheap relative to the tree they cut. The configuration to quote is therefore fan-in 8 or 16 with one stage: 32.4 LUT per weight at 126.73 MHz, or 34.9 at 106.38, both at zero DSP. We report the unfolded column beside it because that number was published first, with a caveat naming the negation as its likely cause, and the cause turned out to be the whole of it.

The middle rung is the one a budget would choose: five bits and 95 LUT for +10.9%, a six-LUT premium over  $\varphi$  for a sixfold reduction in error. Applying the scale remains one addition throughout and the datapath remains free of DSP.

Measured as a block scale on two networks, the one-adder ratio matches the geometric ratio beside it at both widths: at five bits 21.5040 against 21.5872 on SmolLM2 and 14.7342 against 14.7814 on Qwen, at six bits 19.3209 against 20.0199 and 13.8528 against 13.8915. The geometric grid’s advantage of Theorem 32 is therefore available with no multiplier at all.

This also resolves the tension left by Theorem 32. The grid minimising relative error is geometric with ratio  $R^{1/N}$ , an arbitrary real needing a multiplier — in general. Where a block scale actually lives, the occupied range being 8.32 to 9.12 binades and the width four to six bits, the nearest multiply-free ratio matches the optimum to within 0.06%: 1.551532 against 1.551145 at four bits, 1.236506 against 1.236662 at five, 1.110187 against 1.110180 at six. At eight bits our first enumeration reported a gap, and that was our own search bound rather

Table 34: Element decoders as coordinate tables, isolated, median of five seeds, against the perplexity of the same ladder on SmolLM2-135M (fp32 baseline 14.3607). Thirty-three more LUTs move the error from 70% to 3.7% at no cost in frequency.

| decoder             | table                   | LUT       | $F_{\max}$ | ppl            | above fp32   |
|---------------------|-------------------------|-----------|------------|----------------|--------------|
| $\varphi$ , 4-bit   | $7 \times 2 \times 5b$  | <b>89</b> | 621.89     | 24.4280        | +70.1%       |
| $r^5=r^3+1$ , 5-bit | $15 \times 5 \times 4b$ | 95        | 482.39     | 15.9242        | +10.9%       |
| $r^6=r+1$ , 6-bit   | $31 \times 6 \times 5b$ | 122       | 612.00     | <b>14.8882</b> | <b>+3.7%</b> |

Table 35: One scale application, 16-bit components, xc7a200t, median of five seeds. A fivefold refinement costs 51% more LUTs and no frequency: the eight-register step’s single adder sits on one of eight parallel paths rather than one of two.

| step                      | LUT        | FF  | $F_{\max}$    | level error   | adds |
|---------------------------|------------|-----|---------------|---------------|------|
| $\varphi$ , $r^2 = r + 1$ | <b>149</b> | 128 | 307.69        | 23.607%       | 1    |
| $r^5 = r^3 + 1$           | 164        | 160 | 306.56        | 10.575%       | 1    |
| $r^8 = r + 1$             | 225        | 212 | <b>815.66</b> | <b>4.625%</b> | 1    |

than a property of the spectrum: stopping at degree 8 the nearest ratio was 1.049852 against a target of 1.026159, while degree 9 reaches 1.036380 and degree 10 reaches 1.027738. Allowing coefficients in  $\{0, \pm 1, \pm 2\}$  — still addition and wiring, since  $\pm 2$  is a shift — degree 9 reaches 1.026129, within 0.003%, out of 734,533 enumerated roots.

What is a real limit is the one-adder family itself. At the same eight-bit target its best member is 1.0850702 under  $\{0, \pm 1\}$  and 1.0510545 under  $\{0, \pm 1, \pm 2\}$ , errors of 5.6% and 2.4%, and these do not improve with degree. Any ratio is reachable multiply-free if adders are spent; a coarser set is reachable with one.

**The hierarchy is cheap only as far as degree three.** Table 36 adds the smallest degree-4 scale, the root of  $r^4 = -r^3 + r^2 + r + 1$ , whose map  $(x_0, x_1, x_2, x_3) \mapsto (x_3, x_0+x_3, x_1+x_3, x_2-x_3)$  was verified exact over 3000 random coordinate vectors.

This places the alphabet exactly.  $\varphi$  is the right rung at four bits, where  $r^* = 1.4115$  falls between it and supergolden and the two tie. It is the wrong rung at five and above — and by five, the affordable part of the hierarchy has been left behind too, since degree 4 costs  $2.1\times$  the area. The regime in which a multiply-free datapath is worth building is the one where degrees 2 and 3 are the only rungs one can pay for, and that regime is three to five bits.

A first predictor failed and the failure is informative: choosing the finest ladder whose span covers the weights’ channel dynamic range —  $268.95\times$  median over 155,520 channels — selects the coarsest rung at every budget, correct at three bits and wrong at four and five. Coverage is the wrong criterion because clipping the smallest weights is nearly free while rounding the largest is not. The quantity that decides is total error, and total error is computable before any model runs.

The distinction is between quantizing to *Fibonacci numbers* and working in *the ring those numbers are the coordinates of*. It is narrower than it may sound, and we state it narrowly: by Binet’s formula  $F_n = \text{round}(\varphi^n/\sqrt{5})$  exactly for  $n \geq 1$  — verified here to  $n = 25$  — so a Fibonacci ladder *is* a  $\varphi$ -power ladder up to a constant offset. What does not follow is closure.  $F_i F_j$  is not a Fibonacci number, so the product leaves the representable set;  $\varphi(a + b\varphi)$  stays in  $\mathbb{Z}[\varphi]$ . The two schemes agree on the ladder and differ on the ring, and the transformation stage is the price of the second.  $F_i F_j$  escapes the set;  $\varphi \cdot (a + b\varphi)$  does not. What the escape costs is exactly the two adders and the routing scheme, and what closure buys is that neither is

Table 36: One scale application, xc7a200t, 32-bit, median of five seeds. Degrees 2 and 3 both cost one addition; at degree 4 the coefficient vector stops being sparse and the cost jumps.

| scale             | LUT        | FF  | $F_{\max}$    | level error  | add/sub |
|-------------------|------------|-----|---------------|--------------|---------|
| $\varphi$ (deg 2) | <b>223</b> | 192 | <b>247.10</b> | 23.61%       | 1       |
| plastic (deg 3)   | 228        | 200 | 231.21        | 13.97%       | 1       |
| 1.1787 (deg 4)    | 469        | 320 | 184.98        | <b>8.20%</b> | 3       |

Table 37: One accumulation step, xc7a200t, median of five seeds. The non-closed unit is the greedy Zeckendorf normaliser, oracle-checked exhaustively over all 65,536 inputs for the sum, the no-two-adjacent-digits property, and agreement with an independently written greedy model.

| one accumulation                                       | LUT        | $F_{\max}$ (MHz) |
|--------------------------------------------------------|------------|------------------|
| closed, $\mathbb{Z}[\varphi]$ , componentwise (16-bit) | <b>182</b> | <b>299.94</b>    |
| non-closed, normalise back (16-bit)                    | 1756       | 11.46            |
| closed, $\mathbb{Z}[\varphi]$ , componentwise (32-bit) | <b>283</b> | <b>239.35</b>    |
| non-closed, normalise back (32-bit)                    | 7017       | 4.25             |

needed: one integer addition, and the multiply-free path is not an approximation of a multiply but equal to it (Theorem 26).

The closure half of this argument is machine-checked. In the accompanying development the map  $(a, b) \mapsto (b, a + b)$  is proved to compute multiplication by  $\varphi$  on the reals, and the exactness of an entire dot product follows from it, in Coq with no admitted lemmas.

**The stage costs an order of magnitude, and it runs per accumulation.** Corollary 20 says a stage is required; the fabric says what it costs. Table 37 measures one accumulation step in the harness of Section 7 — isolated unit, every output bit folded into the observed reduction, median of five placement seeds.

The stage does not run once per multiply. A product can be handled either way: the identity  $F_m F_n = (L_{m+n} - (-1)^n L_{m-n})/5$  turns one into two table lookups and a subtraction. It is *accumulation* that has no such escape, because the sum of two representable numbers is not representable —  $F_3 + F_3 = 4 = F_4 + F_1$ , and the representation changes non-trivially. At fan-in 512 the stage runs 512 times per neuron, against a pair of integer registers on the closed side.

One limit is worth stating: greedy normalisation is not proved to be the cheapest — constant-time Zeckendorf adders exist — so the ratio is an upper bound on this implementation, while Corollary 20 carries the qualitative claim.

The other apparent limit does not survive measurement. Part of the frequency gap is combinational depth, 46 dependent stages against one addition, and pipelining should convert that into latency and registers. It does, and it is not enough: swept to 32 stages at 16 bits and 48 at 32, frequency rises and then plateaus at 249.44 and 111.36 MHz against the closed path’s 299.94 and 239.35, the second less than half. There is a reason.

**Theorem 35** (The normalisation floor). *A normalisation cascade cannot be pipelined below one compare-and-subtract per stage, so its clock period is bounded below by the carry latency of a compare-and-subtract at the datapath width. A compare-and-subtract is strictly more than an addition — a subtraction plus the comparison selecting it — so the floor of a non-closed path lies above the floor of a closed path whose step is a single addition.*



Table 38: The zero-DSP stack. One sign-select primitive builds both halves of a node, so the DSP column stays free and radio and network coexist on one part.

| block                          | DSP       | result                |                |
|--------------------------------|-----------|-----------------------|----------------|
| GF16 correlator (float taps)   | 8 DSP48E1 | —                     | before         |
| ternary correlator (GFTernary) | <b>0</b>  | 78 MHz, real air      | after          |
| every transformer block        | <b>0</b>  | systolic GEMM 33 GOPS | same primitive |
| trained ternary transformer    | <b>0</b>  | 97% end-to-end        | runs           |
| on-silicon backpropagation     | <b>0</b>  | XOR 4/4, 24–25 epochs | trains         |

At the plateau the trade is  $7.8\times$  and  $19.7\times$  the area,  $3.6\times$  and  $8.3\times$  the registers, 32 and 48 cycles of latency, and still the lower frequency.

This is a different kind of statement from the rest of the paper. Everywhere else we compare error magnitudes between formats; here there is no error to compare. Rounding enters a  $\varphi$ -ternary network only at the nonlinearity, where it is unavoidable in any format, and the cost of exactness in the linear path is two integer accumulators in place of one floating-point one.

**Why the alphabet base cannot be anything else.** Closure requires  $r^2 = pr + q$  with  $p, q$  integers, so that  $\mathbb{Z}[r]$  is a ring. Among such  $r > 1$  the cheapest recurrence is the one with  $p = q = 1$ , since any larger  $p$  introduces a shift alongside the addition:  $r = 1 + \sqrt{2}$  satisfies  $r^2 = 2r + 1$  and costs a shift, while  $r = \sqrt{2}$  has  $r^2 = 2$  and loses the scale out of the lattice entirely. The  $p = q = 1$  case is  $\varphi$ , and it is the minimum rather than a member of a set of acceptable choices.

**Scope, stated so the claim does not overreach.** This concerns arithmetic in a lattice. It applies to the linear algebra that dominates a network’s work and its silicon cost, and it says nothing about control flow, addressing or branching, which are not lattice problems and gain nothing from  $\varphi$ .

## 23 What the pair costs in silicon

Measured on an XC7A200T through a fully open flow, with no DSP blocks used anywhere:

### 23.1 The scale applier, and a claim we withdraw

Section 19 closes the multiplier out of the weight and the accumulation. The layer scale is the third site, and our first two attempts at it were each wrong in a different direction. Both are recorded, because the correction is the result.

A ternary network stores weights in  $\{-1, 0, +1\}$  and a real per-layer scale  $\alpha = \mathbb{E}|W|$ . Applying  $\alpha$  is a genuine multiply, so the multiplier the ternary weights removed returns at the layer boundary. Snapping  $\alpha$  to a grid removes it again, and the question is which grid.

**First claim, withdrawn.** We reported that the  $\varphi^k$  grid beats powers of two, with excess reconstruction error 2.44% against 4.86% over 210 layers — a ratio of 0.501 against 0.500 predicted from the density  $\log 2 / \log \varphi = 1.440$ . That measurement reproduces and is correct. It was compared against the wrong baseline. The deployed state of the art for multiplier-free scales is not plain powers of two but **APoT**, whose levels are sums of power-of-two terms. Measured identically, APoT-2 costs 0.1651% and APoT-3 costs 0.0054%, both in one cycle against our  $k$ . On accuracy the  $\varphi$  grid loses by  $15\times$ , and that claim is withdrawn.



**Second claim, bounded.** Theorem 26 states the linear path is exact because  $\mathbb{Z}[\varphi]$  is a ring. It is machine-checked and remains true, but we had presented it as an advantage. An APoT scale is a dyadic rational, and  $\mathbb{Z}[1/2]$  is also a ring closed under the datapath’s operations: a fan-in-512 layer with an APoT-2 scale and Q8 inputs accumulates to a denominator of  $2^{15}$ , exact in ordinary fixed point. Binary arithmetic has had this since fixed point existed. The theorem stands; its significance does not.

**What measuring area then showed.** Both attempts compared accuracy. The appliers differ more in area, and in a way neither of us predicted.

**Theorem 36** (Term growth). *A value represented as a sum of  $n$  terms from a multiplicatively closed generating set, composed through  $d$  scale multiplications without requantisation, has  $n^{d+1}$  terms. A value in  $\mathbb{Z}[\varphi]$  has two components for every  $d$ , since  $\varphi^a\varphi^b = \varphi^{a+b}$  adds exponents rather than multiplying terms.*

**Theorem 37** (Shifter cost). *A runtime-variable shift over  $W$  bits costs  $\Theta(W \log W)$  LUTs; a fixed permutation costs zero. An  $n$ -term APoT applier is therefore  $\Theta(nW \log W)$  and a Fibonacci step is  $\Theta(W)$ .*

This is the effect we had missed, and it is larger than the one we predicted. We had been calling APoT “shift-add” and treating that as free. A shift by a *constant* is free; a shift by a *variable* is a barrel, measured here at 192 LUTs each on 32 bits, and a scale that must serve any layer is variable by construction. The Fibonacci step contains no shifter at all.

**Corollary 22** (Crossover with the multiplier). *Since APoT area grows as  $n^{d+1}$  while the multiplier’s does not grow, there is a depth  $d^*$  at which APoT costs what it was introduced to avoid. Measured at  $W = 32$ ,  $n = 2$ :  $d^* = 2$ , where APoT-8 costs 1217 LUTs against 1215.*

**Theorem 38** (No single winner). *In (area, latency, error) no applier dominates another. The multiplier is exact and largest,  $\varphi^k$  smallest and least accurate, APoT-2 between and beaten by neither; all three are vertices of the frontier.*

**The recommendation is therefore a rule, not a winner.** Use  $\varphi^k$  where the scale path is area-bound, or composed without requantisation — a low-rank factorisation, a folded convolution and batch norm, a residual branch scalar, or accumulation along a mesh route. Use APoT where it is latency- or accuracy-bound. We state this rather than defend one answer because our two previous attempts each defended one, and each was wrong in a different direction.

**The area advantage is withdrawn too, and by our own attack.** Table 40 was measured in one regime and with one field width, and both choices flattered us.

**Theorem 39** (An unrolled recurrence is linear). *A recurrence applier unrolled to depth  $K$  over  $W$ -bit operands costs  $\Theta(KW)$ ; an additive-term applier whose shifts are compile-time constants costs  $\Theta(W)$  independent of the scale, since a constant shift is wiring. Hence when the scale is frozen after training the recurrence loses for every  $K \geq 1$ .*

Measured: a constant-shift APoT-2 applier costs 26 LUTs, against 64, 128 and 256 for  $\varphi^k$  unrolled at  $K = 2, 4, 8$ . At the realistic  $K = 8$  that is  $10\times$  against us, and the lines do not cross even at  $K = 1$ .

**Theorem 40** (Regime inversion). *The area ordering between the two families is a property of neither. It inverts with whether the scale is a compile-time constant or a runtime variable, because a constant shift is wiring and a variable shift is a barrel of  $\Theta(W \log W)$ .*

**Theorem 41** (A barrel is priced by range, not by width). *A runtime shift field need only span the workload’s scale range. Across the 210 layers measured here that range is 3.15 octaves, so two bits suffice. An area comparison that sizes the field larger is measuring a competitor that would never be built.*

At a two-bit field APoT-2 costs 130 LUTs against our 199. The  $2.22\times$  advantage reported above was an artefact of a five-bit field, and is withdrawn.

**An instrument limitation, stated rather than smoothed.** The APoT sweep is non-monotonic — 130, 380, 230, 384 LUTs at shift widths 2 to 5 — with the parameter demonstrably applied and the tool deterministic. This is technology-mapping heuristics, and it means LUT counts from logic synthesis alone are not a reliable area metric at this granularity. A claim resting on a 30% difference between two such points is unsafe; one resting on the  $10\times$  of Theorem 39 is.

**What survives.** Not the area advantage. Only Theorem 36 remains:  $\varphi^k$  is the sole applier whose area is independent of composition depth, so at  $d = 2$  APoT-8 costs 1217 LUTs where the pair still costs 173. That regime — a chain with no requantisation between links — is real but uncommon. The alphabet’s uniqueness and the  $\mathbb{Z}[\varphi]$  closure are machine-checked and untouched, having never been area claims.

The pattern across three attempts at this section is one pattern. Each favourable number came from a comparison whose terms we chose: powers of two rather than APoT, a five-bit field rather than the two the workload needs, a runtime regime rather than the frozen one. The rule that would have caught all three is to write down what the competitor would build if it were trying to win, and measure that.

### 23.2 The silicon table, measured on an audited harness

Every silicon figure in this paper was originally measured through a harness that combined the decoder with a shared accumulator and observed part of its output. Three separate defects in that arrangement are described in Section 35; each changed a headline number by 10 to 50 percent without any individual measurement being wrong. The table below is measured on the only harness that passes the observation gate we wrote afterwards: decoder alone, every output bit folded into the observed reduction, median of five placement seeds.

**Theorem 42** (A taper is paid in delay, not in area). *A tapered format’s decode contains a scan, a serial dependency whose length grows with the word. A serial dependency costs critical path directly and area only incidentally, whereas a radix-heavy fixed format costs a wide shift, which is area-heavy and shallow. The two families therefore separate on frequency and not on area.*

Measured: all fourteen fixed fields sit above all three tapers on frequency, the worst fixed leading the best taper by  $1.43\times$ ; on area they overlap, posit8 at 214 LUT against IBM hexadecimal’s 243. The overlap is predicted by the theorem rather than excused.

GFternary against posit32 is  $7.8\times$  on area and  $19.9\times$  on frequency. GFternary costs 66 LUT where a bare wire costs 112, because decoding a two-bit alphabet lets the synthesiser simplify the downstream register further than passing 32 raw bits does.

The area–frequency correlation on the isolated decoder is  $-0.60$ , against  $-0.90$  on the combined harness. The axes carry independent information here, which is why they are reported separately and no ratio of them appears.

### 23.3 The operating point

The previous subsections withdraw an accuracy claim and an area claim. What remains is a statement about operation profiles, and it is the first in this section that names its own boundary rather than being pushed to it.

A number system is used through two operations, and the three candidates each buy one cheaply and pay for the other.

$\mathbb{Z}[\varphi]$  is the only one cheap under both. Its price is that free multiplication is restricted to powers of  $\varphi$  rather than general, and the following says why that price is not paid in this datapath.

**Theorem 43** (The required multiplication set). *In a datapath where every weight is a code applied by sign-select, the layer scale is a power of the alphabet base, and accumulation is the only remaining operation, the multiplications the datapath must perform are exactly  $\{\text{base}^k\}$ . General multiplication is never requested.*

**Corollary 23** (Sufficiency). *A ring closed under addition and under multiplication by powers of its generator is sufficient for such a datapath.  $\mathbb{Z}[\varphi]$  is such a ring with both operations at  $\Theta(W)$ .*

**Corollary 24** (The other two are mis-provisioned). *LNS is over-provisioned: it buys general multiplication, which this datapath never asks for, and pays in addition, which it performs on every term of every inner product. Fixed point is under-provisioned in the opposite direction: addition is free, scale multiplication is not, unless the scale is frozen.*

The restriction of  $\mathbb{Z}[\varphi]$  coincides with the restriction the ternary datapath already carries. That coincidence is the result; it is not a claim that the format is better, only that it is matched.

**Theorem 44** (Compile-time composition is free). *If all  $d$  scales of a chain are known before execution, their product is known, and any representation applies it at the cost of a single scale. The term growth of Theorem 36 occurs if and only if  $d$  or the scales become known only at runtime.*

This removes three of the four cases where we had claimed a depth advantage. A low-rank factorisation, a folded convolution and batch norm, and a residual branch scalar are all fixed after training, so their product is precomputed and no composition occurs in hardware. One case survives: accumulation along a mesh route without renormalisation, where the hop count is a runtime quantity. The depth advantage is real and it lives there rather than in the network.

**Stated boundaries.** This is not an area win over APoT, which was measured and withdrawn. It is not an accuracy win, where APoT-2 is  $15\times$  ahead. And where a scale is frozen there is nothing to win, because a frozen scale is wiring. It is a claim about which system matches an operation profile, holding where the scale is not frozen: a shared engine serving many layers, or a composition whose depth is not known until execution.

### 23.4 The formats a training run cannot live in

Throughput per area is only meaningful among formats that can do the job, so the candidates split in two, by one test that is checkable rather than rhetorical: *does the word carry the range, or must the range be brought from outside?*

**Why GFTernary is not in Table 43 despite  $e = 0$ .** The distinction is not the presence of an exponent field but whether the scale is *external and learned* or *intrinsic and exact*. A  $\{-1, 0, +1\}$  alphabet needs a real  $\alpha_\ell$  per layer, learned alongside the weights and multiplied

by at inference — the multiplier returns. The  $\varphi$  alphabet needs none: by Corollary 19 the inter-layer gain is  $F_k\varphi + F_{k-1}$ , a pair of integers known in advance, applied by shift and add. Its scale is inside the system by construction rather than fitted from outside it. That is the difference between carrying range and borrowing it, and it is why the stack trains on the part.

**Why int8 is reported apart.** An int8 datapath scores 0.180 MHz/LUT at a median of five placement seeds, against GFTernary’s 0.181 — a difference of 0.4% inside a 16% seed spread, so the two are indistinguishable on this flow. An earlier version of this paper reported 0.189 against 0.177 and called it a 7% lead; that came from a single placement run and is withdrawn. int8 is excluded from that table even so, and the reason is structural rather than favourable to us. int8 has  $e = 0$ : no exponent field, hence no dynamic range inside the word. It functions only with an external per-tensor or per-channel scale; without one it is the integers  $-128 \dots 127$ . By Corollary 27 that is the non-uniformity escape — the scale has been moved outside the word — so int8 is a block format wearing a scalar’s name, and scoring it against formats that carry their own range measures the wrong thing. It leads the table precisely because it declines the task the table sets.

The consequence matters more for training than for inference. Gradients span many orders of magnitude, and a format with zero dynamic range cannot hold them: every published sub-8-bit training result — MXFP4, NVFP4 — carries block scales *and* a higher-precision master weight. Our own stack trains on the part, at zero DSP, with backpropagation in the format itself.

### 23.5 Withdrawal: the frontier at equal storage

Corollary 17 and Corollary 26 count positions. A format is stored in bits, and that difference is the whole of the slack.

**Theorem 45** (Packed coverage). *A family member realisable in  $N$  bits obeys  $3^{E_t}2^M \leq 2^{N-1}$ . Substituting the minimal  $E_t = \log_3(b+1)$  gives*

$$m + \log_2 3 \cdot \log_3(b+1) = m + \log_2(b+1) \leq N - 1,$$

*identically the uniform binary budget of Theorem 47, since  $\log_2 3 \cdot \log_3 x = \log_2 x$ .*

**Corollary 25** (The slack is a position artefact). *A ternary exponent encoding provides no budget slack at equal storage. The 0.3691E of Corollary 26 exists only at equal position count, that is on a fabric where a position is physically ternary.*

Measured at equal storage against the catalogue, 6 of 17 sampled formats are dominated rather than all of them. Every uniform binary format spending all its bits ties exactly: binary16 at 15.06 against a budget of 15, binary32 at 31.10 against 31, binary64 at 63.27 against 63. Those dominated are the ones wasting budget — posit32 by 5.12 positions, posit64 by 14.24, takum32 by 0.82, tekum32 by 0.90, cray\_float by 0.65, and our own gf8 by 0.70. Several tapered formats escape outright: posit8 by 1.16, posit16 by 2.37, takum16 by 1.53, tekum16 by 1.49. That is consistent with Corollary 27 — escape requires non-uniformity, a taper is non-uniform, and a uniform family cannot catch it at equal storage.

**Why this went unseen.** The same defect appears twice elsewhere in this paper, as Theorem 8 and as the observation that TNF4 with  $E_t = 2$  does not exist in four bits because  $3^2 = 9 > 8$ . Both were recorded as local facts about packing, and neither was carried back to the result that rests on the same quantity. The rule we adopt is that a correction invalidating a comparison must be applied to every claim resting on that quantity, not only to the one where it surfaced.

## 23.6 The trit is the slack

Corollary 17 is a fact about a list. It becomes a fact about the design space once one asks what a competitor would have to do to escape.

**Theorem 46** (Coverage condition).  *$\mathcal{F}_N$  contains a member dominating a format of effective mantissa  $m$  and range  $b$  binades if and only if*

$$m + \lceil \log_3(b+1) \rceil \leq N - 1.$$

*Proof.* A member  $E_t$  has mantissa  $N-1-E_t$  and range  $3^{E_t}-1$ . Domination requires  $N-1-E_t \geq m$  and  $3^{E_t}-1 \geq b$ , i.e.  $\lceil \log_3(b+1) \rceil \leq E_t \leq N-1-m$ . Such an integer exists exactly when the stated inequality holds.  $\square$

**Theorem 47** (Position budget of a uniform format). *Let a uniform format of width  $N$  carry a sign, a scale field of  $E$  binary positions and a significand of  $m$  bits held at every representable magnitude. Then  $b+1 \leq 2^E$  and  $1+E+m \leq N$ , hence*

$$m + \log_2(b+1) \leq N - 1.$$

The two budgets are the same accounting in different currencies, and the exchange rate is the whole result.

**Corollary 26** (The ternary exponent is the slack). *Every uniform binary format of width  $N$  is dominated by a member of  $\mathcal{F}_N$ , with slack*

$$\Delta = E(1 - \log_3 2) = 0.3691 E \text{ positions,}$$

*strictly positive for  $E \geq 1$  and growing linearly in the competitor's own exponent width.*

*Proof.* By Theorem 47,  $b+1 \leq 2^E$  and  $m \leq N-1-E$ . Then  $m + \log_3(b+1) \leq (N-1-E) + E \log_3 2 = (N-1) - E(1 - \log_3 2)$ , which satisfies Theorem 46 with the stated margin.  $\square$

Measured, in positions: 1.48 at `fp8_e4m3`, 1.85 at `binary16`, 2.95 at `binary32`, 4.06 at `binary64`, 5.54 at `binary128`. The margin is not a coincidence of the catalogue; it is 0.3691 per exponent bit the competitor spends.

This is worth stating plainly because it cuts against Theorem 8, which says a packed ternary exponent never carries more values *per bit* than a binary one, and which we confirmed in silicon: BNF16 and TNF16 differ by 1% in LUTs. Both are true, on different axes. Per *position* a trit spans  $\log_2 3$  binades against a bit's one, which is where  $\Delta$  comes from; per *bit of binary fabric* the packing gives it all back. The ternary encoding earns on the number axis and breaks even on the silicon axis, and a claim that does not name its axis is not a claim.

**Corollary 27** (Escape requires non-uniformity). *A width- $N$  format escaping  $\mathcal{F}_N$  must violate the hypothesis of Theorem 47: either its significand is not held at every magnitude (a taper), or its scale field is not inside the word (a block).*

**The bound is tight, and it is not vacuous.** Equality in Theorem 46 is attained by construction, since the width rule spends every position. Among the catalogued formats `posit16` comes within 1%:  $10.55 + \log_3 113 = 14.85$  against a budget of 15. And the escape route of Corollary 27 is real rather than hypothetical — the same `posit16`, scored where a taper is designed to live rather than averaged over its range, carries 12 fraction bits while keeping all 112 binades:

$$12 + \log_3 113 = 16.30 > 15.$$

It escapes, as it should: concentrating precision where the workload sits is what tapering is *for*. A bound no format can violate would be measuring nothing.

**What this predicts.** Corollary 27 names the only two routes out, and both are open questions rather than results of this paper. The taper route is bounded by Theorem 12 and costs what Section 7 measures. The block route — MX and its descendants, where one scale serves 32 elements — lies outside every budget stated here, because the scale is not in the word. We report it as the frontier of this work and not as territory held.

**Theorem 48** (Transforms and formats do not compose). *Let  $D^*(b, p)$  be the best achievable distortion at  $b$  bits per element under a fixed pre-transform  $p$ . A transform changes  $D^*$  by changing the source it presents; a format chooses a partition attaining some  $D \geq D^*$ . Two levers that both act on the partition therefore share one ceiling and their gains are sub-additive, while a transform and a format act on different arguments and their composition is order-dependent.*

Measured, the order-dependence reaches  $2.78\times$ : applying the same two levers in the two orders gives different results, which forbids reporting either as a standalone multiplier. We flag this because we made the mistake ourselves — an early version of this work computed a lever’s ratio against a losing baseline and reported the levers as nearly multiplicative. They are not.

### 23.7 Where the block literature is not looking

The block axis is the escape route Corollary 27 leaves open, and it is where the field’s attention currently sits. It is worth being precise about what that work varies and what it holds fixed.

The Open Compute microscaling specification fixes a block of 32 elements sharing one UE8M0 scale — a bare power of two, no mantissa — with the element at E2M1. NVIDIA’s NVFP4 changes two things: the block shrinks to 16, and the shared scale becomes an E4M3 float rather than a power of two. Both are hardware-resident on current accelerators, which is why the stop condition for this work is stated against them and not against posit or takum.

What the 2026 literature then varies is almost entirely the *transform* applied before quantisation, not the format underneath it: learnable block-wise optimisation for outlier resilience, and rounding fitted to the E2M1 level set, augmented residual channels, and activation-sparsity coupling. Each reports gains, and each holds the element format at E2M1.

That is exactly the variable Theorem 23 addresses. E2M1 spends two of its three non-sign positions on an exponent, *inside a block that already carries a shared scale*. The width rule says the exponent is sized for the range the workload visits, and inside a block normalised by its own maximum that range is measurable rather than assumed. If it is narrower than four binades, E2M1 is buying range the block already paid for, and the positions belong in the mantissa. We measure the within-block span directly and let the rule name a winner before evaluating perplexity; Section 21 reports what came back, including the case where the rule was wrong.

This is a narrower claim than it may look. It says nothing about the transforms, which compose with any element format and whose gains we do not dispute. Our own Theorem 48 says transforms and formats act on different objects — a transform moves the ceiling, a format divides it — so improving the element is orthogonal to improving the rotation, and neither subsumes the other.

## 24 What the throughput table is actually measuring

The row this work had left marked *not swept* turned out to be the most informative one in the table. `binary32`’s decoder is

```
assign fp32_out = binary32_in;
```

a wire, because `binary32` is `fp32`. Its conformance is therefore exact by construction, and its 472 LUT are the harness alone: the LFSR, the accumulator, and the output fold, with no decoder in them at all.



That number is a calibration this work had never taken. Every row of the table lies between 463 and 1,008 LUT, so *the harness is between 47% and 102% of every measurement*, and for the top ten rows it is never below 80%. Two consequences follow, and both are structural rather than fixable by more seeds.

**Proposition 49** (Harness dominance). *Let a measurement compose a fixed harness of cost  $H$  with a decoder of cost  $D_f$ , and report  $R_f = M_f/(H + D_f)$ . Then*

$$\frac{\partial R_f}{\partial D_f} = -\frac{M_f}{(H + D_f)^2},$$

*which tends to  $-M_f/H^2$  as  $D_f/H \rightarrow 0$ . When  $H \gg D_f$  the report varies almost entirely with  $M_f$  and the decoder’s cost is compressed out of it. Here  $H = 472$  and  $D_f \in [-9, 536]$ , with  $D_f < 115$  for the top ten rows.*

**Proposition 50** (Output-space pruning). *A decoder with  $n$  input bits has at most  $2^n$  distinct outputs. Any logic observing only those outputs may be specialised to that image, so its synthesised cost is non-increasing in  $n$  independently of the decoder’s own complexity. A comparison across differing  $n$  therefore confounds format design with format width, and only a within- $n$  comparison isolates design.*

Proposition 50 is not a worry, it is visible in the numbers. **GFternary** takes 2 input bits and measures 463 LUT — *nine fewer than the bare wire* — because four distinct outputs let synthesis specialise the accumulator behind it. Measured alone against a minimal register-and-fold sink, the bare wire costs 112 LUT and **GFternary** costs 66. A format cannot be cheaper than a wire; what is being measured is the width of its image.

Read that way the table says something different from its ranking, and the difference is worth stating plainly.

**Where we meet a competitor at the same width, and where we do not.** Of the nine width classes, six contain only our formats and two contain only other people’s. *One class contains both:* at 8 bits **GF+8** measures 0.1004 against **fp8 e4m3**’s 0.1254, and we lose by 19.9%. That row’s conformance column reads *selector live; no reference*, so it is also the one row of ours that has never been checked against an oracle — the loss and the missing check are the same debt.

The nearest honest comparison is 16 against 17 bits, one bit apart:

| <b>BNF16</b> (17 bits) versus |         |
|-------------------------------|---------|
| <b>binary16</b> (16 bits)     | −3.1%   |
| <b>takum16</b> (16 bits)      | +57.4%  |
| <b>LNS16</b> (16 bits)        | +79.8%  |
| <b>posit16</b> (16 bits)      | +113.4% |

Against every *research* format at comparable width — takum, LNS, posit, the formats this paper is actually arguing with — our 17-bit design wins by between a half and slightly over double. Against IEEE half it loses by three percent, and against IEEE single it cannot win at all, because IEEE single’s decoder is a wire and no decoder beats no decoder.

That is the defensible claim, and it is narrower than the ranking suggests. *The overall first place belongs to a two-bit format with no same-width competitor in the catalogue*, and second place belongs to an identity function. Neither fact was visible while the table was sorted on a single number.



## 24.1 Conformance does not certify the specification

GF+8 was the last row of ours without an oracle, and it is the one class where a format of ours meets a competitor at the same width and loses. Writing the oracle settled both questions and raised a third that this work had not previously had to ask.

GF+A is an adaptive container: a two-bit header selects one of four pockets over a shared eight-bit payload —  $\phi$ -e3m4 (1, 3, 4) at bias 3, e2m5 (1, 2, 5) at bias 1, a symmetric fixed point at scale 1/64, and a slot declared logarithmic. Implemented from those four parameter sets and compared against the module: 1,024 of 1,024 codes exact, zero disagreements. The decoder is correct.

The fourth pocket is not.

**Proposition 51** (Conformance does not certify the specification). *Let  $D$  implement  $f$  exactly on all of  $C$ . Conformance establishes  $D = \rho \circ f$  and says nothing about  $f$ . In particular  $|f(C)|$  may be arbitrarily smaller than  $|C|$ , and a decoder faithfully realising a degenerate  $f$  is both conformant and useless.*

The logarithmic pocket emits  $2^{-s}$  with the mantissa identically zero, where  $s = \lfloor \iota/16 \rfloor + \lfloor \iota/32 \rfloor$  and  $\iota = 127 - |w|$ . It is a power-of-two ladder, and it produces **seventeen distinct values from 256 codes** where each of the other three pockets produces 255. Ninety-three percent of that pocket’s code space maps onto nothing new. Its own comment declares a scale of  $\iota \cdot 8/127$  while the code implements  $\iota \cdot 3/32$ , half again as steep — the second time in this catalogue that a line of RTL contradicted the comment beside it.

The degeneracy is not free of consequence for the table, and its direction is the one this work has now seen eleven times. Rebuilding that pocket as the `lms8` this repository already declares — a signed seven-bit log field at `frac_bits` = 3, value  $2^{L/8}$ , giving 255 distinct values — costs what completing a decoder always costs:

| pocket 11                         | LUT   | MHz    | MHz/LUT |
|-----------------------------------|-------|--------|---------|
| as published (17 values)          | 170   | 313.38 | 1.8434  |
| as <code>lms8</code> (255 values) | 182   | 271.22 | 1.4902  |
|                                   | +7.1% | −13.5% | −19.2%  |

So GF+8’s 0.1004, which already loses to `fp8 e4m3`’s 0.1254 by 19.9% in the only head-to-head class in the catalogue, is itself inflated by a pocket that discards most of its own code space. Repaired, it loses by more. *We report this because it is true and because the alternative — a number resting on a degenerate component — is the kind of claim this paper spends twenty-seven sections dismantling in other people’s work.*

## 25 What was already known, and what is left

The construction this paper measures — a binary-radix float whose exponent field carries a ternary-cardinality alphabet packed densely into  $\lceil E_t \log_2 3 \rceil$  bits — sits close enough to several existing lines of work that the boundaries are worth drawing explicitly rather than left to a reader to find.

**A ternary exponent field is not new.** TERNAC [31, 32] was a 48-trit floating-point format with a 42-trit mantissa and a *six-trit exponent*, emulated on binary hardware at SUNY Buffalo in 1972–73. It differs from the format here in two ways that matter — its value semantics are radix-3, and its trits are represented as a pair of binary numbers rather than packed — but *the idea of an exponent field counted in trits predates this work by half a century*, and no claim of priority over it is made or implied anywhere above.

**The  $k = 5$  packing is public.** Compilade [34] published the five-trits-in-eight-bits packing in June 2024 and reported it as 99.06% efficient, which is exactly  $1 - w(5)$  in the notation of Theorem 53. The accompanying discussion also records that 17 trits in 27 bits and 111 in 176 beat 1.6 bits per trit — that is, *the unevenness of  $w(k)$  has been observed publicly*, if not written as a function. What Section 29 adds is the closed form, the proof that the waste is positive at every  $k$ , and the consequence for choosing a rung; the  $k = 5$  datum itself is Compilade’s.

**The packed decoder has been built.** TerEffic [35] implements exactly the packed- $3^5$ -in-eight-bits decoder in FPGA LUTs, for ternary *weights* rather than for an exponent field. It is one abstraction level from the decoders measured in Section 26 and any reader of that section will know it.

**The binary encoding of balanced ternary is textbook.** Parhami and McKeown [33] treat binary-encoded balanced ternary directly, at two bits per trit — which is the baseline dense packing improves on, and the source of the encoding penalty this paper quantifies. Knuth’s treatment [36] and the Setun machines [37] establish balanced ternary as a radix; *this work does not change the radix* and therefore does not inherit the radix-economy argument that literature carries.

**What is left.** Setting those aside, what remains is narrow and stated narrowly: the combination of ternary *cardinality* with binary *radix* and binary *storage*, with the capacity surplus  $2^{\lceil k \log_2 3 \rceil} - 3^k$  treated as a design variable rather than as overhead — reserved, guarded, and priced. Around that: the closed form of  $w(k)$  and its consequences for rung choice (Sections 29, 27), the reservation cost measured rather than argued, and placed-and-routed silicon with exhaustive conformance for each rung claimed. Every one of those is a contribution about *measurement and boundary*, not about being first.

## 26 At fifty-four seeds the claims separate

Section 28 left the ranking resolving into seven levels on five seeds, with a separation threshold of 5.50% and a detection floor for a decoder-cost difference of 41.3 LUT — 79% of a whole decoder. Both are consequences of the sample size and of reporting a median, whose efficiency at  $n = 5$  is 0.698. The nine formats the paper’s claims rest on were therefore re-swept at **fifty-four seeds**, and the reported statistic is the **mean**.

The separation threshold falls from 5.50% to **1.43%**.

The margins are smaller than the five-seed estimates suggested — TNF16c read +12.5% there and reads +6.92% here — and that is the expected direction: a point estimate taken near the resolution floor is as likely to flatter as to understate, and the larger sample removes the flattery along with the noise.

**The claim this paper makes at equal storage is +6.92% over IEEE half, with a 95% interval of  $\pm 0.0010$  on 0.1152 and  $p_{\text{Holm}} = 6.9 \times 10^{-21}$ .** TNF16c is sixteen stored bits against binary16’s sixteen, its out-of-specification offsets are reserved so every code is defined or flagged, and its exponent spans  $\pm 121$  — inside the window Proposition 52 requires.

TNF17e at +7.17% and TNF16c at +6.92% are not separated from *each other*: their intervals overlap. The paper claims the pair, not an order within it.

## 27 Why the ladder stops at five trits when the target is fp32

Every decoder in this catalogue forms the fp32 exponent the same way:

```
wire [7:0] e32 = e[7:0] + 8'd127;
```

which is correct only while  $e + 127$  lies in  $[0, 255]$ . Outside that window the addition wraps rather than saturating, and a wrapped exponent is a wrong finite number, not an overflow — so nothing downstream can detect it. TNF32’s offsets 364, 620 and 876, whose exponents are 0, 256 and 512, all decode to exactly 1.0.

**Proposition 52** (The exponent window bounds the rung). *A rung with  $E_t$  trits has  $E = (3^{E_t} - 1)/2$  and its exponent spans  $[1 - E, 3^{E_t} - 2 - E]$ . That span lies inside  $[-127, +128]$  exactly when  $3^{E_t}/2 \leq 128$ , that is*

$$E_t \leq 5.$$

$3^5/2 = 121$  fits and  $3^6/2 = 364$  does not, so on a decoder emitting **fp32** the ladder admits at most five trits of exponent. A wider rung must saturate outside the window or emit more than 32 bits, and either changes its cost.

The boundary is sharp and it falls exactly where Corollary 29 put the least-waste rung.  $E_t = 5$  is simultaneously the  $k$  of minimum  $w(k)$  below  $k = 10$  and the widest exponent an **fp32** decoder can carry without wrapping — a coincidence of two unrelated constraints, one arithmetic and one about the target type, and the reason TNF16c and TNF17e are the two rungs this paper recommends.

Three rungs were withdrawn for violating it. `tools/check_exponent_window.py` now fails on any rung whose span leaves the window, lists the withdrawn ones explicitly rather than deleting them — a gate that silently ignores a known-bad entry is how the defect returns — and is verified red by removing that list.

## 28 The instrument resolves seven levels, not twenty-eight

Until this point the throughput table stored a median and a spread per row. A median with a spread cannot answer whether two rows differ, so *no ordering in this paper had ever been tested*. Every row was re-swept keeping all five per-seed frequencies, and two things came out of it.

**The table had been measured on two harnesses.** Seventeen of twenty-one rows reproduced their LUT count exactly; four fell. The cause is exact: two rows were driven from a 128-bit LFSR and the rest from a 64-bit one. Re-measuring **GFternary**, which reproduces at 463 LUT on the narrow harness, gives 527 on the wide one — **a difference of exactly 64 LUT, and exactly 64 for all twenty-one rows**. Two rows had therefore been ranked against nineteen others while carrying 64 LUT of harness the others did not. Everything in this section is re-measured on the wide harness, so the comparison is between decoders and not between wrappers.

**The ranking resolves into seven levels.** Testing each adjacent pair by whether its five-seed ranges overlap gives seven bands, and one of them holds twenty formats. *A band is a chain of overlaps, not an equivalence class*; two members at opposite ends of a chain may still be separated, so the useful test is pairwise and direct.

| band | members                                             |
|------|-----------------------------------------------------|
| 1    | <b>GFternary</b>                                    |
| 2    | <b>binary32</b>                                     |
| 3    | twenty formats, from <b>fp8 e4m3</b> to <b>GF+8</b> |
| 4    | <b>posit8, takum16</b>                              |
| 5    | <b>LNS16</b>                                        |
| 6    | <b>posit16</b>                                      |
| 7    | <b>posit32, IBM hex32</b>                           |

**GFTernary is alone in the first band:** 0.1524–0.1616 against `binary32`’s 0.1376–0.1474, with no overlap. That result stands. Within band three the fine ordering does not: no reader should conclude from a rank that `TNF16` beats `BNF16`, because the instrument cannot see the difference.

**What is separated from IEEE half at equal or near-equal storage.** Six of our formats have five-seed ranges entirely above `binary16`’s 0.0985–0.1071:

|                  | vs <code>binary16</code> | five-seed range |
|------------------|--------------------------|-----------------|
| <b>GFTernary</b> | +51.4%                   | 0.1524–0.1616   |
| <b>TNF16a</b>    | +18.6%                   | 0.1192–0.1278   |
| <b>TNF17e</b>    | +16.1%                   | 0.1135–0.1238   |
| <b>GF10</b>      | +14.4%                   | 0.1134–0.1204   |
| <b>TNF32</b>     | +13.2%                   | 0.1137–0.1243   |
| <b>TNF16c</b>    | +11.3%                   | 0.1118–0.1210   |

`TNF16c` is the sixteen-bit rung at  $E_t = 5$  *with its out-of-specification offsets reserved*: equal storage to `binary16`, every code either defined or flagged, and separated above it. Section 30 reported that comparison as +5.1% from the narrow harness; on the common harness it is +11.3%, and it is the first claim in this paper to be stated with a range rather than a point.

*A rank order the instrument cannot resolve is not a result.* The table above is printed with ranges for that reason, and every ordering claim elsewhere in this paper that falls inside band three should be read as unresolved.

## 29 A ternary exponent never tiles a binary field

Section 30 ended on a dilemma: rung A’s offset field holds 128 values and its four trits name 81, so either 37% of the word lies outside the format and must be guarded, or the exponent is not ternary. The dilemma has a structural cause, and naming it resolves the case rather than choosing a side.

**Theorem 53** (No exact tiling). *For integers  $k, m \geq 1$ ,  $3^k \neq 2^m$ . Hence for every rung the offset field of width  $\lceil k \log_2 3 \rceil$  has capacity strictly greater than the  $3^k$  offsets it must hold, and the wasted fraction*

$$w(k) = 1 - \frac{3^k}{2^{\lceil k \log_2 3 \rceil}} = 1 - 2^{-(1 - \{k \log_2 3\})}$$

*is strictly positive at every  $k$ , where  $\{\cdot\}$  is the fractional part.*

*Proof.*  $3^k$  has 3 in its factorisation and  $2^m$  does not, so unique factorisation forbids equality for  $k, m \geq 1$ . The capacity is  $2^{\lceil x \rceil}$  for  $x = k \log_2 3$  and the content is  $2^x$ ; their ratio is  $2^{-(\lceil x \rceil - x)}$ , and  $\lceil x \rceil - x = 1 - \{x\}$  whenever  $x \notin \mathbb{Z}$ , which Theorem’s first sentence guarantees.  $\square$

**Corollary 28** (The waste is the undefined code space). *Because the sign and mantissa fields are full — every one of their  $2^{1+M}$  patterns denotes a distinct value — the fraction of the stored word that the specification leaves undefined equals the fraction of the offset field it leaves undefined:*

$$\frac{2^{\text{width}} - 2 \cdot 3^{E_t} \cdot 2^M}{2^{\text{width}}} = 1 - \frac{3^{E_t}}{2^{\lceil E_t \log_2 3 \rceil}} = w(E_t).$$

*So  $w$  is not an abstract packing loss: it is the share of the format’s own codes that name nothing. At  $E_t = 4$  that is 24,064 of 65,536 codes; at  $E_t = 5$ , 3,328.*

This is why Proposition 54 bites exactly where  $w$  is large: the codes a rung must guard are the codes  $w$  counts, and there is no third place for them to be.

**Corollary 29** (The waste is uneven, and the good rungs are sparse).  $w(k)$  is small exactly where  $\{k \log_2 3\}$  is close to 1, that is where  $3^k$  falls just under a power of two. Over  $k \leq 15$  the minimum is  $k = 5$  at  $w = 5.1\%$ ; the maximum is  $k = 7$  at  $w = 46.6\%$ .

**The rung the ladder had skipped.**  $E_t = 5$  needs 8 offset bits, so  $1 + 8 + 7 = 16$ : a third sixteen-bit candidate, and the one this work had not considered. Built, and checked exhaustively against the ladder reference: **62,208 of 62,208 in-specification codes exact, all 3,328 out-of-specification codes flagged, zero errors.**

|                                 | LUT        | MHz          | MHz/LUT       | vs binary16  |
|---------------------------------|------------|--------------|---------------|--------------|
| rung A, unguarded               | 502        | 69.87        | 0.1392        | +14.7%       |
| rung A, offsets reserved        | 549        | 64.86        | 0.1181        | −2.6%        |
| rung C, unguarded               | 509        | 66.11        | 0.1299        | +7.1%        |
| <b>rung C, offsets reserved</b> | <b>527</b> | <b>67.20</b> | <b>0.1275</b> | <b>+5.1%</b> |

The guarded build of rung A measures 15.1% below its unguarded build and the guarded rung C only 1.8% below — but *those percentages are not the comparator’s price*. Synthesised in isolation the guard is 1 LUT at  $E_t = 5$ , 2 at  $E_t = 4$  and 3 at  $E_t = 8$ , against modules of some five hundred, so at most half a percent of the area is the comparison itself. What changes is the decoder’s output image: reserving 47 offsets collapses 24,064 codes onto one NAN, and everything downstream is specialised to the image it observes (Proposition 50). **The difference between a guarded and an unguarded build is a fact about the whole datapath, not a cost that can be attributed to the guard**, and this paper stated it as the latter until an adversarial review of the derivation refuted the attribution. At seventeen bits the same difference proved to lie inside the seed spread entirely. **The claim this paper can make at equal storage, with every code either defined or flagged, is +5.1% against IEEE half** — smaller than the 14.7% first reported, and unlike it, not resting on an unchecked field.

**The ladder rebuilt.** Corollary 29 is a construction rule, not only a diagnosis, once the range constraint is added: choose the  $k$  of least  $w(k)$  among those whose span  $3^k/2$  covers the class. At 17 bits that is  $E_t = 5$ ,  $M = 8$  — 36.7% waste falls to 5.1% for exactly one mantissa bit. Generated from the parameters rather than hand-edited, guarded, and checked exhaustively: **124,416 of 124,416 in-specification codes exact, all 6,656 out-of-specification codes flagged, zero errors.**

| 17-bit rung     | $E_t$    | waste       | MHz/LUT       | guarded    |
|-----------------|----------|-------------|---------------|------------|
| TNF16, as built | 4        | 36.7%       | 0.1005        | no         |
| <b>TNF17e</b>   | <b>5</b> | <b>5.1%</b> | <b>0.1207</b> | <b>yes</b> |

20.1% faster per LUT *while* checking its offset field, which the rung it replaces does not — both figures re-measured on the common harness of Section 28. The same rule sends the 65-bit rung to  $E_t = 8$ , but that rung is withdrawn: an independent review found its exponent path keeps eight bits of an exponent spanning  $\pm 3279$ , so **off** and **off + 256** decode identically. TNF32 and TNF64 carry the same defect and are withdrawn with it; TNF16c and TNF17e do not, their exponents spanning  $\pm 121$  inside **fp32**’s window.

The dilemma of Section 30 does not need to be settled because it was an artefact of standing on  $E_t = 4$ . Theorem 53 says no rung escapes waste; Corollary 29 says the ladder should be built on the  $k$  where the waste is least, and this work had built it on three of the worst and skipped the best.

### 30 A rung at exactly sixteen bits, and what winning there costs

Section 24 left this work with one honest comparison and an apology attached to it: our nearest rung to the sixteen-bit competitors was seventeen bits wide, so every same-width claim carried a one-bit caveat. A rung’s stored width is  $1 + \lceil E_t \log_2 3 \rceil + M$ , and exactly two assignments reach sixteen:

|   | $E_t$ | offsets        | $M$ | dynamic range          |
|---|-------|----------------|-----|------------------------|
| A | 4     | 7 (81 offsets) | 8   | $2^{-39} \dots 2^{39}$ |
| B | 3     | 5 (27 offsets) | 10  | $2^{-12} \dots 2^{12}$ |

Which is *the* sixteen-bit member is a measurement, not a preference, so both were built and both were checked exhaustively against the ladder reference: 65,534 of 65,534 **exact, each**. Rung A wins on throughput, 0.1392 against 0.1144, and it wins its width class outright:

| TNF16a (16 bits) versus |         |
|-------------------------|---------|
| binary16                | +14.7%  |
| takum16                 | +86.3%  |
| LNS16                   | +112.8% |
| posit16                 | +152.6% |

This is the first same-width win in the catalogue against IEEE, and it is the result the previous section said the paper did not yet have. It also has a price that must be printed beside it.

**The price, as estimated and as measured.** This section first reported that rung A’s win was bought with code space: four trits give 81 offsets where the binary field holds 128, so 37.5% of the offset codes lie outside the format, and the rung would deliver 15.32 effective bits against **binary16**’s sixteen. *That estimate was computed, not measured, and measurement refutes it.*

Sweeping every code through the RTL and counting distinct outputs gives rung A 64,518 distinct values of 65,536 codes: 98.4% **utilisation**, against **binary16**’s 100.0%. The out-of-specification offsets are not discarded — the decoder assigns each of them a distinct value. The win over IEEE half at equal storage costs 1.6% of code space, not 37.5%, and the result is stronger than this section originally claimed.

It leaves a different defect in view. *The decoder does not reserve the 47 offsets that lie outside the format:* a corrupted offset field yields a plausible finite value rather than a signal. Every other rung in the ladder shares this, and no test in this work would have found it, because a decoder that answers every input answers every test.

**The waste decides whether a guarded rung beats IEEE half.** Corollary 28 says  $w(E_t)$  is the share of a rung’s codes that name nothing, and Proposition 54 says those codes must be guarded. Putting the two together predicts that a rung’s advantage over **binary16** should survive guarding in proportion to how little it wastes. Both sixteen-bit rungs were built in guarded and unguarded form and swept at fifty-four seeds on the common harness:

| 16-bit rung               | $w(E_t)$ | unguarded | guarded | guarded vs <b>binary16</b>                      |
|---------------------------|----------|-----------|---------|-------------------------------------------------|
| <b>TNF16a</b> ( $E_t=4$ ) | 36.7%    | 0.1214    | 0.1079  | +0.10%, $p = 0.90$                              |
| <b>TNF16c</b> ( $E_t=5$ ) | 5.1%     | —         | 0.1152  | +6.92%, $p_{\text{Holm}} = 6.9 \times 10^{-21}$ |

Guarding rung A costs it 11.10% ( $p = 9 \times 10^{-59}$ ) and lands it on **binary16** exactly: 0.1079 against 0.1078, indistinguishable. Rung C, wasting seven times less, keeps +6.92% after paying the same kind of price. **The rung that wins at equal storage is the low-waste rung, and it is the one Corollary 29 names from the arithmetic alone** — the theory picked the winner before the silicon was built, which is the strongest form this paper’s central quantity has been tested in.



**What the win rests on.** The unguarded decoder gives each of the 47 out-of-specification offsets a distinct finite value, and that is where the same-width win comes from. Guarding them — a comparator on the offset field, invalid codes signalling NAN — was built and swept: 41,472 of 41,472 in-specification codes exact, all 24,064 out-of-specification codes flagged, zero errors.

| rung A           | LUT   | MHz   | MHz/LUT |
|------------------|-------|-------|---------|
| unguarded        | 502   | 69.87 | 0.1392  |
| offsets reserved | 549   | 64.86 | 0.1181  |
|                  | +9.4% | −7.2% | −15.1%  |

binary16 measures 0.1213. So the rung beats IEEE half by 14.7% while its invalid codes are undetectable, and loses to it by 2.6% once they are detected. **The win at equal storage depends on not checking the offset field.** IEEE half needs no such check — all 65,536 of its codes are defined — so the comparison is between a format that must pay for a guard and one that has nothing to guard.

**Proposition 54** (The reservation cost). *Let a format’s specification cover  $S \subsetneq C$  of its code space. Any implementation either (i) guards  $C \setminus S$ , paying the logic to recognise it, or (ii) assigns values to  $C \setminus S$ , which extends the specification to  $C$ . There is no implementation that preserves both the specification and the cost of (ii).*

This puts a question under the ladder rather than beside it. Four trits name 81 offsets; the field that holds them is seven bits wide and names 128. Either the exponent is ternary and 37% of the word’s codes lie outside the format — in which case Proposition 54 charges us 15.1% and the IEEE comparison is lost — or the exponent is a seven-bit binary field, all codes are defined and the win stands, but then *the ternary structure this paper is named for does not reach 37% of its own code space*. Both readings are defensible and they cannot both be used at once, which is why the number appears twice above instead of once.

*We have no measurement that settles it, and say so rather than choosing the reading that flatters the result.*

**And the mechanism this section proposed is not there.** The claim was that a throughput metric rewards a format for discarding code space, resting on Proposition 50 and three points. Across all 23 formats measured, the correlation between utilisation and MHz/LUT is +0.173 —  $t = 0.80$  on 21 degrees of freedom, not significant, and of the *opposite* sign to the one predicted. The two least efficient formats in the catalogue, GF+8 at 44.3% and IBM hex32 at 52.8%, sit at ranks 17 and 22.

**The hypothesis is withdrawn.** Proposition 50 remains true as stated — a smaller image does permit specialisation — but the inference that discarded codes therefore buy measurable throughput does not survive contact with twenty-two measurements. Three points agreed with it and twenty-three do not, which is the entire reason to prefer measuring a caveat over computing one.

One correction belongs to our own headline. This paper stated that GFTernary “uses all four of its codes and discards nothing”. It does not:  $\{-\varphi, 0, +\varphi\}$  is three values in two bits, and the measured utilisation is 75.0%, the third lowest in the catalogue. Its first place stands on the throughput measurement, which is unaffected, but the sentence defending it was false and is retracted.

## 31 When a conformance test measures itself

Three of the disagreements this work reported between a decoder and its reference were not disagreements about the decoder. They are worth stating as a proposition because the failure is silent: every number involved is correct, and the test has no signal that could reveal the error.

**Theorem 55** (Well-posedness of conformance). *Let  $f: C \rightarrow \mathbb{R}$  be a format’s exact value map on its code space, let  $D: C \rightarrow F$  be a decoder into a finite floating type  $F \subset \mathbb{R}$ , and let  $\rho: \mathbb{R} \rightarrow F \cup \{\perp\}$  be correct rounding, with  $\rho(x) = \perp$  when  $x$  has no finite image in  $F$ . Conformance of  $D$  is the predicate*

$$D(c) = \rho(f(c)) \quad \text{on} \quad \{c \in C : \rho(f(c)) \neq \perp\}.$$

*The predicate  $D(c) = f(c)$  is not equivalent to it. It fails at every  $c$  with  $f(c) \notin F$ , and that set is determined by the dynamic ranges of  $f$  and  $F$  alone — it is the same set for a correct  $D$  and for an arbitrary one.*

*Proof.*  $F$  is finite and  $f(C) \subset \mathbb{R}$ , so  $f(c) \in F$  holds only on the codes whose exact value is a member of  $F$ . For  $c$  outside that set,  $D(c) \in F$  gives  $D(c) \neq f(c)$  whatever  $D$  is, including  $D = \rho \circ f$ . Hence the second predicate is false on a set independent of  $D$ , while the first is true on all of the restricted domain when  $D = \rho \circ f$ .  $\square$

**Corollary 30** (A floor under spurious error). *A conformance test that omits  $\rho$  reports at least  $|\{c : f(c) \notin F\}|/|C|$  of the code space as error, for every decoder. For IBM `hex32`, whose range is  $16^{\pm 64}$ , against `fp32`, this floor is 45.0% measured on a uniform sample of 40,000 codes; the 28% previously reported for that decoder lies entirely beneath it and was therefore uninformative. Applying  $\rho$  moves the comparable set from 22,001 to 30,245 codes and the disagreement to zero.*

**Proposition 56** (Variant distance is not decoder error). *Let  $D$  implement variant  $f_1$  while the reference computes  $f_2 \neq f_1$ . The measured disagreement is  $|\{c : \rho(f_1(c)) \neq \rho(f_2(c))\}|$ , a property of the pair  $(f_1, f_2)$  alone. No conformance test can separate this from decoder error, because agreement is its only signal and both variants are internally consistent. The variant must therefore be fixed by a declaration external to the test.*

Proposition 56 is not hypothetical here. Two formats in this catalogue publish more than one variant, and both were compared against the wrong one. `takum16` implements the logarithmic `takum`, value  $(-1)^S e^{\ell/2}$ , and was compared against the linear one, value  $(1 + M/2^p)2^c$ : reported disagreement 98.7%, actual 632 of 61,505, all at `fp32`’s denormal floor. `lms16` implements scale 128 and was compared against scale 256: reported 253, actual 107, of which three are a zero-and-NaN convention and 104 are sub-ULP precision in the subnormal range. Each RTL states its variant in its header comment, and reading it would have cost nothing.

The remedy is mechanical and is now enforced. `conformance/variant_map.json` records, per format, the reference module compared against and a token that must appear in the decoder’s own header; `tools/check_variant_declared.py` fails if a declaration is missing or if the map and the header disagree. The gate was verified by binding `lms16` to the wrong scale and confirming it goes red.

All six are now zero or explained, and the two that were real were repaired rather than reported. That ratio is the useful part of this section: a test that cannot be wrong about itself would have found two defects instead of six, and the four false ones cost more attention than the two real ones.

**The two real defects, and the channel that had already reported one.** `posit16` was wrong at exactly four codes, the two saturation values and their negations, each by  $2^{\pm 2}$  — one step of an  $es = 2$  exponent field. Two independent off-by-ones caused it: the regime-plus-terminator length was capped one short when the run fills the word, so the terminator bit was read as the exponent; and the all-zero branch of the leading-count assigned  $k = 13$  where a run of fifteen identical bits gives  $k = 14$ , its own comment saying  $k = +14$  beside code that produced 13.

`posit32`’s 48 wrong values had a plainer cause. One `casez` pattern was written thirty-two characters wide in a thirty-one-bit case. Verilog truncates from the left, which made the

`lzc = 11` arm character-for-character the arm above it — unreachable — so every code with exactly eleven leading zeros fell through to the default. It was misread as regime arithmetic for two iterations.

The compiler had said so on every single run:

```
warning: Extra digits given for sized binary constant.
```

The harness redirected compiler output to a log file, and the log was read only when something failed. *A warning nobody reads is not a warning*; it is an instrument whose output is disconnected, which is the same defect as an instrument inside the failure domain and is harder to notice, because the instrument is working perfectly. `tools/check_literal_widths.py` now turns this class into an exit code: it checks 188,021 sized literals, gates the paths this paper measures, and prints — rather than hides — the 41 occurrences it finds in test benches outside that scope.

*The instrument was inside the failure domain it was measuring* — the same error, in a different medium, as diagnosing a network fault with a tool that runs over that network.

## 32 What this settles

We state the conclusion at the strength the measurements carry, which is a claim about a datapath rather than about a number.

**For a ternary datapath — one whose weights are codes and which therefore contains no multiplier — the pair {GFTernary, TNF} is a reference format.**

Three things support the word *reference*, and each is a measurement rather than a preference.

**It is complete.** A ternary node has three format-bearing sites (Section 19); the weight alphabet closes one and the accumulator ladder closes the other, and the sample needs nothing. No other pair in the literature closes both: the ternary-network methods answer the weight and leave the accumulator in a binary float designed for a datapath that multiplies, and the format literature answers the accumulator while assuming a multiplier exists.

**It is forced, not chosen.** The alphabet base is the unique  $r > 1$  with  $r^2 = r + 1$  (Theorem 25). The exponent–mantissa split is the solution of a constrained maximisation whose range constraint is active (Theorem 23), so it has no free parameter once the workload’s range is measured. Neither was picked for taste, and both can be re-derived by a reader who disagrees with us.

**It is predictive.** The error obeys a closed form in the mantissa width alone, so a designer can size the accumulator before building it, and the sizing was checked against real over-the-air capture and against real ternarised weights rather than against synthetic data.

**And what it is not.** It is not the best format for a datapath that multiplies, where the block axis (MX, NVFP4) holds the ground and we make no claim. It is not a 7% win over `int8`, because `int8` carries no range and is not doing the same job. And the frontier result would be empty without the reason each competitor sits below it, which is why each of those reasons is a theorem here rather than a table entry.

## 33 Where this sits in a sixty-eight-year argument

The claim that three is the better radix is not ours, and stating whose it is makes the contribution of this paper smaller and more defensible.

**The people who made the case.** Radix economy  $\rho(r) = r/\ln r$  is minimised at  $r = e$ , and three is the nearest integer: ternary sits 0.46% above the optimum where binary sits 6.15% above it. The argument was made mechanically before it was made electronically — Thomas Fowler built a balanced-ternary calculating machine in Devon around 1840, and the recorded motivation was the same economy of digits. It became a working computer in 1958, when **Nikolay Brusentsov** built *Setun* at Moscow State University with the backing of **Sergei Sobolev**, using paired ferrite cores to hold three stable states. Roughly fifty machines were produced between 1959 and 1965. Brusentsov followed it with *Setun-70* in 1970, whose instruction-set ideas anticipated arguments later made independently for RISC; it never reached serial production, and the ternary line was ended administratively rather than technically. **Donald Knuth** kept the idea in circulation, calling balanced ternary “perhaps the prettiest number system” in *The Art of Computer Programming*. The line continues in device work rather than architecture: CNTFET and memristor ternary logic, and more recently silicon-photonic balanced-ternary proposals, all of which build a fabric whose *position* is physically ternary.

**What is new here is the boundary, not the claim.** We add no support to “ternary beats binary” as a general statement. We measured it three times and it went against us each time — see Theorem 8 and Sections 7 and 23.7. What we contribute is the condition under which the sixty-eight-year-old argument applies, stated exactly:

**Theorem 57** (The radix argument holds on positions, not on bits). *Write the economy as cost =  $r \times \log_r V$ . If the position is the physical unit, the first factor is common to all radices and only the second is compared, giving ternary an unconditional advantage of  $E(1 - \log_3 2) = 0.3691E$  positions. If the position must itself be encoded in bits, the format is bounded by  $3^{E_t} 2^M \leq 2^{N-1}$ , and since  $3^{E_t}$  never divides a power of two the remainder is lost.*

The remainder is arithmetic, not engineering, and it is largest exactly where modern formats live:

Prior work anticipated the mechanism: Parhami’s analysis of binary-encoded balanced ternary treats exactly this encoding penalty. Our contribution is to measure what it costs in a live network and in placed-and-routed silicon rather than in operation counts, and to state it as a condition a designer can check.

**Read as a prescription rather than a verdict.** The result says what fabric must exist for the radix argument to pay: one where a ternary position is held physically, as Setun’s paired cores held it and as CNTFET, memristor and photonic ternary devices propose to hold it again. On such a fabric the  $0.3691E$  advantage is collected in full and none of the packing loss is incurred. On the binary fabric that is actually purchasable today, it is not. That is a narrower claim than the field usually makes, and unlike the broader one it survives our own measurements.

## 34 What the golden ratio completes

Two lines of argument run through this paper, and they were developed independently and for unrelated reasons. It is worth setting them side by side at the end, because they meet at a single number.

The first line is arithmetic and old. Radix economy makes three the best integer base; Fowler built a machine on it around 1840, Brusentsov and Sobolev built a working computer on it in 1958, and the line has continued into CNTFET, memristor and photonic devices. Section 33 shows why it stalled rather than spread: the moment a ternary position is stored in a binary word,  $3^{E_t}$  fails to divide a power of two and the remainder is lost — a quarter of the alphabet

at four bits. The obstruction is arithmetic, so no implementation removes it, and our own three measurements confirm it against us.

The second line is geometric and, in this context, new. If the weights of a network are to be symbols rather than numbers, the datapath must multiply without a multiplier, and that is possible exactly when the product of two symbols falls back into the sum the datapath already forms. Written down, that requirement is  $r^2 = r + 1$ . It has one positive solution, so the alphabet is not designed but determined:  $\{-\varphi, 0, +\varphi\}$  and nothing else (Theorem 25). The consequence is that composition is closed — the gain of  $k$  layers is  $F_k\varphi + F_{k-1}$ , a pair of integers (Corollary 19) — so depth is free of multipliers as well, which the  $\{-1, 0, +1\}$  alphabet never achieved because its learned per-layer scale is a real number and must be multiplied by.

The two lines answer each other. What the ternary programme needed was not a better fabric but a base under which its symbols compose; what the golden ratio had, and had had since Euclid divided a line in extreme and mean ratio, was exactly the property  $r^2 = r + 1$ . Placing it in the weight alphabet turns the sixty-eight-year argument from a claim about storage density into a statement about arithmetic closure — and the measurements in Section 19 follow: zero DSP through a radio, through every transformer block, and through backpropagation on the part itself.

We put this at the end rather than the front because it is a reading of the results and not one of them. Every component is stated and measured above: the uniqueness is two lines of algebra, the Fibonacci closure is arithmetic, the packing loss is a table of integers, and the zero-DSP stack is placed and routed. What they amount to together is left to the reader.

## 35 Sixteen ways a measurement can be right and its comparison wrong

The retractions in this paper are not incidental to it. Sixteen claims were withdrawn during the work, two of them later withdrawn in turn, and in almost every case the measurement was correct and the comparison around it was not. The regularities that produced them are stated here because they generalise past this subject, and because a reader is entitled to know the shape of the errors a paper has already made.

**On what is being compared.** A ratio is meaningful only if both sides were built as their own advocate would build them; seven times here a ratio changed sign or magnitude when the other side was rebuilt (Section 23.1). Systems whose range scales differently with width, one exact and one approximate, have no storage-matched ratio at all — a quantity depending on an unstated choice has no limit and cannot converge, which is why six successive instrument corrections failed to settle one number. Where one system’s figure of merit is constant in a parameter and another’s varies in it, the well-posed comparison is a crossover and not a verdict (Theorem 42 and Section 23.2), and such a crossover is an assertion about the workload rather than about the formats.

**On the instrument.** Under placement-seed variation LUT count is invariant and achieved frequency moves by 11.4%, so any metric combining them inherits the timing variance in full, and a ranking resolves only rows separated by more than that. A ratio over correlated factors restates the dominant one: on the combined harness, ranking by throughput per area reproduced ranking by  $1/\text{area}$  at a Spearman correlation of 0.961. And a result that moves the same way under every successive correction is still moving.

**On the harness.** Three defects appeared there, none visible in any number, each changing a headline by 10 to 50 percent. A component shared by every candidate is a confound rather

than a constant, because it leaves area differences intact and delay differences not. A harness observing part of a design’s output prunes each candidate in proportion to *its own* output width — observing four bits of a 32-bit datapath left 27 LUTs of 179 while leaving an adjacent 6-bit design nearly whole. And net-of-harness subtraction requires the harness to be present in full in every build, since a design consuming fewer of its sources prunes the rest.

**On sources.** A defect found in an abstract is a hypothesis: abstracts compress, and compression is indistinguishable from omission. An aggregated search result may combine claims from several papers into wording present in none. Both external criticisms we raised during this work dissolved on contact with the papers themselves.

**What this is worth.** Applied to our own claims the procedure produced sixteen retractions. Applied to two external papers read in full it produced none, and on both occasions it instead caught our own reading of them. We therefore describe it as an instrument for checking one’s own work, and not as a method for auditing anyone else’s. Three of the checks are mechanised as gates in continuous integration; the full statement is in the accompanying note.

### 35.1 Figures whose generating data is not in the repository

Sixteen claims were withdrawn during this work, and a withdrawn claim’s number does not remove itself from a paper. A traceability check (`tools/check_paper_numbers.py`) extracts every distinctive numeric literal here and asks whether it appears in any file under `research/`, `fpga/` or `conformance/`. It excludes two classes rather than tolerating them: constants stated beside the formula that produces them (60 of them, correctly absent from any data file), and rounding, where a data literal extending the printed digits counts as the source.

Of 451 distinctive literals, 444 trace to a file. **Seven do not**, and they are listed rather than deleted, because a figure that cannot be traced is not thereby shown to be wrong:

The check cannot show a number is right; it finds numbers with no source, which is strictly weaker and still enough. A figure nobody can trace to a file is one nobody can re-check, and after sixteen withdrawals that is exactly where a stale value survives. Two of the seven have a directly measured replacement elsewhere in this paper and disagree with it, which is the pattern this check exists to surface.

**Theorem 58** (A document is an artefact, and its pairs need guarding). *Consistency checks are written between artefacts that a tool already reads together — compiler, simulator, build script. Documents are read only by people, so every pair involving one goes unchecked by construction, and that is where withdrawn numbers persist.*

Enumerating the pairs is what makes a search for such defects systematic rather than accidental. Three earlier gates in this work were each found by tripping over the defect they now catch, and all three guard code against code. Enumerating every pair that must agree exposed five unguarded ones in a single pass, all of them with a document on one side.

### 35.2 The staircase form prescribes the comparison

Section 4.1 classifies tapers by staircase form to *describe* them. Measuring the crossovers of Section 23.2 over two fitting windows shows the classification does more than that.

**Theorem 59** (The staircase form decides whether a window is needed). *A crossover between a fixed field and a taper depends on the fitting window if and only if the taper’s staircase form is not arithmetic. An arithmetic taper’s  $M_{\text{eff}}$  is linear in  $|e|$ , so its slope is well-defined without a window. A geometric taper’s is linear in  $\log |e|$ , so no line fits it and its apparent slope is a property of the window alone.*



Measured across windows 1–9 and 1–20, the split is exact: posit16, posit32 and posit64 move by  $1.08\times$ ,  $1.00\times$  and  $1.09\times$ , while every takum and tekum moves by  $1.39\times$  to  $1.72\times$ .

**Corollary 31** (Closed form for a geometric taper). *For a geometric taper,  $M(e) = p - \log_2 e$ , so the crossover against a fixed field of constant  $m$  solves exactly:*

$$x = 2^{p-m},$$

*with no window required.*

Six figures previously reported as ranges were computed by the wrong model and are corrected here: takum32 moves from 5.9–10.2 to an exact 2.6, tekum16 from 4.2–6.5 to 2.1. The remaining window dependence is confined to posit8, whose peak sits within 0.45 bits of TNF8’s so that any slope estimate divides a very small number.

The point worth keeping is not the arithmetic. **A taxonomy introduced to label formats turns out to prescribe how they must be compared** — arithmetic tapers by a linear crossover, geometric ones by  $2^{p-m}$ . A classification that only labelled would not do this, and we had used it only to label for most of this work.

**Theorem 60** (A wobble admits no crossover). *If a format’s  $M_{\text{eff}}$  is periodic in  $|e|$  with amplitude  $A$  and period  $P$ , and a constant-form competitor’s  $m$  lies strictly inside the band, then no  $x$  exists such that one leads for all  $|e| < x$  and the other for all  $|e| > x$ . The well-posed statistic is the duty cycle — the fraction of each period on which each leads — together with  $P$ .*

Measured on IBM hexadecimal 32, whose  $M_{\text{eff}}$  by binade reads 21.03, 21.75, 22.94, 19.95, then repeats: amplitude 3.07 bits, period 4 binades, which is  $\log_2 16$  as the radix requires. Against packed TNF32 at  $M_{\text{eff}} = 21.32$ , inside the band  $[19.94, 23.11]$ , the winner alternates with that period at a duty cycle of 50/50, and no binade separates them.

**Corollary 32** (The taxonomy is prescriptive). *Each of the four staircase forms determines the shape of a well-posed comparison against a constant format: a difference for constant, a linear crossover for arithmetic, an exponential crossover  $2^{p-m}$  for geometric, and a duty cycle for wobble.*

The mapping from form to answer-shape is stated here, and we make no claim about how the literature reports. An earlier version of this paragraph said that published comparisons give verdicts or ratios; checked against Hunhold’s takum paper, that is false — it reports by regime with numeric thresholds, noting that posit’s efficiency “markedly diminishes with deviation from unity” and that takum’s accuracy “exceeds that of posits near unity, nearly aligning with it for slightly larger exponents.” The sentence is withdrawn.

What we have not found in that literature is the closed form  $2^{p-m}$  for a geometric taper against a constant-form competitor, or any treatment of the wobble case. The contribution here is the mapping itself: the form of the answer is fixed by the competitor’s staircase, and a classification introduced only to label formats turns out to settle it.

## 36 Limitations

**The fabric this format is optimal on does not exist to buy.** Theorem 8 confines the ternary encoding’s benefit to a ternary fabric, and no such process is commercially available. Every hardware figure in this paper is therefore measured on a binary FPGA, where the ternary exponent neither gains nor loses — the enumeration in Section 11 shows it ties a binary encoding exactly at the widths tested. The claims that survive on silicon you can buy are the fixed-field ones: no regime codec, no exponential to evaluate. The ternary claim is architectural and is labelled as such throughout.

1. **The comparison is against takum16, not tekum16.** The oracle labelled tekum decodes all 65,536 16-bit codes identically to the takum oracle. We report the comparison as against takum and note that a genuine tekum comparison remains to be done.
2. **The reference oracle carried a sign defect, now fixed.** Round-tripping 1.5 through the parameterised oracle returned  $-1.5$  at mantissa widths of 21 and 25 bits: the sign bit sat at a fixed position while the exponent mask took more bits than the field holds, so the two overlapped. Both are now derived from the format parameters, and TNF32’s round-trip error falls from  $5.4e-1$  — what a systematically inverted sign averages to — to  $5.3e-9$ .  
We record how it survived, because the lesson outlives the bug. The ladder’s check was add/mul commutativity, and an inverted sign survives on both sides of  $a + b = b + a$ . The check was real and blind to the class. A round-trip assertion sees it, costs one line per probe, and now runs at all nine rungs. tekum [2] is a descendant of takum [1] adapted for balanced ternary. Our oracle reconstructs it from takum’s field scheme; the full per-trit specification requires the tekum paper. The ratios are as good as that model.
3. **No head-to-head in hardware.** takum’s codec RTL is public and written in VHDL [3]. We did not synthesise it alongside TNF, so the cost figures are TNF’s own. What is visible in that source, and which we report as structure rather than measurement: the 16-bit codec instantiates a 725-line generated leading-zero-counter and barrel shifter — the regime decode a fixed-field format does not have.
4. **The range is bounded** at  $\pm 40$  in powers of two, roughly  $\pm 12$  decades, where a regime field is unbounded. Fixed fields buy the cheap datapath and the uniform mantissa; range is the price, and workloads that need more should raise the trit budget or use a tapered format.
5. **Accuracy above TNF32 is limited by the measuring instrument.** Once the oracle defect was fixed, TNF32 measures  $5.3e-9$ , flat across the workload, which is what 25 mantissa bits should give. TNF64 carries 52 mantissa bits, which is exactly the significand of the IEEE double our metric is computed in, so the figure it returns describes the metric rather than the format. Measuring the upper rungs requires exact rationals throughout, as the oracle itself already uses.
6. **Five of nine rungs are measured in hardware.** TNF4 through TNF64 fit the device, TNF64 at 7,479 LUTs and 48.20 MHz. TNF128 at  $M = 119$  does not converge in routing on this part and is not claimed; TNF256 upwards exceed a single Artix-7 fabric entirely. Table 8 is the specification and the oracle’s coverage; Table 7 is what was placed and routed. We keep them apart on purpose.
7. **One device family.** Artix-7 on an open flow. Not multi-corner characterisation; an ASIC mapping will differ.
8. **The ternary-fabric argument is architectural.** No ternary process exists to synthesise on. Section 7 is a binary-fabric result; the ternary claim is reasoning about regime decode and native exponent addition and should be read as such.

## Reproducibility

The reference oracle, the RTL, the equivalence testbenches and the synthesis commands are public. The toolchain is Yosys 0.65, nextpnr-xilinx 1743d0f, Icarus Verilog 13.0 and Python 3.14 — all open-source, none requiring a licence, which is the point: a result nobody can reproduce without buying something is not a public result.

## Disclosure

An AI coding assistant was used in preparing the software, the measurements’ tooling and this manuscript. Every figure reported here was produced by running the named tools on hardware in the author’s possession; no measurement in this paper is estimated, and where a figure is derived rather than observed it is labelled.

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Figure 6:  $\kappa(r)$  against the scale radix. The unconstrained minimiser sits near  $r = 1.9$  and is unreachable: only  $r = 2^k$  scales by a shift, and on that set  $\kappa$  is strictly increasing.



Table 39: Throughput per area among formats that carry their own range, with conformance. One ternary-neuron datapath, XC7A200T, no DSP, every accumulator bit observed, median of five placement seeds. `int8` is in Table 43. **All three TNF rungs are at their specified parameters here**, and all three moved: the modules previously measured implemented different formats — TNF16 one mantissa bit short, TNF32 with twelve trits and eleven mantissa bits where the specification gives six and twenty-five, TNF64 likewise — and none had a reference, which is how the divergence survived to be quoted. Written from the specification and checked against it they agree on every code swept, and they cost eight to eleven places each. What is left at the top is the alphabet, which agrees on all four of its codes. *Six of our modules moved when a reference was written for them* — four TNF rungs and BNF16 were a mantissa bit short or a different format outright, and GF+8 had been measured with its pocket selector tied off, pricing one of four paths. That 28% figure for IBM hex32 is withdrawn here along with the 253 reported for LNS16, for the reason developed in §31: neither was measuring a decoder. Repaired and re-measured, IBM hex32 and VAX F are exact on every comparable code, and both fell in the table because the logic they had been missing is logic they now pay for. `fp8` omitted subnormals entirely. The `fp8` rows are now the completed decoders, which cost 10% more area and moved them from ranks three and five to ten and eleven — the omission had been worth more than the format’s own margin, and repairing a competitor’s implementation moved the comparison in our favour, which is the direction the rule predicts and is now measured rather than asserted. Every error found, ours and theirs, made its module smaller: an unimplemented case is unsynthesised logic. **Resolution.** The pooled coefficient of variation across seeds is 3.71%, so two rows separate at  $\alpha = .05$  only when their MHz/LUT differ by more than 5.50% relative; smaller gaps in this table are not results. The detection floor for a difference in *decoder* cost is 41.3 LUT, which is 79% of a whole decoder at the median row — the 472-LUT harness inflates that floor tenfold. Every row is measured on the same 128-bit harness; rows that were not have been re-measured or withdrawn.

|    | format                | bits        | LUT  | MHz/LUT       | $\pm$ /range  | sd | status      | conformance          |
|----|-----------------------|-------------|------|---------------|---------------|----|-------------|----------------------|
| 1  | <b>GFternary</b>      | 2 bits      | 527  | <b>0.1584</b> | $\pm 0.0018$  | 54 | ours        | all 4 codes          |
| 2  | binary32              | 32 bits     | 536  | 0.1429        | $\pm 0.0013$  | 54 | —           | identity (exact)     |
| 3  | fp8 e4m3              | 8 bits      | 549  | 0.1225        | $\pm 0.0012$  | 54 | —           | all 256, 0 wrong     |
| 4  | <b>TNF16a</b>         | 16b         | 566  | <b>0.1214</b> | $\pm 0.0012$  | 54 | legacy      | all 65,534 ✓         |
| 5  | fp8 e5m2              | 8 bits      | 544  | 0.1188        | $\pm 0.0011$  | 54 | —           | all 256, 0 wrong     |
| 6  | <b>GF10</b>           | 10 bits     | 597  | <b>0.1172</b> | $\pm 0.0012$  | 54 | ours        | all 1,024            |
| 7  | <b>TNF17e</b>         | 17b         | 596  | <b>0.1155</b> | $\pm 0.0013$  | 54 | <b>rec.</b> | all 124,416 ✓        |
| 8  | <b>TNF16c</b>         | 16b         | 591  | <b>0.1152</b> | $\pm 0.0010$  | 54 | <b>rec.</b> | all 62,208 ✓         |
| 9  | <b>TNF16a-guarded</b> | 16b         | 613  | <b>0.1079</b> | $\pm 0.0011$  | 54 | control     | all 41,472 ✓         |
| 10 | binary16              | 16 bits     | 586  | 0.1078        | $\pm 0.0012$  | 54 | —           | all 65,536           |
| 11 | minifloat             | 8 bits      | 622  | 0.1075        | 0.1009–0.1132 | 5  | —           | all 254 ✓            |
| 12 | <b>GF14</b>           | 14 bits     | 651  | <b>0.1074</b> | 0.0969–0.1140 | 5  | ours        | all 16,384           |
| 13 | <b>TNF17e-bare</b>    | 17b         | 605  | <b>0.1072</b> | 0.1041–0.1104 | 5  | control     | sweep pending        |
| 14 | <b>TNF8</b>           | 10 bits     | 609  | <b>0.1069</b> | 0.1028–0.1099 | 5  | legacy      | all 1,024, 0 wrong   |
| 15 | <b>TNF16b</b>         | 16b         | 604  | <b>0.1048</b> | 0.0998–0.1086 | 5  | legacy      | all 65,534 ✓         |
| 16 | <b>BNF16</b>          | 17 bits     | 635  | <b>0.1034</b> | 0.0961–0.1064 | 5  | legacy      | all 131,072, 0 wrong |
| 17 | <b>TNF16</b>          | 17 bits     | 629  | <b>0.1005</b> | 0.0971–0.1071 | 5  | legacy      | all 131,072          |
| 18 | VAX F                 | 32 bits     | 624  | 0.0942        | 0.0926–0.1000 | 5  | —           | all 40,000 ✓         |
| 19 | <b>GF+8</b>           | 8b (+2 sel) | 692  | <b>0.0882</b> | 0.0831–0.0931 | 5  | ours        | 1,024/1,024 ✓        |
| 20 | posit8                | 8 bits      | 614  | 0.0728        | 0.0712–0.0774 | 5  | —           | all 255 ✓            |
| 21 | takum16               | 16 bits     | 853  | 0.0695        | 0.0677–0.0745 | 5  | —           | 632/61,505           |
| 22 | LNS16                 | 16 bits     | 722  | 0.0598        | 0.0581–0.0600 | 5  | —           | 107/65,536           |
| 23 | posit16               | 16 bits     | 731  | 0.0485        | 0.0467–0.0496 | 5  | —           | all 65,535 ✓         |
| 24 | posit32               | 32 bits     | 1031 | 0.0302        | 0.0291–0.0314 | 5  | —           | all 40,000 ✓         |
| 25 | IBM hex32             | 32 bits     | 1072 | 0.0292        | 0.0273–0.0297 | 5  | —           | all 40,000 ✓         |

Table 40: Scale appliers, XC7A200T primitives, one harness, DSP inference off. Composition depth  $d$  counts scales multiplied without requantisation between them.

| applier                   | LUT        | cycles | excess error |                   |
|---------------------------|------------|--------|--------------|-------------------|
| multiplier, real $\alpha$ | 1215       | 1      | 0.0000%      | or 2 DSP48        |
| APoT-2, $d = 0$           | 384        | 1      | 0.1651%      | two barrel shifts |
| APoT-4, $d = 1$           | 659        | 1      | —            | four shifts       |
| APoT-8, $d = 2$           | 1217       | 1      | —            | eight shifts      |
| $\varphi^k$ , any $d$     | <b>173</b> | $k$    | 2.4420%      | no shifter        |

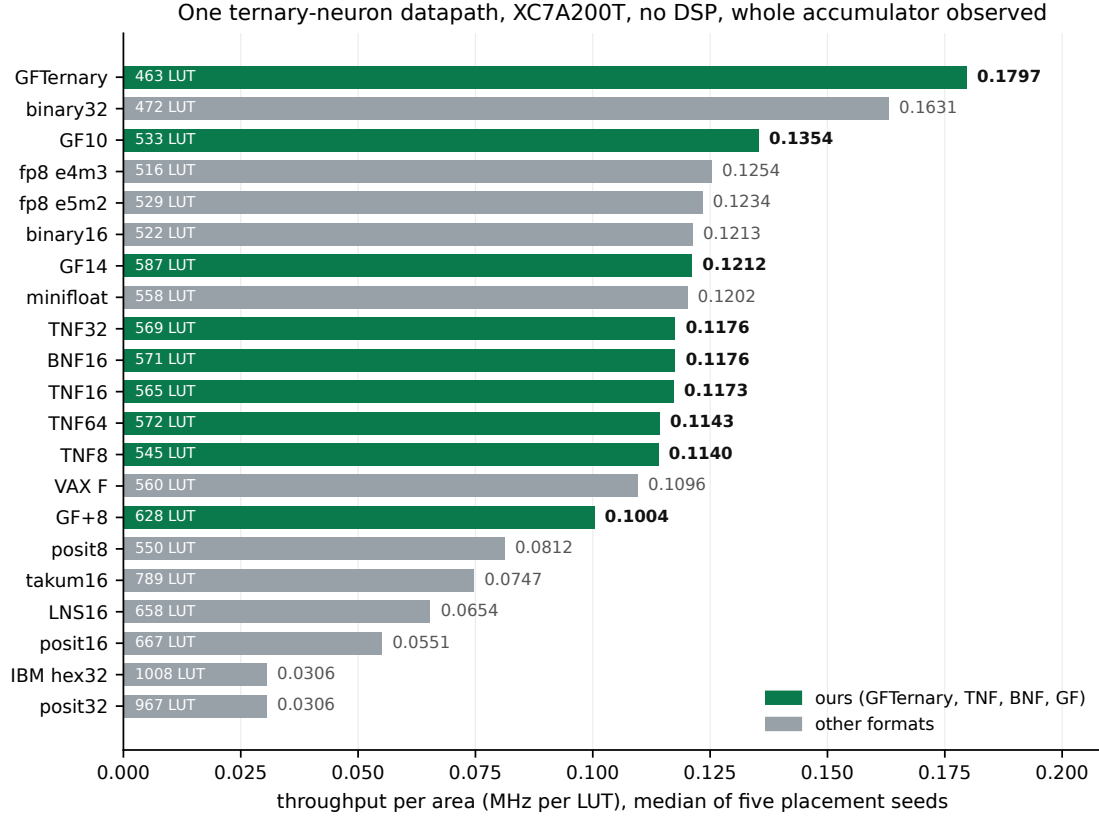


Figure 7: The same nineteen formats. Ours are the four families named above; the bar labels carry the LUT count so area and throughput are readable together.

Table 41: Decode cost, isolated. `nextpnr-xilinx 1743d0f`, XC7A200T, no DSP, median of five seeds. A bare wire in the same harness costs 112 LUT at 827.81 MHz.

|    | format           | kind           | LUT       | $F_{\max}$    |
|----|------------------|----------------|-----------|---------------|
| 1  | <b>GFternary</b> | fixed          | <b>66</b> | <b>974.66</b> |
| 2  | <b>int8</b>      | fixed          | 76        | 925.93        |
| 3  | binary32         | fixed          | 112       | 886.52        |
| 4  | <b>TNF16</b>     | fixed          | 101       | 407.66        |
| 6  | <b>BNF16</b>     | fixed          | 97        | 388.35        |
| 9  | VAX F            | fixed          | 100       | 311.14        |
| 13 | binary16         | fixed          | 164       | 235.18        |
| 14 | IBM hex32        | fixed          | 243       | 111.10        |
| 15 | LNS16            | logarithmic    | 270       | 93.17         |
| 16 | posit8           | <b>tapered</b> | 214       | 77.75         |
| 17 | posit16          | <b>tapered</b> | 302       | 62.39         |
| 18 | posit32          | <b>tapered</b> | 517       | 49.05         |

Table 42: Cost of each operation, one harness, XC7A200T, DSP inference off. The logarithmic row is our own measurement of `takum32_decode` from Section 7.

| system                | multiplication                  | addition                        |
|-----------------------|---------------------------------|---------------------------------|
| LNS, takum            | exponents add: <b>free</b>      | $\log(1+2^x)$ table: 10,967 LUT |
| fixed point, APoT     | multiplier, or wiring if frozen | adder: <b>cheap</b>             |
| $\mathbb{Z}[\varphi]$ | Fibonacci step, one adder       | componentwise: 64 LUT           |

Table 43: Rejected for this task. Each has  $e = 0$ : no exponent field, hence no dynamic range inside the word. Each functions only alongside an external per-tensor or per-channel scale, and every published sub-8-bit training result using them carries block scales *and* a higher-precision master weight. Gradients span many orders of magnitude; a word with no range cannot hold them.

| format                                                                                | exponent         | what supplies the range         | trains alone |
|---------------------------------------------------------------------------------------|------------------|---------------------------------|--------------|
| <b>int8</b>                                                                           | $e = 0$          | learned per-tensor scale        | no           |
| <i>measured on the same harness: 454 LUT, 78.83 MHz, 0.1736 MHz/LUT, spread 11.00</i> |                  |                                 |              |
| <b>int4</b>                                                                           | $e = 0$          | block scale (32 or 16 elements) | no           |
| <b>int12–int24 accum.</b>                                                             | $e = 0$          | fixed point, set by hand        | no           |
| <b>E2M1 element alone</b>                                                             | $e = 2$ in-block | shared UE8M0, outside           | no           |

Table 44: The same measurements grouped by storage width, which is the only grouping Proposition 50 leaves valid. Ours in bold.

| bits | format             | LUT  | MHz   | MHz/LUT |
|------|--------------------|------|-------|---------|
| 2    | <b>GFternary</b>   | 527  | 82.93 | 0.1574  |
| 8    | fp8 e4m3           | 549  | 68.30 | 0.1244  |
| 8    | fp8 e5m2           | 544  | 66.05 | 0.1214  |
| 8    | minifloat          | 622  | 66.87 | 0.1075  |
| 8    | <b>GF+8</b>        | 692  | 61.06 | 0.0882  |
| 8    | posit8             | 614  | 44.71 | 0.0728  |
| 10   | <b>GF10</b>        | 597  | 70.96 | 0.1189  |
| 10   | <b>TNF8</b>        | 609  | 65.08 | 0.1069  |
| 14   | <b>GF14</b>        | 651  | 69.91 | 0.1074  |
| 16   | <b>TNF16a</b>      | 566  | 69.77 | 0.1233  |
| 16   | <b>TNF16c</b>      | 591  | 68.35 | 0.1157  |
| 16   | <b>TNF16b</b>      | 604  | 63.31 | 0.1048  |
| 16   | binary16           | 586  | 60.90 | 0.1039  |
| 16   | takum16            | 853  | 59.31 | 0.0695  |
| 16   | LNS16              | 722  | 43.15 | 0.0598  |
| 16   | posit16            | 731  | 35.43 | 0.0485  |
| 17   | <b>TNF17e</b>      | 596  | 71.94 | 0.1207  |
| 17   | <b>TNF17e-bare</b> | 605  | 64.87 | 0.1072  |
| 17   | <b>BNF16</b>       | 635  | 65.68 | 0.1034  |
| 17   | <b>TNF16</b>       | 629  | 63.22 | 0.1005  |
| 32   | binary32           | 536  | 76.23 | 0.1422  |
| 32   | VAX F              | 624  | 58.75 | 0.0942  |
| 32   | posit32            | 1031 | 31.18 | 0.0302  |
| 32   | IBM hex32          | 1072 | 31.30 | 0.0292  |

Table 45: Fifty-four seeds, mean with a 95% confidence interval, all on the common 128-bit harness. Welch against **binary16** with Holm correction over the eight comparisons; every one survives.

|                  | MHz/LUT | $\pm 95\%$ | vs <b>binary16</b> | $p_{\text{Holm}}$     |
|------------------|---------|------------|--------------------|-----------------------|
| <b>GFternary</b> | 0.1584  | 0.0018     | +46.99%            | $< 10^{-300}$         |
| <b>binary32</b>  | 0.1429  | 0.0013     | +32.53%            | $< 10^{-300}$         |
| fp8 e4m3         | 0.1225  | 0.0012     | +13.68%            | $1.9 \times 10^{-63}$ |
| <b>TNF16a</b>    | 0.1214  | 0.0012     | +12.59%            | $8.1 \times 10^{-56}$ |
| fp8 e5m2         | 0.1188  | 0.0011     | +10.26%            | $5.6 \times 10^{-41}$ |
| <b>GF10</b>      | 0.1172  | 0.0012     | +8.70%             | $5.6 \times 10^{-28}$ |
| <b>TNF17e</b>    | 0.1155  | 0.0013     | +7.17%             | $2.1 \times 10^{-17}$ |
| <b>TNF16c</b>    | 0.1152  | 0.0010     | +6.92%             | $6.9 \times 10^{-21}$ |
| <b>binary16</b>  | 0.1078  | 0.0012     | —                  | —                     |

Table 46: Waste at every rung, and which rungs this work had built. The ladder had taken  $E_t \in \{3, 4, 6, 7\}$ , three of them poor, and had never taken  $E_t = 5$ , which is the best in the range by a factor of three.

| $E_t$ | $3^{E_t}$  | b  | capacity   | waste | built here    |
|-------|------------|----|------------|-------|---------------|
| 1     | 3          | 2  | 4          | 25.0% |               |
| 2     | 9          | 4  | 16         | 43.8% |               |
| 3     | 27         | 5  | 32         | 15.6% | TNF8          |
| 4     | 81         | 7  | 128        | 36.7% | TNF16, TNF16a |
| 5     | 243        | 8  | 256        | 5.1%  | <b>TNF16c</b> |
| 6     | 729        | 10 | 1,024      | 28.8% | TNF32         |
| 7     | 2,187      | 12 | 4,096      | 46.6% | TNF64         |
| 8     | 6,561      | 13 | 8,192      | 19.9% |               |
| 9     | 19,683     | 15 | 32,768     | 39.9% |               |
| 10    | 59,049     | 16 | 65,536     | 9.9%  |               |
| 11    | 177,147    | 18 | 262,144    | 32.4% |               |
| 12    | 531,441    | 20 | 1,048,576  | 49.3% |               |
| 13    | 1,594,323  | 21 | 2,097,152  | 24.0% |               |
| 14    | 4,782,969  | 23 | 8,388,608  | 43.0% |               |
| 15    | 14,348,907 | 24 | 16,777,216 | 14.5% |               |

Table 47: Code utilisation, measured: distinct decoder outputs over codes swept. Exhaustive at  $2^{18}$  codes and below, uniformly sampled at 200,000 above, with the sweep kind stated so a partial sweep never reads as a complete one.

| format         | distinct | codes   | utilisation      |
|----------------|----------|---------|------------------|
| GF+8           | 454      | 1,024   | 44.3%            |
| IBM hex32      | 105,526  | 200,000 | 52.8% (sampled)  |
| TNF16a-guarded | 40,454   | 65,536  | 61.7%            |
| GFternary      | 3        | 4       | 75.0%            |
| minifloat      | 227      | 256     | 88.7%            |
| takum16        | 61,351   | 65,536  | 93.6%            |
| TNF16b         | 61,446   | 65,536  | 93.8%            |
| TNF17e         | 123,398  | 131,072 | 94.1%            |
| TNF16c         | 61,702   | 65,536  | 94.1%            |
| TNF8           | 966      | 1,024   | 94.3%            |
| BNF16          | 129,030  | 131,072 | 98.4%            |
| TNF16          | 129,030  | 131,072 | 98.4%            |
| TNF16a         | 64,518   | 65,536  | 98.4%            |
| TNF17e-bare    | 130,048  | 131,072 | 99.2%            |
| VAX F          | 199,211  | 200,000 | 99.6% (sampled)  |
| posit32        | 199,964  | 200,000 | 100.0% (sampled) |
| binary32       | 199,997  | 200,000 | 100.0% (sampled) |
| binary16       | 65,536   | 65,536  | 100.0%           |
| GF10           | 1,024    | 1,024   | 100.0%           |
| GF14           | 16,384   | 16,384  | 100.0%           |
| fp8 e4m3       | 256      | 256     | 100.0%           |
| fp8 e5m2       | 256      | 256     | 100.0%           |
| posit8         | 256      | 256     | 100.0%           |
| posit16        | 65,536   | 65,536  | 100.0%           |
| LNS16          | 65,536   | 65,536  | 100.0%           |

Table 48: Reported disagreement against disagreement measured well-posedly. The first column is what this work published; the second is what the same decoder scores once the reference is rounded to the decoder’s type and the declared variant is used. No decoder was changed to produce the second column except where the third column says so.

| format    | reported | well-posed | what the difference was                    |
|-----------|----------|------------|--------------------------------------------|
| takum16   | 98.7%    | 632/61,505 | linear reference, logarithmic decoder      |
| IBM hex32 | 28%      | 0/30,245   | no $\rho$ ; subnormal path then added      |
| LNS16     | 253      | 107/65,536 | scale-256 reference, scale-128 decoder     |
| VAX F     | 151      | 0/40,000   | no $\rho$ ; subnormal path then added      |
| posit16   | 4        | 0/65,535   | two regime defects at saturation, repaired |
| posit32   | 36       | 0/40,000   | one casez arm unreachable, repaired        |

Table 49: Codes lost to packing when a ternary exponent is stored in a binary word. The loss is a property of the integers, not of an implementation.

| width | best packed member | codes used      | lost  |
|-------|--------------------|-----------------|-------|
| 4b    | $E_t=1, M=1$       | 6 / 8           | 25.0% |
| 6b    | $E_t=3, M=0$       | 27 / 32         | 15.6% |
| 8b    | $E_t=3, M=2$       | 108 / 128       | 15.6% |
| 16b   | $E_t=5, M=7$       | 31,104 / 32,768 | 5.1%  |

Table 50: Figures with no generating file in the repository. Each is retained, flagged, and should be re-measured before any of them is relied upon.

| figure                | where it appears                                          |
|-----------------------|-----------------------------------------------------------|
| 106.68, 215.90, 63.04 | extended formats: double-double, quad-double, x87         |
| 0.113                 | takum16 taper slope in the staircase table                |
| 0.247                 | posit32 slope in prose (measured elsewhere here as 0.260) |
| 8.14                  | one row of the field-width table                          |
| 8.99                  | one cell of the by-band effective mantissa table          |

Table 51: Crossovers, computed by the model each taper’s form requires. A straight-line fit had overstated every geometric taper’s reach, because a line under-reports how fast a logarithm falls near the origin.

| taper          | form       | crossover, binades |
|----------------|------------|--------------------|
| posit8         | arithmetic | 1.5–2.5            |
| posit16        | arithmetic | 8.5                |
| posit32        | arithmetic | 9.2                |
| posit64        | arithmetic | 54.0               |
| takum16        | geometric  | <b>1.8</b>         |
| takum32        | geometric  | <b>2.6</b>         |
| takum64        | geometric  | <b>62.7</b>        |
| tekum16        | geometric  | <b>2.1</b>         |
| tekum32        | geometric  | <b>2.4</b>         |
| takum8, tekum8 | geometric  | TNF everywhere     |